

BST C1200 Data Sheet

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Index

1 Introduction.....	1
1.1 Block diagram.....	2
1.2 Ordering information	2
2 Abbreviated Terms	4
3 Features Summary	5
3.1 CPU.....	5
3.2 GPU	5
3.3 DSP	5
3.4 Image sensor Interface.....	6
3.5 ISP	6
3.6 Computer Vision	7
3.7 Neural NET	7
3.8 ASIL-D Safety subsystem	8
3.9 ASIL-D MCU.....	8
3.10 CODEC	9
3.11 Display.....	9
3.12 Video Output Interface	10
3.13 Switch subsystem	10
3.14 High-speed IO Interface	11
3.15 Memory Interface	11
3.16 Universal Peripherals	11
3.17 Process and Package	12
3.18 ISO26262 Information.....	12
4 Device Comparison.....	13
5 Electrical characteristics	15
5.1 Chip-level conditions.....	15
5.1.1 Absolute maximum ratings	15
5.1.2 ESD damage immunity	17
5.1.3 Thermal resistance.....	17
5.1.4 Operating ranges.....	17
5.1.5 External clock sources.....	19
5.1.6 On-chip oscillator	19
5.1.7 Maximum supply currents	20
5.1.8 Power modes	22
5.2 Power supplies requirements and restrictions	22
5.2.1 Power-up sequence	22
5.2.2 Power-down sequence	25
5.3 System modules timing.....	25

5.3.1 Reset timings parameters	25
5.4 I/O DC parameters	25
5.4.1 General purpose I/O (GPIO) DC parameters	26
5.4.2 SDIO DC parameters.....	26
5.4.3 DDR I/O DC electrical characteristics	27
5.4.4 LVDS IO DC parameters	32
5.5 I/O AC parameters.....	32
5.5.1 General purpose I/O AC parameters.....	33
5.5.2 DDR I/O AC parameters.....	33
5.6 External peripheral interface parameters	37
5.6.1 JTAG timing parameters	37
5.6.2 OSPI timing parameters	38
5.6.3 CAN FD timing parameters.....	39
5.6.4 GMAC timing parameters	40
5.6.5 UART timing parameters.....	43
5.6.6 I2C timing parameters.....	44
5.6.7 PWM_OUT timing parameters.....	45
5.6.8 ERR_RPT timing parameters	45
5.6.9 I3C timing parameters.....	45
5.6.10 SPI timing parameters.....	48
5.6.11 I2S timing parameters	49
5.6.12 TDM timing parameters	50
5.6.13 VIN timing parameters	51
5.6.14 VOUT timing parameters	52
5.6.15 SDEMMC timing parameters.....	54
5.6.16 LIN timing parameters.....	55
5.6.17 FlexRay timing parameters.....	56
5.6.18 ADC parameters	56
5.6.19 10G Serdes PHY parameters	58
5.6.20 UFS MIPI MPHY parameters.....	73
5.6.21 LVDS PHY parameters.....	76
5.6.22 DP/eDP PHY parameters	76
5.6.23 MIPI DSI/CSITX DPHY parameters	80
5.6.24 MIPI RX CDPHY parameters	81
5.6.25 USB3 PHY parameters	85
5.6.26 USB2 PHY parameters	88
5.6.27 PCIE PHY parameters	89
5.6.28 USB3 PHY DC electrical characteristics	93
5.6.29 USB2 PHY DC electrical characteristics	96
5.6.30 PCIe PHY DC electrical characteristics.....	96
5.6.31 MIPI RX CDPHY DC electrical characteristics.....	97
5.6.32 MIPI MPHY electrical characteristics.....	99
5.6.33 DDR	99

6 Boot configuration	103
6.1 Boot device interface allocation	103
6.2 Boot strap pin configuration	104
7 PCB Guideline or Requirement	105
7.1 PCB Guideline.....	105
8 Package information and contact assignments.....	106
8.1 Package outline Dimension	106
8.2 Supplies contact assignments and functional contact assignments	108
8.3 Pinmux and Ball map	110
9 Revision history	112

Tables

Table 1-1	Ordering information	3
Table 2-1	Abbreviated Terms	4
Table 4-1	Device Comparison.....	13
Table 5-1	Absolute maximum ratings	15
Table 5-2	ESD damage immunity	17
Table 5-3	Thermal Resistance Characteristics	17
Table 5-4	Supply Operating ranges.....	17
Table 5-5	Temperature Operating ranges	19
Table 5-6	Specification of oscillator	19
Table 5-7	Maximum supply currents	20
Table 5-8	Power domain	22
Table 5-9	A Power-up Sequence Example.....	22
Table 5-10	Table Reset timing parameter	25
Table 5-11	Recommended Operating Conditions	26
Table 5-12	DC Characteristics	26
Table 5-13	Recommended Operating Conditions	26
Table 5-14	DC Characteristics	26
Table 5-15	Common DC Input Conditions	27
Table 5-16	Common DC Input Conditions for Flyover Bypass.....	27
Table 5-17	Common DC Input Conditions for BP_PWROK	28
Table 5-18	LPDDR4 DC Input Conditions	28
Table 5-19	LPDDR4X DC Input Conditions	28
Table 5-20	LPDDR5 DC Input Conditions	29
Table 5-21	Common DC Output Parameters	29
Table 5-22	LPDDR4 DC Output Parameters	30
Table 5-23	LPDDR4X DC Output Parameters	30
Table 5-24	LPDDR5 DC Output Parameters	31
Table 5-25	Common DC Output Conditions for Flyover Bypass1	31
Table 5-26	Common DC Output Conditions for SEC.....	31
Table 5-27	Recommended Operating Conditions	32
Table 5-28	DC Characteristics	32
Table 5-29	Maximum frequency of operation.....	33
Table 5-30	Common AC Input Conditions	33
Table 5-31	LPDDR4 AC Input Conditions	33
Table 5-32	LPDDR4X AC Input Conditions	34
Table 5-33	LPDDR5 AC Input Conditions	35
Table 5-34	Common AC Output Parameters	36
Table 5-35	Output Timing Characteristics	36
Table 5-36	Timing Requirements for IEEE 1149.1 JTAG	37
Table 5-37	Switching Characteristics Over Recommended Operating Conditions for IEEE 1149.1 JTAG	37

Table 5-38	OSPI SDR mode timing parameter	38
Table 5-39	OSPI DDR mode timing parameter	39
Table 5-40	CANFD timing parameter.....	40
Table 5-41	GMAC signals map	40
Table 5-42	RGMII signal switching specifications.....	41
Table 5-43	RMII signal switching specifications.....	42
Table 5-44	MDIO signal switching specifications.....	43
Table 5-45	I2C timing parameters.....	44
Table 5-46	PWM output timing parameters.....	45
Table 5-47	I3C timing parameters.....	47
Table 5-48	SPI master timing parameter.....	48
Table 5-49	SPI slave timing parameter	49
Table 5-50	I2S master timing parameter.....	50
Table 5-51	I2S slave timing parameter	50
Table 5-52	TDM timing parameter	51
Table 5-53	VIN signals map	51
Table 5-54	VIN timing parameters	52
Table 5-55	VOUT signals map	52
Table 5-56	VOUT timing parameters.....	53
Table 5-57	DS/HS/SDR12/SDR25/SDR50 mode timing parameter	54
Table 5-58	EMMC bus speed modes summarizes	54
Table 5-59	Timing parameters of EMMC(SDR Mode).....	55
Table 5-60	Timing parameters of EMMC(HS200 Mode).....	55
Table 5-61	LIN timing parameter	56
Table 5-62	Flexray timing parameter.....	56
Table 5-63	Timing parameter of ADC	56
Table 5-64	Reference Clock Input Requirements	58
Table 5-65	Reference Clock External Pad Input Parasitics.....	59
Table 5-66	Reference Resistor Requirements	60
Table 5-67	AC Coupling Capacitors - Target Protocol.....	62
Table 5-68	External Component Requirements	62
Table 5-69	Package Guidelines for RX and TX Differential Routes	62
Table 5-70	Driver DC specification	64
Table 5-71	Receiver DC specification	64
Table 5-72	Driver AC specification	64
Table 5-73	Receiver AC specification.....	65
Table 5-74	Transmitter characteristics for 10GBASE-KR.....	65
Table 5-75	Transmitter output waveform requirements related to coefficient update	70
Table 5-76	Transmitter output waveform requirements related to coefficient status	71
Table 5-77	Receiver characteristics for 10GBASE-KR.....	71
Table 5-78	10GBASE-KR interference tolerance parameters	72
Table 5-79	Reference clock Random Jitter (RJ) Characteristics.....	73
Table 5-80	Reference Clock Electrical Characteristics.....	73

Table 5-81	Model of Reference Clock Pad Input Load	74
Table 5-82	Table Reference Resistor Requirements	75
Table 5-83	Example Supply Bypass Capacitors	75
Table 5-84	External Component Requirements	75
Table 5-85	LVDS timing parameters	76
Table 5-86	Reference Clock Requirements: Random and Deterministic Jitter.....	77
Table 5-87	Reference Clock Electrical Characteristics	78
Table 5-88	Model of Reference Clock Pad Input Load	79
Table 5-89	Reference Resistor Requirements	79
Table 5-90	Example Supply Bypass Capacitors	80
Table 5-91	External Component Requirements	80
Table 5-92	Maximum Pad/Package/Board Requirements	80
Table 5-93	Supported Power Supply Limits at the PHY Macro Bumps.....	80
Table 5-94	TX DPHY High Speed Clock Timings	80
Table 5-95	MIPI RX DPHY High Speed Clock Timings.....	81
Table 5-96	MIPI RX DPHY Calibration and Global Timing Parameters	82
Table 5-97	MIPI RX CPHY Calibration and Global Timing Parameters	84
Table 5-98	MIPI RX DPHY Receiver Eye Diagram Details	84
Table 5-99	MIPI RX CPHY Receiver Eye Diagram Details	84
Table 5-100	Reference Clock Requirements: Random and Deterministic Jitter.....	85
Table 5-101	Reference Clock Electrical Characteristics	86
Table 5-102	Model of Reference Clock Pad Input Load	87
Table 5-103	Reference Resistor Requirements	87
Table 5-104	AC Coupling Capacitors	88
Table 5-105	Example Supply Bypass Capacitors	88
Table 5-106	External Component Requirements	88
Table 5-107	Top-Level Pin Settings for Supported Reference Clocks (in functional mode)...	88
Table 5-108	Top-Level Pin Settings for Supported Reference Clocks in TESTBURNIN Mode	89
Table 5-109	Reference Clock Jitter Requirements for PCIe 1.1,2.1,3.1 and 4.0.	89
Table 5-110	Reference Clock Amplitude Requirements	89
Table 5-111	Reference Clock Electrical Characteristics	90
Table 5-112	Reference Clock External Pad Input Parasitics.....	91
Table 5-113	Reference Resistor Requirements	92
Table 5-114	AC Coupling Capacitors - - Target Protocols	93
Table 5-115	External Component Requirements	93
Table 5-116	Supported Power Supply Limits at the PHY Macro Bumps	93
Table 5-117	Maximum Power Supply Slew Rates.....	96
Table 5-118	Maximum Pad/Package/Board Requirements.....	96
Table 5-119	Supported Power Supply Limits at the PHY Macro Bumps	96
Table 5-120	MIPI RX DPHY and CPHY Input DC Specifications	97
Table 5-121	MIPI RX DPHY HS Line Receiver DC Specifications	97
Table 5-122	MIPI RX CPHY HS Line Receiver DC Specifications	98

Table 5-123	MIPI RX DPHY and CPHY LP Line Receiver DC Specifications.....	98
Table 5-124	MIPI RX DPHY and CPHY Contention Line Receiver DC Specifications	99
Table 5-125	Recommended Operating Conditions ¹	99
Table 6-1	Interface allocation during boot	103
Table 6-2	Boot strap pin configuration	104
Table 8-1	Supplies contact assignments.....	108
Table 9-1	Revision history	112

Figures

Figure 1-1	Block Diagram	2
Figure 1-2	Marking Diagram	2
Figure 5-1	Reset timing diagram	25
Figure 5-2	Load circuit for output.....	32
Figure 5-3	Output transition time waveform	33
Figure 5-4	JTAG Timing.....	38
Figure 5-5	SDR mode input timing.....	38
Figure 5-6	SDR mode output timing.....	38
Figure 5-7	DDR mode timing	39
Figure 5-8	RGMII transmit signal timing diagram.....	42
Figure 5-9	RGMII receive signal timing diagram.....	42
Figure 5-10	RMII transmit signal timing diagram.....	42
Figure 5-11	RMII receive signal timing diagram.....	42
Figure 5-12	MDIO transmit signal timing diagram	43
Figure 5-13	MDIO receive signal timing diagram.....	43
Figure 5-14	Serial data format	43
Figure 5-15	I2C timing	44
Figure 5-16	PWM timing.....	45
Figure 5-17	Error report pin timing	45
Figure 5-18	I3C START Timing	46
Figure 5-19	I3C STOP Timing.....	46
Figure 5-20	Controller SDR Timing	47
Figure 5-21	Controller DDR Timing	47
Figure 5-22	SPI timing.....	48
Figure 5-23	I2S bus stereo frame format	49
Figure 5-24	Timing for I2S transmitter.....	50
Figure 5-25	Timing for I2S receiver	50
Figure 5-26	TDM interface timing.....	51
Figure 5-27	VIN timing diagram	52
Figure 5-28	VOUT timing diagram	53
Figure 5-29	SD input timing.....	54
Figure 5-30	SD output timing	54
Figure 5-31	Data input/output timing	55
Figure 5-32	Timing Waveform of ADC	57

Figure 5-33	Timing Waveform of RSTN Function.....	57
Figure 5-34	VDREF_CLK.....	59
Figure 5-35	VDREF_CLK_ALT	59
Figure 5-36	Model of Reference Clock Pad Input Load.....	59
Figure 5-37	External Reference Resistor Connectivity for a Single PMA	61
Figure 5-38	External Resistor Sharing.....	61
Figure 5-39	Reference data and clock circuit	63
Figure 5-40	Driver Clock and Data Alignment.....	64
Figure 5-41	Transmit test fixture for 10GBASE-KR.....	66
Figure 5-42	Transmitter differential peak-to-peak output voltage definition.....	67
Figure 5-43	Minimum differential output return loss	68
Figure 5-44	Minimum common-mode output return loss.....	68
Figure 5-45	Transmit equalizer example	69
Figure 5-46	Transmitter output waveform.....	70
Figure 5-47	Model of Reference Clock Pad Input Load.....	75
Figure 4-48	LVDS timing diagram	76
Figure 5-49	Model of Reference Clock Pad Input Load.....	79
Figure 5-50	Model of Reference Clock Pad Input Load.....	87
Figure 5-51	Model of Reference Clock Pad Input Load.....	91
Figure 5-52	Specified DC Limits.....	94
Figure 5-53	vp/vptxX/vpdig AC Limits at the Die	94
Figure 5-54	Specified AC Limits.....	94
Figure 5-55	Specified DC Limits.....	95
Figure 5-56	vph AC Limits at the Die.....	95
Figure 5-57	Specified AC Limits	95
Figure 5-58	Core Voltage Tolerance for Minimum VDD Voltage	102
Figure 5-59	Core Voltage Tolerance for Maximum VDD Voltage.....	102
Figure 5-60	Core Voltage Tolerance for Minimum VDDQ Voltage	103
Figure 8-1	Package outline dimension.....	108

1 Introduction

BST Wudang C1200 is a family of SoCs, comprising multiple different series. Currently, it includes the C129X and C123X series, focusing respectively on cross-domain fusion scenarios and NOA scenarios. The C1296 and C1236 are representative chips from these two series. In the following Data Sheet descriptions, unless otherwise specified, references to C1200 default to the specifications of the C1296, while detailed comparisons of the differences among other chips will be provided in Chapter 4.

As a high-performance, low-power image sensing, real-time computer vision and neural network processing ASIC platform, C1200 is designed for a wide range of embedded applications with a central focus on camera-based sensing and AI computing for autonomous driving systems and IVI systems, automotive gateway systems.

BST C1200 deploys a heterogeneous multi-core compute architecture consists of:

- Application processor cluster based on multi-core ARM Cortex-A family processor IPs
- multi-core Cortex-R5F running in DCLS to perform Decision and Control.

BST C1200 includes dual-32-bit LPDDR5 high performance memory interfaces.

To meet decision and control requirements, BST C1200 integrates an extended MCU sub-system which has its own power supply and safety requirements.

To meet automotive safety and security requirements, BST C1200 deploys a dedicated safety and security sub-system that meets ASIL-D requirements. Several other modules involved in sensing for autonomous driving are designed according to ASIL-B requirements.

1.1 Block diagram

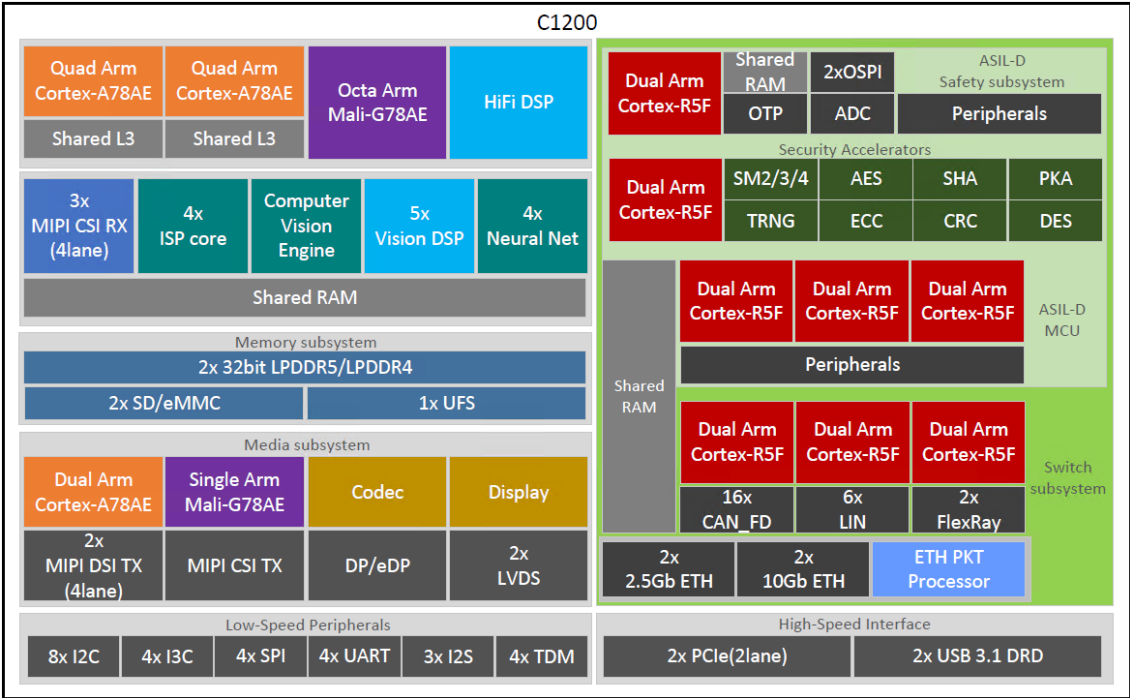


Figure 1-1 Block Diagram

1.2 Ordering information

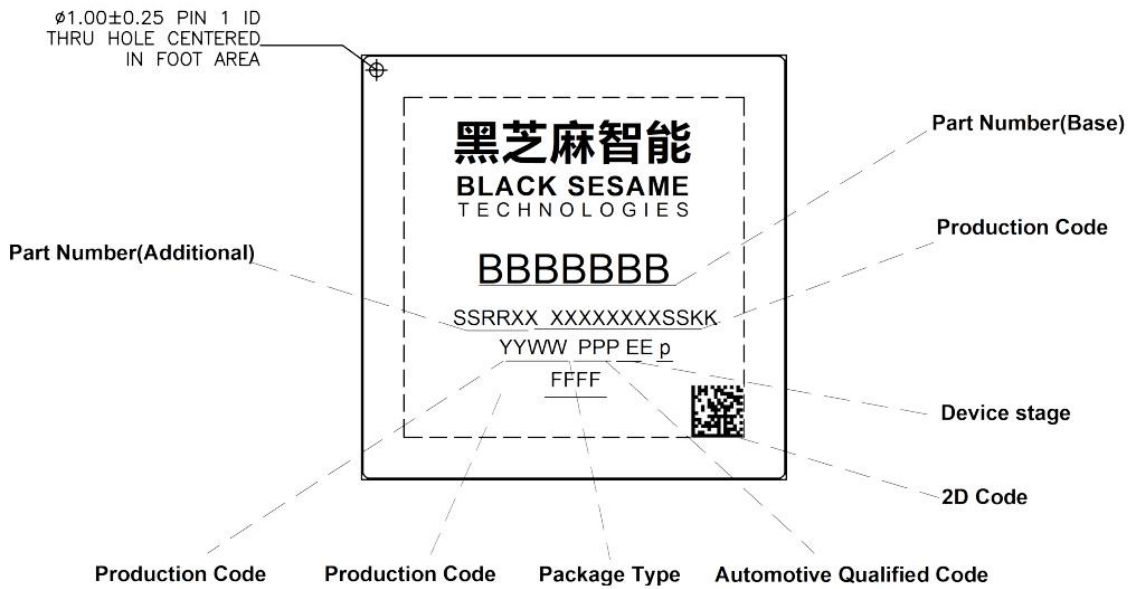


Figure 1-2 Marking Diagram

Table 1-1 Ordering information

Field	Field Parameter	Field Description	Values	Value Description
Part Number (Base)	BBBBBBBB	Base part number	C1200	
Part Number (Additional)	SS	Device revision	As Marked	
	RR	Device revision	As Marked	
	Z	Speed grade	As Marked	
	Y	Device type	As Marked	
Production Code	XXXXXXXXSSKK	Production code 1	As Marked	
	YY	Production code 2	As Marked	Year Code
	WW	Production code 3	As Marked	Week Code
	FFFF	Production code 4	As Marked	
Package Type	PPP	Package Type	FBA	FCBGA
Automotive qualified code	EE	Automotive qualified	Q1	AEC-Q100 grade 2 qualified
Device stage	p	Device stage	e/x	

2 Abbreviated Terms

Table 2-1 Abbreviated Terms

Abbreviation	Description
ASIL	Automotive Safety Integrity Level
BIST	Built In Self Test
CAN	Controller Area Network
CAN FD	CAN with Flexible Data-Rate
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
CV	Computer Vision
CODEC	Coder/Decoder
DDR	Double Data Rate
DMA	Direct Memory Access
DSP	Digital Signal Processing
DP/eDP	DisplayPort/embedded DisplayPort
ESD	Electro-Static Discharge
GPIO	General Purpose I/O
GPU	Graphics Processing Unit
I2C	Inter-Integrated Circuit
I3C	Improved Inter Integrated Circuit
I2S	Integrated Interchip Sound
ISP	Image Signal Processing
JTAG	Joint Test Action Group
LVDS	Low-Voltage Differential Signaling
MIPI	Mobile Industry Processor Interface
OTP	One Time Password
OSPI	Octal Serial Peripheral Interface
PCIe	PCI Express
PLL	Phase Lock Loop
PWM	Pulse Width Modulation
RAM	Random Access Memory
ROM	Read Only Memory
SOC	System-on-chip
SPI	Serial Peripheral Interface
SERDES	SERializer/DESerializer
UART	Universal Asynchronous Receiver Transmitter
VIN	Video input
VOU	Video output
PVT	Process, voltage and temperature sensor

3 Features Summary

3.1 CPU

- 2x Quad-core ARM® Cortex®-A78AE(1.7GHz)
 - 64KB I- and D- L1 cache
 - 512KB L2 cache per core
 - 2MB On-Chip L3 RAM with ECC per Quad-core cluster
- 1x Dual-core ARM Cortex-A78AE(1.0GHz)
 - 64KB I- and D- L1 cache
 - 512KB L2 cache per core
- Total 8x Dual ARM® Cortex®-R5F(1.2GHz) with floating point extensions and optional lockstep operation
 - 32KB I- and 32KB D- L1 cache per core
 - 32KB ATCM + 2x16KB BTCM per core
 - On-Chip RAM with ECC dedicated for R5F
- Single core ARM® Cortex®-R5F(400MHz) with floating point extensions
 - 32KB ATCM
 - On-Chip RAM with ECC dedicated for R5F

3.2 GPU

- 1x Octa-core ARM® Mali®-G78AE(900MHz)
 - OpenCL up to 2.2
 - OpenGL ES 1.1,2.0 and 3.2
 - Safety critical drivers for OpenGL SC1.0 and 2.0
 - Vulkan 1.0 and 1.1
 - Up to 28.8Gtex/s Texture Fill Rate
 - Up to 14.4Gpix/s Pixel Fill Rate
 - Up to total 460 GFLOPs(FP32)
- 1x Single-core ARM® Mali®-G78AE(900MHz)
 - OpenCL up to 2.2
 - OpenGL ES 1.1,2.0 and 3.2
 - Safety critical drivers for OpenGL SC1.0 and 2.0
 - Vulkan 1.0 and 1.1
 - Up to 3.6Gtex/s Texture Fill Rate
 - Up to 1.8Gpix/s Pixel Fill Rate
 - Up to 57.6 GFLOPs(FP32)

3.3 HiFi DSP

- Single-core HIFI DSP
 - 32KB ICAHCHE & 64KB DCACHE
 - 128KB IRAM & 256KB DRAM

- 19.2 Float GOPs
- 22.4 Int GOPs
- Single-core NNE: 128KB UBUF and 204GOPs with 128 8-bit MAC

3.4 Image sensor Interface

- 3x MIPI CSI2 Rx Ports
 - D-PHY v2.1: 4.5Gbps/lane on 4 data-lanes per port, 18 Gbps/port, up to 54Gbps total
 - C-PHY v1.2: 7.98Gbps/trio on 3 trios per port, 23.94Gbps/port, up to 71.82Gbps total
- Support 4 virtual channel per port, up to 12 virtual channels total
- Up to 12 camera streams with a typical high-performance configuration of
 - Stereo (2 streams) 4K@30fps, up to 20-bit raw input for far distance vision (> 150m)
 - 2 streams of 1080P@30fps for DMS and OMS
 - 8 streams of 1080P@30fps for near distance vision (<50m)
- 16-bit Parallel input, up to 1080p
- SLVS-EC and HiSPi sensor supported via external bridges

3.5 ISP

- 4 ISP cores
- Each ISP core runs at 600MHz, with one-pixel/cycle processing capability
- Each ISP core has 3 views output to DDR
 - View0 support YUV422/420 output with down scale
 - View1 support YUV422/420 output with down scale
 - View2 support YUV422/420 output with crop
- Adjustable AE and AWB
- Adjustable 33 knee point inversed PWL
- RGGGB/RCCCB/RYYB/RGB-IR sensor support, 4C sensor support
- Lens shading correction, dynamic color shading correction
- Defect pixel correction
- Color correction
- Raw DNS and UV DNS
- Bayer CFA de-mosaicing
- Edge-adaptive sharpening
- Gamma correction and contrast adjustment
- Tone mapping
- Cropping, resizing
- Multi-exposure HDR
 - up to 3 exposures per stream
 - 120dB dynamic range
- Night vision enhancement
- LED flick suppression

3.6 Computer Vision

Computer vision acceleration is realized using a combination of dedicated HW acceleration modules and Vision DSP. The following functions are implemented in dedicated HW acceleration module:

- GWARDP: Affine/perspective transform
 - 1000MHz running frequency
 - Support bilinear, bicubic interpolation
 - Support RGB/YUV422/420 input/output
 - LUT based warp engine
- Scaler: Upscale/Downscale acceleration engine
 - Support RGB/YUV422/420 input/output
 - Support polyphase and pyramid mode
- Shareram: 6MBytes with two banks running at 900MHz frequency

In addition, there are four high-performance wide SIMD Vision DSP processors, each capable of performing 32-way or 64-way computations. They are used to support the following most commonly Used CV kernels or functions:

- 4 Vision DSP running at 1000MHz frequency
 - 64KB ICACHE & 256KB DRAM per core
 - Fixed-Point 256GMACs per core
 - Floated-Point 32GFLOPs per core
- Support OpenVX, eigen and other CV library for following APIs
 - Stereo vision
 - Optical flow/SFM
 - SLAM
 - Image pyramid
 - Feature extraction (HOG, Harris, ORB)

3.7 Neural NET

- Dedicated DSP for control logic and custom operator
- HW based CNN with chaining architecture
- Optimized for big image processing
 - True 32 TOPS @ 1GHz
 - 3-D MAC array with >85% utilization
 - Spatial mode and group convolution mode
 - Double IO throughput
 - Shared SRAM with DSP to further reduce IO pressure
 - Structured weight pruning and compression
 - Up-to 4x sparsity computation speed up
 - INT8/INT16 data precision
- Programmable vectored math engine
 - Generic math functions (add/sub/mul/div)
 - Non-linear functions (ReLU, sigmoid, tanh, exp, sqrt)

- Special layer processing (softmax, etc.)

3.8 ASIL-D Safety subsystem

- Security
 - 1.2GHz dual-core dedicated secure ARM Cortex-R5F processor configured as DCLS (dual-core lockstep)
 - 512KB SRAM
 - Secure OTP and security boot ROM
 - HW true random number generator
 - Watchdog timer for secure MCU
 - Encryption/Decryption Engine: AES, SHA2(SHA-224/SHA-256), CRC, SM2/3/4, RSA/ECC (elliptic curve)
- Safety
 - 1.2GHz dual-core ARM Cortex-R5F safety processor configured as DCLS (dual-core lockstep)
 - 1MB SRAM
 - Run-time functional and BIST self-test control
 - Watchdog timer
 - PVT monitor
 - PMU status control
 - Safety DMA
- Shared SRAM with ECC protection
- Peripheral Interface(Max Num by Pinmux)
 - Support 1x RGMII or RMII
 - 2x CAN-FD
 - 2x XIP capable OSPI interface
 - 1x SPI master or slave
 - 2x UART
 - 2x I2C
 - 1x ADC with 12 channels

3.9 ASIL-D MCU

- 6-core ARM Cortex-R5F(1.2GHz) configured as DCLS(dual-core lockstep)
- Safety DMA
- Safety SRAM(total 8MB) for the cpu running
- Watchdog timer
- Peripheral Interface(Max Num by Pinmux)
 - 6x CAN-FD
 - 2x LIN
 - 2x FlexRay
 - 2x SPI master or slave
 - 4x I2C
 - 2x I3C

- 4x UART
- 1x I2S
- 1x TDM
- 16x PWM OUT

3.10 CODEC

- Dual 3core CODEC
- Support up to 4K@60 Encode or 4K@60 Decode per 3core CODEC
- Encode standards supported:
 - HEVC (H.265) Main.
 - HEVC (H.265) Main 10 profile.
 - H.264 Baseline Profile (BP).
 - H.264 Main Profile (MP).
 - H.264 High Profile (HP).
 - VP8.
 - VP9 Profile 0.
 - VP9 Profile 2 at 10-bit depth.
 - JPEG, baseline sequential, YUV420 and YUV422.
- Decode standards supported:
 - AVS Part 16 (AVS+, Guandongian).
 - AVS2 Part 2 (GY/T 299.1) Main Picture/Main/Main10 bit profile.
 - HEVC (H.265) Main/Main10.
 - H.264 BP/MP/HP.
 - VP8.
 - VP9 Profile 0.
 - MPEG4 SP/ASP.
 - MPEG2 MP.
 - H.263 Profile 0.
 - JPEG, baseline sequential, YUV420 and YUV422

3.11 Display

- 2*Dual PIPE Display and 1*Single PIPE Display
- Each PIPE support max 4K@60 resolution ratio
- Layer pre-processing
 - Input supports various RGB and YUV pixel formats.
 - Image rotation by 90,180, and 270 degrees, flip horizontally and vertically.
 - Image cropping, Image scaling and enhancement, color space conversion, gamma transformation.
- Multi-layer composition:
 - Composition of up to eight layers.
 - Programmable Z-order of display layers before scaling.
 - Support layer alpha
- Rich post-processing features:

- Scaling and memory-writeback of composition result.
 - Dithering, color space conversion, gamma transformation.
- Support of display formats:
 - Dual link output and side-by-side processing.
 - Video and Command mode displays. Tearing effect support.

3.12 Video Output Interface

- 2x MIPI DSI Port
 - D-PHY v1.2: 2.5Gbps/lane on 4 data-lanes per port, 10 Gbps/port, up to 20Gbps total
- 1x DisplayPort 1.4a Port
 - up to 8.1Gbps/lane, max 32.4Gbps/port
 - 1Mbps AUX channel
 - Embedded Display Port (eDP) 1.4b
 - Support HDCP 2.2, HDCP 1.3
- 2x LVDS Port
 - 1-channel clk and max 5-channel data lane per port
 - Support RGB888 1080P60 per port

3.13 Switch subsystem

- 6-core ARM Cortex-R5F(1.2GHz) configured as DCLS(dual-core lockstep)
- Total 4MB SRAM
- Watchdog timer
- Encryption/Decryption Engine
- Ethernet Packet Processor
 - External SERDES Port: 2x 10Gbps + 2x 2.5Gbps
 - Internal MAC Port: 1x 10Gbps + 3x 1Gbps
 - Internal Host port: 1x host port support 8 virtual MAC
 - L2 switch
 - VLAN support
 - Classifier
 - Configurable ACL
 - IPsec
 - MACsec
 - Qos
- Support TSN features:
 - PTP(1588 V2),gPTP(802.1as)
 - 802.1QAV
 - 802.1QBV
 - 802.1CB
- Peripheral Interface(Max Num by Pinmux)
 - 16x CAN-FD
 - 6x LIN

- 2x FlexRay

3.14 High-speed IO Interface

- 2x USB3.1/USB2.0 DRD
 - Support host mode or device mode
 - Support USB3.1 SSP(10Gbps) mode
 - Support USB3.0 SuperSpeed(5Gbps) mode
 - Support USB2.0 high-speed (480Mbps) mode
 - Support USB2.0 full-speed (12Mbps) mode
 - Support USB2.0 low-speed (1.5Mbps) mode only in host mode
- PCIE Gen4 interface
 - Support RC or EP mode
 - Configurable to 2x 2lane or 1x 4lane mode
 - Support max speed Gen4(16GT/s), compliant with Gen3(8GT/s), Gen2(5GT/s), Gen1(2.5GT/s)

3.15 Memory Interface

- 2x 32-bit LPDDR5 or LPDDR4/4X memory interfaces
 - Support LPDDR5 up to 6400Mbps
 - LPDDR5 support link-ecc/inline-ecc
 - Support LPDDR4/4X up to 4266Mbps
 - LPDDR4/4x support inline-ecc
- 2x eMMC/SD interfaces
 - 4-bit/8-bit eMMC5.1 host support legacy, high speed SDR and HS200 modes.
 - 4-bit SD4.2 host support default speed(DS), high speed(HS), SDR12, SDR25,SDR52,SDR104 modes
- 1x UFS3.1 interfaces
 - Support 2 lanes, compliant with 1 lane
 - Support HS-G4(11.648Gbps)/HS-G3(5.824Gbps)/HS-G2(2.912Gbps)/and HS-G1(1.456Gbps)
 - Support PWM-G4(24~72Mbps)/PWM-G3(12~36Mbps)/PWM-G2(6~18Mbps)/PWM-G1(3~9Mbps)

3.16 Universal Peripherals

(Max Num by Pinmux)

- 8x I2C
- 4x I3C
- 4x UART
- 2x SPI master or slave
- 3x I2S
- 4x TDM

- 2x FlexRay
- 16x PWM_OUT
- GPIOs

3.17 Process and Package

- 7nm Technology
- 31mm x 31mm, 0.8mm pitch, 1355-pin BGA

3.18 ISO26262 Information

Function safety support up to ASIL D.

4 Device Comparison

Table 4-1 Device Comparison

FEATURES	REFERENCE NAME	C1296	C1236
PROCESSORS AND ACCELERATORS			
MAIN MPU	Arm A78AE	2 x Quad-core ARM® Cortex®-A78AE(1.7GHz) 2MB On-Chip L3 RAM with ECC per Quad-core cluster	
ISOLATED MPU	Arm A78AE	1 x Dual-core ARM Cortex-A78AE(1.0GHz)	NA
Arm Cortex-R5F	Safety	Dual-core ARM Cortex-R5F (1.2GHz)	
	Security	Dual-core ARM Cortex-R5F (1.2GHz)	
	Realtime	Six-core ARM Cortex-R5F(1.2GHz)	
	Switch	Six-core ARM Cortex-R5F(1.2GHz)	
	DMA R5F	Single core ARM® Cortex®-R5F(400MHz)	
	Lockstep	Yes	
DSP	HiFi5 DSP	1	NA
	P6 Vision DSP	5 x 1GHz	
Deep Learning Accelerator	DynamAI NN Engine	1 x 1GHz (32TOPS)	
MAIN GPU	G78AE	1 x Octa-core ARM® Mali®-G78AE(900MHz)	1 x Quad-core ARM® Mali®-G78AE(900MHz)
ISOLATED GPU	G78AE	1 x Single-core ARM® Mali®-G78AE(900MHz)	NA
Hardware Computer Vision Accelerators	HWCV	1 x Scaler+2 x GWARP	
Image Signal Processing	NeuralIQ ISP	4 x 600MHz	
Video Encoder / Decoder	Codec	4K@120fps	
PROGRAM AND DATA STORAGE			
On-Chip Shared Memory (RAM)	Switch	4MB	
	Realtime	8MB	
	Security	512KB	
	Safety	1MB	
DDR Subsystem	DDR	2 x 32-bit LPDDR5 up to 6400MHz or LPDDR4/4X up to 4266MHz	
PERIPHERALS			
Display	PIPE Display	2 x Dual + 1 x Single	1 x Dual
Video Output Interface	eDP 4L	1	NA
	DSI 4L TX	2	
	LVDS	2	NA

CAN-FD	Safety	2
	Realtime	6
	Switch	16
LIN	Realtime	2
	Switch	6
Flexray	Realtime	2
	Switch	2
	Main	2
I2C	Safety	2
	Realtime	4
	Main	8
I3C	Realtime	2
	Main	4
ADC	Safety	1 x ADC with 12 channels
I2S	Realtime	1
	Main	3
TDM	Realtime	1
	Main	4
CSI2	Main	3 x 4 Lane D-PHY/ 3 trios C-PHY
Ethernet Switch	Switch	Yes
Ethernet Interfaces	Switch	2 x 10Gbps
	Switch	2 x 2.5Gbps
	Safety	1 x RGMII or RMII
eMMC/SD	Main	2 x eMMC/SD
UFS	Main	1 x UFS3.1
OSPI	Safety	2
SPI	Safety	1
	Realtime	2
	Main	2
PCIe	Main	2 x 2Lane or 1 x 4Lane
PWM_OUT	Realtime	4
	Main	16
UART	Safety	2
	Realtime	4
	Main	4
USB3.1	Main	2 x USB3.1



Notes:

Digital Peripherals is the maximum number by pinmux.

5 Electrical characteristics

5.1 Chip-level conditions

5.1.1 Absolute maximum ratings

Stresses beyond those listed under Table 5-1 may affect reliability or cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated in the operating ranges or parameters tables is not implied.

Table 5-1 Absolute maximum ratings

Symbol	Min(V)	Max(V)	Notes
AVDD_ADC_1V8	0	1.98	
AVDD_CSI_0V75	0	0.82	
AVDD_CSI_1V8	0	1.98	
AVDD_CSIVDD_0V75	0	0.82	
AVDD_DDR0_1V8	0	1.98	
AVDD_DDR0_VDD2H	0	LP5: 1.12V LP4X: 1.17V LP4: 1.17V	
AVDD_DDR0_VDDQ	0	LP5: 0.37V/0.57V LP4X: 0.65V LP4: 0.82V	
AVDD_DDR1_1V8	0	1.98	
AVDD_DDR1_VDD2H	0	LP5: 1.12V LP4X: 1.17V LP4: 1.17V	
AVDD_DDR1_VDDQ	0	LP5: 0.37V/0.57V LP4X: 0.65V LP4: 0.82V	
AVDD_DP_0V75	0	0.82	
AVDD_DP_1V8	0	1.98	
AVDD_DSI_1V8	0	1.98	
AVDD_DSIVDD_0V75	0	0.82	
AVDD_DSIVP_0V75	0	0.82	
AVDD_DSIVPL_0V75	0	0.82	
AVDD_EFUSE_1V8	0	1.98	
AVDD_ETH0_1V8	0	1.98	
AVDD_ETH1_1V8	0	1.98	
AVDD_ETH_0V75	0	0.82	
AVDD_OSC_1V8	0	1.98	
AVDD_OTP0_1V8	0	1.98	
AVDD_PCIE0_1V8	0	1.98	
AVDD_PCIE1_1V8	0	1.98	
AVDD_PCIE_0V75	0	0.82	
AVDD_PLL0_0V75	0	0.82	
AVDD_PLL0_1V8	0	1.98	

AVDD_PLL1_0V75	0	0.82	
AVDD_PLL1_1V8	0	1.98	
AVDD_PLL2_0V75	0	0.82	
AVDD_PLL2_1V8	0	1.98	
AVDD_PLL_LJ0_0V75	0	0.82	
AVDD_PLL_LJ0_1V8	0	1.98	
AVDD_PLL_LJ1_0V75	0	0.82	
AVDD_PLL_LJ1_1V8	0	1.98	
AVDD_PLL_LJ2_0V75	0	0.82	
AVDD_PLL_LJ2_1V8	0	1.98	
AVDD_PLL_LJ3_0V75	0	0.82	
AVDD_PLL_LJ3_1V8	0	1.98	
AVDD_PVT0_1V8	0	1.98	
AVDD_PVT1_1V8	0	1.98	
AVDD_PVT2_1V8	0	1.98	
AVDD_PVT3_1V8	0	1.98	
AVDD_PVT4_1V8	0	1.98	
AVDD_PVT5_1V8	0	1.98	
AVDD_PVT6_1V8	0	1.98	
AVDD_PVT7_1V8	0	1.98	
AVDD_UFS_0V75	0	0.82	
AVDD_UFS_1V8	0	1.98	
AVDD_USB0_0V75	0	0.82	
AVDD_USB0_1V8	0	1.98	
AVDD_USB0_3V3	0	3.63	
AVDD_USB0_VDDH	0	3.63	
AVDD_USB1_0V75	0	1.98	
AVDD_USB1_1V8	0	1.98	
AVDD_USB1_3V3	0	3.63	
AVDD_USB1_VDDH	0	3.63	
DVDD_AON_0V75	0	0.82	
DVDD_DDR0_0V75	0	0.82	
DVDD_DDR1_0V75	0	0.82	
DVDD_ISPCV_0V75	0	0.82	
DVDD_SOC_0V75	0	0.82	
DVDD_ST_0V75	0	0.82	
VDDIO_AON_1V8	0	1.98	
VDDIO_RGMII	0	3.3	
VDDIO_SDIO_0	0	3.3	
VDDIO_SDIO_1	0	3.3	
VDDIO_SOC_1V8_0	0	1.98	
VDDIO_SOC_1V8_1	0	1.98	
VDDIO_ST_1V8	0	1.98	
DVDD_SOC_SENSE	0	0.82	
DVDD_AON_SENSE	0	0.82	

DVDD_ISPCV_SENSE	0	0.82	
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5.1.2 ESD damage immunity

Table 5-2 ESD damage immunity

	Min	Max	Unit	Notes
Human Body Model (HBM)	-	2000	V	
Charged Device Model (CDM)	-	5	A	

5.1.3 Thermal resistance

Table 5-3 provides the thermal resistance characteristics for the package used on this device. Natural convection (Four-layer board 2s2p).

Table 5-3 Thermal Resistance Characteristics

PARAMETER	DESCRIPTION	°C/W
R _{ΘJC}	Junction to case	0.167
R _{ΘJB}	Junction to board	1.44
R _{ΘJA}	Junction to free air	10.05
	Junction to moving air	6.9 ¹
Ψ _{JT}	Junction to package top	0.052 ¹
Ψ _{JB}	Junction to board	1 ¹



Note:

The air flow is 2m/s.

5.1.4 Operating ranges

Table 5-4 provides the operating ranges. Typical voltage values should be used.

Table 5-4 Supply Operating ranges

Symbol	Min(V)	Typ(V)	Max(V)	Notes
AVDD_ADC_1V8	1.719	1.8	1.881	
AVDD_CSI_0V75	0.7125	0.75	0.795	
AVDD_CSI_1V8	1.71	1.8	1.908	
AVDD_CSIVDD_0V75	0.7125	0.75	0.795	
AVDD_DDR0_1V8	1.71	1.8	1.908	
AVDD_DDR0_VDD2H	1.0185	1.05	1.1025	LP5
	1.067	1.1	1.155	LP4X
	1.067	1.1	1.155	LP4
AVDD_DDR0_VDDQ	0.285/0.475	0.3/0.5	0.318/0.53	LP5
	0.576	0.6	0.636	LP4X
	0.72	0.75	0.795	LP4
AVDD_DDR1_1V8	1.71	1.8	1.908	
AVDD_DDR1_VDD2H	1.0185	1.05	1.1025	LP5
	1.067	1.1	1.155	LP4X
	1.067	1.1	1.155	LP4
AVDD_DDR1_VDDQ	0.285/0.475	0.3/0.5	0.318/0.53	LP5
	0.576	0.6	0.636	LP4X
	0.72	0.75	0.795	LP4

AVDD_DP_0V75	0.7125	0.75	0.795	
AVDD_DP_1V8	1.728	1.8	1.908	
AVDD_DSI_1V8	1.746	1.8	1.908	
AVDD_DSIVDD_0V75	0.7275	0.75	0.795	
AVDD_DSIVP_0V75	0.7275	0.75	0.795	
AVDD_DSIVPL_0V75	0.7275	0.75	0.795	
AVDD_EFUSE_1V8	1.728	1.8	1.872	In normal function mode EFUSE is read-only,VQPS can be 0V or floating or 1.8V, suggest to be floating.
AVDD_ETH0_1V8	1.71	1.8	1.908	
AVDD_ETH1_1V8	1.71	1.8	1.908	
AVDD_ETH_0V75	0.72	0.75	0.795	
AVDD_OSC_1V8	1.71	1.8	1.908	
AVDD_OTP0_1V8	1.71	1.8	1.908	
AVDD_PCIE0_1V8	1.71	1.8	1.908	
AVDD_PCIE1_1V8	1.71	1.8	1.908	
AVDD_PCIE_0V75	0.72	0.75	0.795	
AVDD_PLL0_0V75	0.7275	0.75	0.795	
AVDD_PLL0_1V8	1.71	1.8	1.908	
AVDD_PLL1_0V75	0.7275	0.75	0.795	
AVDD_PLL1_1V8	1.71	1.8	1.908	
AVDD_PLL2_0V75	0.7275	0.75	0.795	
AVDD_PLL2_1V8	1.71	1.8	1.908	
AVDD_PLL_LJ0_0V75	0.7275	0.75	0.795	
AVDD_PLL_LJ0_1V8	1.71	1.8	1.908	
AVDD_PLL_LJ1_0V75	0.7275	0.75	0.795	
AVDD_PLL_LJ1_1V8	1.71	1.8	1.908	
AVDD_PLL_LJ2_0V75	0.7275	0.75	0.795	
AVDD_PLL_LJ2_1V8	1.71	1.8	1.908	
AVDD_PLL_LJ3_0V75	0.7275	0.75	0.795	
AVDD_PLL_LJ3_1V8	1.71	1.8	1.908	
AVDD_PVT0_1V8	1.71	1.8	1.908	
AVDD_PVT1_1V8	1.71	1.8	1.908	
AVDD_PVT2_1V8	1.71	1.8	1.908	
AVDD_PVT3_1V8	1.71	1.8	1.908	
AVDD_PVT4_1V8	1.71	1.8	1.908	
AVDD_PVT5_1V8	1.71	1.8	1.908	
AVDD_PVT6_1V8	1.71	1.8	1.908	
AVDD_PVT7_1V8	1.71	1.8	1.908	
AVDD_UFS_0V75	0.7125	0.75	0.795	
AVDD_UFS_1V8	1.71	1.8	1.908	
AVDD_USB0_0V75	0.7125	0.75	0.795	
AVDD_USB0_1V8	1.728	1.8	1.908	
AVDD_USB0_3V3	3.135	3.3	3.498	
AVDD_USB0_VDDH	3.135	3.3	3.498	

AVDD_USB1_0V75	1.71	1.8	1.908	
AVDD_USB1_1V8	1.728	1.8	1.908	
AVDD_USB1_3V3	3.135	3.3	3.498	
AVDD_USB1_VDDH	3.135	3.3	3.498	
DVDD_AON_0V75	0.6975	0.75	0.8025	
DVDD_DDR0_0V75	0.7125	0.75	0.795	
DVDD_DDR1_0V75	0.7125	0.75	0.795	
DVDD_ISPCV_0V75	0.6975	0.75	0.8025	
DVDD_SOC_0V75	0.6975	0.75	0.8025	
DVDD_ST_0V75	0.6975	0.75	0.8025	
VDDIO_AON_1V8	1.674	1.8	1.926	
VDDIO_RGMII	1.674/2.79	1.8/3.0	1.926/3.21	
VDDIO_SDIO_0	1.674/2.79	1.8/3.0	1.926/3.21	
VDDIO_SDIO_1	1.674/2.79	1.8/3.0	1.926/3.21	
VDDIO_SOC_1V8_0	1.674	1.8	1.926	
VDDIO_SOC_1V8_1	1.674	1.8	1.926	
VDDIO_ST_1V8	1.674	1.8	1.926	

Note: DVDD_SOC_SENSE/DVDD_AON_SENSE/DVDD_ISPCV_SENSE are specified SENSE Balls, should not be used as power supply.

Table 5-5 Temperature Operating ranges

	Min	Typ	Max	Unit	Notes
Temperature sensor accuracy	-	-	± 7	°C	
Junction temperature	-40	-	125	°C	



Note:

The power design should according to the typical values.

5.1.5 External clock sources

Chip needs a 25 MHz clock “XTAL_IN” and another 25 MHz clock “XTAL_INB” to used as primary clock source. An internal 1MHz clock "Ref_clk" can be used as a monitoring clock for primary clock, when a fault of primary clock occurs, error will be indicated via error pins.

- Reference clock stability need be less than 50ppm

5.1.6 On-chip oscillator

A 1MHz is used as a monitoring clock for primary clock.

Table 5-6 Specification of oscillator

Parameter	Min	Typ	Max	Unit	Comments
Frequency(after calibration)	0.975	1	1.02	MHz	
Initial settling time		0.36	0.53	ms	Output frequency is within specified range after this time

Period Jitter (RMS)		0.32	1.04	ns	Worst-case 12-Cycle jitter will be $12 \times 1.04 = 12.48\text{ns}$
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5.1.7 Maximum supply currents

Power consumption is highly dependent on the application. Estimating the maximum supply currents required for power supply design is difficult because the use cases that requires maximum supply current is not a realistic use case. The results provided are intended to be used as guidelines for power supply design.

Table 5-7 Maximum supply currents

Power Rail	Normal DC Voltage(V)	Max DC Current(A)	Max Power(W)
AVDD_ADC_1V8	1.8	0.012	0.0216
AVDD_CSI_0V75	0.75	0.111	0.08325
AVDD_CSI_1V8	1.8	0.015	0.027
AVDD_CSI_VDD_0V75	0.75	0.114	0.0855
AVDD_DDR0_1V8	1.8	0.001	0.0018
AVDD_DDR0_VDD2H	1.8	0.014	0.0252
AVDD_DDR0_VDDQ	LP5: 0.3/0.5 LP4X: 0.6 LP4: 0.75	0.5	LP5: 0.15/0.25 LP4X: 0.3 LP4: 0.375
AVDD_DDR1_1V8	1.8	0.001	0.0018
AVDD_DDR1_VDD2H	1.8	0.014	0.0252
AVDD_DDR1_VDDQ	LP5: 0.3/0.5 LP4X: 0.6 LP4: 0.75	0.5	LP5: 0.15/0.25 LP4X: 0.3 LP4: 0.375
AVDD_DP_0V75	0.75	0.088	0.066
AVDD_DP_1V8	1.8	0.058	0.1044
AVDD_DSI_1V8	1.8	0.02	0.036
AVDD_DSI_VDD_0V75	0.75	0.012	0.009
AVDD_DSIVP_0V75	0.75	0.044	0.033
AVDD_DSIVPL_0V75	0.75	0.01	0.0075
AVDD_EFUSE_1V8	1.8	0.05	0.09
AVDD_ETH_0V75	0.75	0.408	0.306
AVDD_ETH0_1V8	1.8	0.106	0.1908
AVDD_ETH1_1V8	1.8	0.106	0.1908
AVDD_OSC_1V8	1.8	0.001	0.0018
AVDD_OTP0_1V8	1.8	0.01	0.018
AVDD_PCIE_0V75	0.75	0.656	0.492
AVDD_PCIE0_1V8	1.8	0.125	0.225
AVDD_PCIE1_1V8	1.8	0.125	0.225
AVDD_PLL_LJ0_0V75	0.75	0.018	0.0135
AVDD_PLL_LJ0_1V8	1.8	0.025	0.045

AVDD_PLL_LJ1_0V75	0.75	0.018	0.0135
AVDD_PLL_LJ1_1V8	1.8	0.025	0.045
AVDD_PLL_LJ2_0V75	0.75	0.018	0.0135
AVDD_PLL_LJ2_1V8	1.8	0.025	0.045
AVDD_PLL_LJ3_0V75	0.75	0.079	0.05925
AVDD_PLL_LJ3_1V8	1.8	0.03	0.054
AVDD_PLL0_0V75	0.75	0.122	0.0915
AVDD_PLL0_1V8	1.8	0.01	0.018
AVDD_PLL1_0V75	0.75	0.244	0.183
AVDD_PLL1_1V8	1.8	0.02	0.036
AVDD_PLL2_0V75	0.75	0.671	0.50325
AVDD_PLL2_1V8	1.8	0.055	0.099
AVDD_PVT0_1V8	1.8	0.001	0.0018
AVDD_PVT1_1V8	1.8	0.001	0.0018
AVDD_PVT2_1V8	1.8	0.001	0.0018
AVDD_PVT3_1V8	1.8	0.001	0.0018
AVDD_PVT4_1V8	1.8	0.001	0.0018
AVDD_PVT5_1V8	1.8	0.001	0.0018
AVDD_PVT6_1V8	1.8	0.001	0.0018
AVDD_PVT7_1V8	1.8	0.002	0.0036
AVDD_UFS_0V75	0.75	0.091	0.06825
AVDD_UFS_1V8	1.8	0.061	0.1098
AVDD_USB0_0V75	0.75	0.116	0.087
AVDD_USB0_1V8	1.8	0.053	0.0954
AVDD_USB0_3V3	3.3	0.004	0.0132
AVDD_USB0_VDDH	3.3	0.016	0.0528
AVDD_USB1_0V75	0.75	0.116	0.087
AVDD_USB1_1V8	1.8	0.053	0.0954
AVDD_USB1_3V3	3.3	0.004	0.0132
AVDD_USB1_VDDH	3.3	0.016	0.0528
DVDD_AON_0V75	0.75	0.27	0.2025
DVDD_DDR0_0V75	0.75	2.86	2.145
DVDD_DDR1_0V75	0.75	2.86	2.145
DVDD_ISPCV_0V75	0.75	15.6	11.7
DVDD_SOC_0V75	0.75	90.67	68.0025
DVDD_ST_0V75	0.75	6	4.5
VDDIO_AON_1V8	1.8	0.33	0.594
VDDIO_RGMII	1.8/3.0	0.12	0.36
VDDIO_SDIO_0	1.8/3.0	0.1	0.3
VDDIO_SDIO_1	1.8/3.0	0.1	0.3
VDDIO_SOC_1V8_0	1.8	0.23	0.414
VDDIO_SOC_1V8_1	1.8	0.18	0.324
VDDIO_ST_1V8	1.8	0.47	0.846



Note:

The Max DC Current is the max avg current value, not the peak value.

5.1.8 Power modes

Table 5-8 Power domain

Power Domain	Power rail	Description
AON domain	VDD_AON	always on
ISP_CV domain	VDD_ISP_CV	CV Cores and ISP Cores
ST domain	VDD_ST	Safety subsystem and Realtime subsystem
SOC domain	VDD_SOC	All other subsystems

5.2 Power supplies requirements and restrictions

5.2.1 Power-up sequence

- For power supply in the same power domain, the IO power supply VDDIO_XXX_1V8 should be powered on earlier than the digital power supply DVDD_XXX_0V75;
- DVDD_AON_0V75, VDDIO_AON_1V8 should be powered on first, and RST_N can not be de-assert before the power supply is stable;
- DVDD_SOC_0V75/DVDD_ISPCV_0V75/DVDD_ST_0V75 should be powered on simultaneously and not earlier than DVDD_AON_0V75;
- The power supply demand of Analog Powers needs to comply with the following standards:
 - AVDD_DDR*_1V8 / DVDD_DDR*_0V75 / AVDD_DDR*_VDD2H / AVDD_DDR*_VDDQ have no timing requirements, and their corresponding power sequence needs to meet the DDR particle requirements
 - AVDD_ADC_1V8 should not earlier than DVDD_ST_0V75
 - AVDD_OTP_1V8 should not earlier than DVDD_ST_0V75
 - AVDD_CSIVDD_0V75 /AVDD_CSI_0V75 should not earlier than DVDD_ISP_0V75, and finally AVDD_CSI_1V8 is powered on
 - AVDD_DSIVDD_0V75/AVDD_DSIVP_0V75 / AVDD_DSIVPL_0V75 should not earlier than DVDD_SOC_0V75, and finally AVDD-DSI_1V8 should be powered on

Table 5-9 A Power-up Sequence Example

Power Rail	Voltage (Unit:V)	Recommend Power Up Sequence					Power Up Sequence Constraints
		T0	T1	T2	T3	T4	
AVDD_ADC_1V8	1.8						Ramp up after DVDD ST_0V75
AVDD_CSI_0V75	0.75						Same with DVDD ST_0V75
AVDD_CSI_1V8	1.8						Ramp up after DVDD ST_0V75
AVDD_CSIVDD_0V75	0.75						Same with DVDD ST_0V75

AVDD_DDR0_1V8	1.8			no requirement
AVDD_DDR0_VDD2H	1.05/ 1.1			no requirement
AVDD_DDR0_VDDQ	0.5/0.6/1. 1			no requirement
AVDD_DDR1_1V8	1.8			no requirement
AVDD_DDR1_VDD2H	1.05/ 1.1			no requirement
AVDD_DDR1_VDDQ	0.5/0.6/ 1.1			no requirement
AVDD_DP_0V75	0.75			Same with DVDD_ST_0V75
AVDD_DP_1V8	1.8			Same with DVDD_ST_0V75
AVDD_DSI_1V8	1.8			Same with AVDD_DSIVP_0V75
AVDD_DSIVDD_0V75	0.75			Same with DVDD_ST_0V75
AVDD_DSIVP_0V75	0.75			Ramp up after AVDD_DSIVDD_0V75
AVDD_DSIVPL_0V75	0.75			Same with AVDD_DSIVP_0V75
AVDD_EFUSE_1V8	1.8			Ramp up after DVDD_ST_0V75
AVDD_ETH0_1V8	1.8			Same with DVDD_ST_0V75
AVDD_ETH1_1V8	1.8			Same with DVDD_ST_0V75
AVDD_ETH_0V75	0.75			Same with DVDD_ST_0V75
AVDD_OSC_1V8	1.8			no requirement
AVDD_OTP0_1V8	1.8			Ramp up after DVDD_ST_0V75
AVDD_PCIE0_1V8	1.8			Same with DVDD_ST_0V75
AVDD_PCIE1_1V8	1.8			Same with DVDD_ST_0V75
AVDD_PCIE_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL0_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL0_1V8	1.8			no requirement
AVDD_PLL1_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL1_1V8	1.8			no requirement
AVDD_PLL2_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL2_1V8	1.8			no requirement
AVDD_PLL_LJ0_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL_LJ0_1V8	1.8			no requirement
AVDD_PLL_LJ1_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL_LJ1_1V8	1.8			no requirement
AVDD_PLL_LJ2_0V75	0.75			Same with DVDD_ST_0V75

AVDD_PLL_LJ2_1V8	1.8			no requirement
AVDD_PLL_LJ3_0V75	0.75			Same with DVDD_ST_0V75
AVDD_PLL_LJ3_1V8	1.8			no requirement
AVDD_PVT0_1V8	1.8			no requirement
AVDD_PVT1_1V8	1.8			no requirement
AVDD_PVT2_1V8	1.8			no requirement
AVDD_PVT3_1V8	1.8			no requirement
AVDD_PVT4_1V8	1.8			no requirement
AVDD_PVT5_1V8	1.8			no requirement
AVDD_PVT6_1V8	1.8			no requirement
AVDD_PVT7_1V8	1.8			no requirement
AVDD_UFS_0V75	0.75			Same with DVDD_ST_0V75
AVDD_UFS_1V8	1.8			Same with DVDD_ST_0V75
AVDD_USB0_0V75	0.75			Same with DVDD_ST_0V75
AVDD_USB0_1V8	1.8			Same with DVDD_ST_0V75
AVDD_USB0_3V3	3.3			Same with DVDD_ST_0V75
AVDD_USB0_VDDH	3.3			Same with DVDD_ST_0V75
AVDD_USB1_0V75	0.75			Same with DVDD_ST_0V75
AVDD_USB1_1V8	1.8			Same with DVDD_ST_0V75
AVDD_USB1_3V3	3.3			Same with DVDD_ST_0V75
AVDD_USB1_VDDH	3.3			Same with DVDD_ST_0V75
DVDD_AON_0V75	0.75			Ramp up second after VDDIO_AON_1V8
DVDD_DDR0_0V75	0.75			Same with DVDD_ST_0V75
DVDD_DDR1_0V75	0.75			Same with DVDD_ST_0V75
DVDD_ISPCV_0V75	0.75			Same with DVDD_ST_0V75
DVDD_SOC_0V75	0.75			Same with DVDD_ST_0V75
DVDD_ST_0V75	0.75			Same with DVDD_ST_0V75
VDDIO_AON_1V8	1.8			Ramp up First
VDDIO_RGMII	1.8/3.0			Ramp up before DVDD_ST_0V75
VDDIO_SDIO_0	1.8/3.0			Ramp up before DVDD_ST_0V75
VDDIO_SDIO_1	1.8/3.0			Ramp up before DVDD_ST_0V75
VDDIO_SOC_1V8_0	1.8			Ramp up before DVDD_ST_0V75
VDDIO_SOC_1V8_1	1.8			Ramp up before DVDD_ST_0V75

VDDIO_ST_1V8	1.8			Ramp up before DVDD ST 0V75
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 Note:

RST_N need keep low at least 3ms after power up

5.2.2 Power-down sequence

Preferred:

Turn off the VDDIO_SOC/VDDIO_AON/ VDDIO_SDIO_0/VDDIO_SDIO_1 last.

Acceptable:

Turn off all powers at the same time.

Not Recommended:

Turn off the VDDIO_SOC/VDDIO_AON/VDDIO_SDIO_0/VDDIO_SDIO_1 first.

 Note:

RST_N must be low before power off.

5.3 System modules timing

5.3.1 Reset timings parameters

The following figure shows the reset timing and table lists the timing parameters.

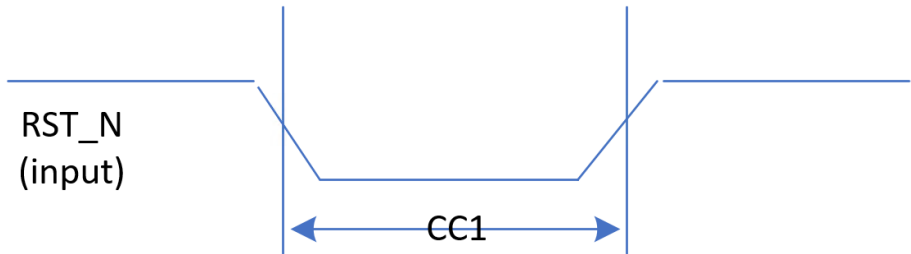


Figure 5-1 Reset timing diagram

Table 5-10 Table Reset timing parameter

ID	Parameter	Min	Max	Unit
CC1	Duration of RST_N to be qualified as 0.	60	-	ns

5.4 I/O DC parameters

This section includes the DC parameters of the following I/O types:

- General Purpose I/O (GPIO)DC parameters
- SDIO DC parameters
- DDR I/O DC parameters
- LVDS IO DC parameters

5.4.1 General purpose I/O (GPIO) DC parameters

5.4.1.1 Recommended Operating Conditions

Table 5-11 Recommended Operating Conditions

Parameter		Min.	Nom.	Max.	Units
VDD	Pre-Driver Voltage	0.72	0.75	0.82	V
VDDPST	Post-Driver Voltage	1.62	1.8	1.98	V
VIMAX	Maximum Input Voltage			1.98	V



Permanent damage could occur if the operation exceeds the ranges listed in Table 5-11.

Table 5-12 DC Characteristics

Parameter		Min.	Nom.	Max.	Units
V _{IL}	Input Low Voltage	-0.3		0.35* VDDIO	V
V _{IH}	Input High Voltage	0.65* VDDIO		VDDIO+0.3	V
V _{OL}	Output Low Voltage			0.45	V
V _{OH}	Output High Voltage	VDDIO-0.45			V
R _{PU}	Pull-up Resistor	20k	32k	49k	Ω
R _{PD}	Pull-down Resistor	20k	30k	46k	Ω

5.4.2 SDIO DC parameters

5.4.2.1 Recommended Operating Conditions

Table 5-13 Recommended Operating Conditions

Parameter		Min.	Nom.	Max.	Units
VDD	Pre-Driver Voltage	0.72	0.75	0.82	V
VDDPST	Post-Driver Voltage, 1.8V mode	1.62	1.8	1.98	V
VDDPST	Post-Driver Voltage, 3.0V mode	2.7	3.0	3.3	V
VIMAX	Maximum Input Voltage, 1.8V mode			2	V
VIMAX	Maximum Input Voltage, 3.0V mode			VDDIO+ 0.3	V



Permanent damage could occur if the operation exceeds the ranges listed in Table 5-13.

Table 5-14 DC Characteristics

Parameter		Min.	Nom.	Max.	Units
V _{IL}	Input Low Voltage, 1.8V mode	-0.3		0.58	V
V _{IH}	Input High Voltage, 1.8V mode	1.27		2	V
V _{OL}	Output Low Voltage, 1.8V mode			0.45	V

V_{OH}	Output High Voltage, 1.8V mode	1.4			V
V_{IL}	Input Low Voltage, 3.0V mode	-0.3		0.25* VDDIO	V
V_{IH}	Input High Voltage, 3.0V mode	0.625* VDDIO		VDDIO +0.3	V
V_{OL}	Output Low Voltage, 3.0V mode			0.125* VDDIO	V
V_{OH}	Output High Voltage, 3.0V mode	0.75* VDDIO			V
R_{PU}	Pull-up Resistor	18k	23k	28k	Ω
R_{PD}	Pull-down Resistor	18k	22k	27k	Ω

5.4.3 DDR I/O DC electrical characteristics

The following tables list parameters that are common across all memory interface standards. Unless otherwise specified, all input specifications (both common and DDR standard specific) are measured at the host PHY input pins.

Subsequent sections list parameters for specific memory interface standards.

5.4.3.1 DDR Input I/O DC electrical characteristics

Table 5-15 Common DC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
BP_DAT input leakage current	IIZ	uA			173	1
Input high voltage for BP_DAT	VIH-DC	V	Vref + 0.020			2,3
Input low voltage for BP_DAT	VIL-DC	V			Vref – 0.020	2,3
Differential input voltage for abs(DQS _t – DQS _c)	VID-DC	V	0.100			4



Notes:

- (1) Leakage current is measured when the pin is configured to a high-impedance state with all on-die termination disabled. Leakage is valid for any input except for Vref over the range: $0 \leq V_{IN} \leq V_{DDQ}$. All pins not under test = VSS or VDDQ.
- (2) The DC values indicate the voltage levels at which the final logic state of the receiver is unambiguously defined.
- (3) After running offset cancellation training sequence
- (4) The differential voltage VID is the difference between the true and complement signals. Absolute value taken so that minimum spec applies to both high and low differential inputs.
- (5) BP_DAT refers to DQS_t*, DMI_t* and DQ_t*.

Table 5-16 Common DC Input Conditions for Flyover Bypass

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Input high voltage for all BP *	VIH_DC	V	0.7 * VDD			1
Input low voltage for all BP *	VIL_DC	V			0.3* VDD	1



Notes:

- (1) BP_MEMRESET_L and BP_ZN do not support Flyover ByPass mode

Table 5-17 Common DC Input Conditions for BP_PWROK

Parameter	Symbol	Unit	Minimum	Typical	Maximum
Input high voltage	VIH_DC	V	0.7 * VDD2H		
Input low voltage	VIL_DC	V			0.3* VDD2H

Table 5-18 LPDDR4 DC Input Conditions

Parameter	Symbol	Unit	Mini.	Typical	Max.	Notes
On-die termination (ODT) programmable resistances	RTT	ohm		120,60,40,30		1,2,3
ODT mid-level (0.1*VDDQ) large- signal resistance	RODT-L/RTT		0.8	1.0	1.1	4
ODT high-level (0.33*VDDQ) large- signal resistance	RODT-M/RTT		0.9	1.0	1.1	4,5
ODT high-level (0.5*VDDQ) large- signal resistance	RODT-H/RTT		0.9	1.0	1.3	4



Notes:

- (1) The ODT level is programmable based on DIMM and DRAM configuration.
 (2) For LPDDR4, ODT is a pull-down to VSS.
 (3) IO operation is supported only with ODT enabled.
 (4) ODT large-signal resistance is defined for an applied input voltage VIN and a measured input current of IIN as.

$$R_{ODT} = \frac{V_{IN}}{I_{IN}}$$

- (5) Calibrated large signal resistance targets RODT-L/RTT = 1.0 across PVT. Min/Max values account for calibration resolution, mismatch, and other effects.

Table 5-19 LPDDR4X DC Input Conditions

Parameter	Symbol	Unit	Mini.	Typical	Max.	Notes
On-die termination (ODT) programmable resistances	RTT	ohm		Open 120,60,40,30		1,2
ODT mid-level (0.2*VDDQ) large- signal resistance	RODT-L/RTT		0.8	1.0	1.1	3
ODT Trid-level (0.5*VDDQ) large- signal resistance	RODT-M/RTT		0.9	1.0	1.1	3,4
ODT high-level (0.75*VDDQ) large- signal resistance	RODT-H/RTT		0.9	1.0	1.3	3



Notes:

- (1) The ODT level is programmable based on DIMM and DRAM configuration.

- (2) For LPDDR4X, ODT is a pull-down to VSS.
- (3) ODT large-signal resistance is defined for an applied input voltage V_{IN} and a measured input current of I_{IN} as.

$$R_{ODT} = \frac{V_{IN}}{I_{IN}}$$

- (4) Calibrated large signal resistance targets $RODT-L/RTT = 1.0$ across PVT. Min/Max values account for calibration resolution, mismatch, and other effects.

Table 5-20 LPDDR5 DC Input Conditions

Parameter	Symbol	Unit	Mini.	Typical	Max.	Notes
On-die termination (ODT) programmable resistances	RTT	ohm		120,60,40,30		1,2
ODT low-level (0.2*VDDQ) large- signal resistance	RODT-L/RTT		0.8	1.0	1.1	3
ODT high-level (0.5*VDDQ) large- signal resistance	RODT-M/RTT		0.9	1.0	1.1	3,4
ODT high-level (0.75*VDDQ) large- signal resistance	RODT-H/RTT		0.9	1.0	1.3	3



Notes:

- (1) The ODT level is programmable based on DIMM and DRAM configuration.
- (2) For LPDDR5, ODT is a pull-down to VSS.
- (3) ODT large-signal resistance is defined for an applied input voltage V_{IN} and a measured input current of I_{IN} as.

$$R_{ODT} = \frac{V_{IN}}{I_{IN}}$$

- (4) calibrated large signal resistance targets $RODT-L/RTT = 1.0$ across PVT. Min/Max values account for calibration resolution, mismatch, and other effect

5.4.3.2 DDR Output I/O DC electrical characteristics

Table 5-21 Common DC Output Parameters

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Output driver pull-down impedance: SE, DIFF	ROnPd	ohm		120, 60, 40,30		1
Output driver pull-up impedance: SEC	ROnSEC Pu	ohm		400,100,66.6,50		2
Output driver pull-down impedance: SEC	ROnSEC Pd	ohm		400,100,66.6,50		2
Large-signal SEC pull-up driver variation	ROnSEC Pu05 /		0.5	1.0	1.8	3

at 0.5*VDD2H	R _{On}					
Large-signal SEC pull-down driver variation at 0.5*VDD2H	R _{OnDQP d05} / R _{On}		0.5	1.0	1.8	4



Notes:

- (1) Calibrated at VOH.
- (2) SEC output driver is uncalibrated and minimum/maximum impedance value is process, voltage, and temperature dependent.
- (3) Large-signal output resistance is defined for an applied voltage V_{OUT} and a measured current of I_{OUT} as:

$$R_{out} = \frac{V_{DD2} - V_{out}}{I_{out}}$$

- (4) Large-signal pull-down output resistance is defined for an applied voltage V_{OUT} and a measured current of I_{OUT} as

$$R_{ODT} = \frac{V_{IN}}{I_{IN}}$$

Table 5-22 LPDDR4 DC Output Parameters

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
SE,DIFF pull-up output driver characteristics,with ZQ calibration:VOHPU,nom=VDDQ_DRAM/2.5	VOH,nom	mv		440		1
	VOHPU,nom	VOH,nom	0.9	1.0	1.1	
SE,DIFF pull-up output driver characteristics,with ZQ calibration:VOHPU,nom=VDDQ_DRAM/3	VOH,nom	mv		367		
	VOHPU,nom	VOH,nom	0.9	1.0	1.1	



Notes:

- (1) All values are after ZQ Calibration. Without ZQ Calibration VOH,nom values are ± 30%
- (2) VOH,nom (mV) values are based on a nominal VDDQ_DRAM = 1.1V

Table 5-23 LPDDR4X DC Output Parameters

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
SE,DIFF pull-up output driver characteristics,with ZQ calibration:VOHPU,nom=VDDQ*0.5	VOH,nom	mv		300		1,2
	VOHPU,nom	VOH,nom	0.9	1.0	1.1	1,2

SE,DIFF pull-up output driver characteristics, with ZQ calibration: VOHPU, nom = VDDQ*0.6	VOH, nom	mv		360		1,2
	VOHPU, nom	VOH, nom	0.9	1.0	1.1	1,2



Notes:

- (1) All values are after ZQ Calibration. Without ZQ Calibration VOH, nom values are $\pm 30\%$
- (2) VOH, nom (mV) values are based on a nominal VDDQ=0.6V

Table 5-24 LPDDR5 DC Output Parameters

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
SE,DIFF pull-up output driver characteristics, with ZQ calibration: VOHPU, nom = VDDQ*0.5	VOH, nom	mv		250		1,2
	VOHPU, nom	VOH, nom	0.9	1.0	1.1	1,2



Notes:

- (3) All values are after ZQ Calibration. Without ZQ Calibration VOH, nom values are $\pm 30\%$
- VOH, nom (mV) values are based on a nominal VDDQ=0.5V

Table 5-25 Common DC Output Conditions for Flyover Bypass1

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Note
Output high voltage	VOH_DC	V	$0.8 * VDDQ$			1,2,3,4
Output low voltage	VOL_DC	V			$0.2 * VDDQ$	1,2,3,4



Notes:

- (1) This timing parameter applies to all address, command, and control signals.
- (2) For VOH / VOL DC testing with current load, use the following formula: maximum loading $< (0.2 * VDDQ / R_{on}) * 0.9$, R_{on} can be found in the “Common DC Output Parameters” table where $R_{on} = R_{onPu}$ or R_{onPd} depending on signal direction.
- (3) When atpg_mode = 1, all output drivers are forced to maximum drive strength.
- (4) BP_ZN does not support Flyover Bypass mode

Table 5-26 Common DC Output Conditions for SEC

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Note
Output high voltage	VOH_DC	V	$0.8 * VDD2H$		VDD2H	1,2,3,4
Output low voltage	VOL_DC	V	VSS		$0.2 * VDD2H$	1,2,3,4



Notes:

- (1) SEC drives full CMOS level without termination
- (2) For VOH / VOL DC testing with current load, use the following formula: maximum loading $< (0.2 * VDD2H / R_{on}) * 0.9$, R_{on} can be found in the “Common DC Output Parameters” table where $R_{on} = R_{onPu}$ or R_{onPd} depending on signal direction.
- (3) The BP_MEMRESET_L driver is forced to maximum strength for all modes.

(5) SEC is used for BP_MEMRESET_L, BP_DTO, LPDDR5-CS, and LPDDR4-CKE interfaces

5.4.4 LVDS IO DC parameters

5.4.4.1 Recommended Operating Conditions

Table 5-27 Recommended Operating Conditions

Parameter		Min.	Nom.	Max.	Units
VDD	Core Digital Supply	0.72	0.75	0.82	V
VDDIO	IO Supply for LVDS	1.62	1.8	1.98	V

Table 5-28 DC Characteristics

Parameter		Min.	Nom.	Max.	Units
V_{IL}	Input Low Voltage			$0.3 \cdot V_{DDIO}$	V
V_{IH}	Input High Voltage	$0.7 \cdot V_{DDIO}$			V
V_{OL}	Output Low Voltage			0.2	V
V_{OH}	Output High Voltage	$V_{DDIO} - 0.2$			V

5.5 I/O AC parameters

This section includes the AC parameters of the following I/O types:

- General Purpose I/O (GPIO) AC parameters
- SDIO AC parameters
- DDR I/O AC parameters

The GPIO and DDR I/O load circuit and output transition time waveforms are shown in Figure 5-2 and Figure 5-3.

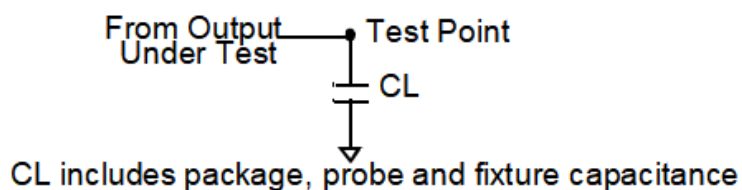


Figure 5-2 Load circuit for output

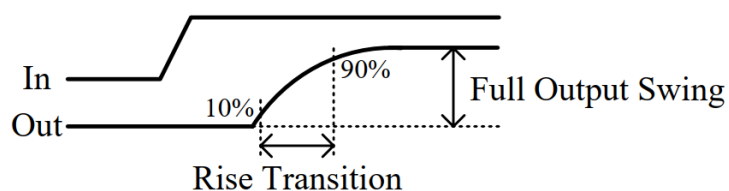


Figure 5-3 Output transition time waveform

5.5.1 General purpose I/O AC parameters

This section presents the I/O AC parameters for GPIO in different modes.

Table 5-29 Maximum frequency of operation

IO Type	Frequency(MHz)	Voltage(V)	C load(pF)
GPIO	10	1.8	35.6
	20	1.8	35.6
	25	1.8	35.6
	50	1.8	35.6
	100	1.8	35.6
	165	1.8	21.8
SDIO	25	1.8	32.6
	100	1.8	16
	125	1.8	16
	208	1.8	14.9
	50	3.0	14.9

5.5.2 DDR I/O AC parameters

The following tables list parameters that are common across all memory interface standards. Unless otherwise specified, all input specifications (both common and DDR standard specific) are measured at the host PHY input pins.

Subsequent sections list parameters for specific memory interface standards.

5.5.2.1 DDR Input I/O AC electrical characteristics

Table 5-30 Common AC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
input capacitance SE	C _{pad} SE	pF	1.170		1.215	1,2
input capacitance DIFF	C _{pad} DIFF_T	pF	1.184		1.219	1,2
	C _{pad} DIFF_C	pF	1.180		1.215	1,2

 Notes:

- (1) This is specified at the top level of the SE and DIFF hard macros. It does not include any customer-supplied routing between these blocks and the C4 bump.
- (2) Effective parallel capacitance when ODT of 40 ohm is enabled.

Table 5-31 LPDDR4 AC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Low-speed input high voltage for BP_DAT (logic 1)	VIH-ACLS	V	V _{ref} + 0.050		VDDQ + 0.15	1,2,6,8,9

Low-speed input low voltage for BP_DAT (logic 0)	VIL-ACLS	V	-0.15		Vref - 0.050	1,2,6,8,9
High-speed input high voltage for BP_DAT (logic 1)	VIH-ACHS	V	Vref + 0.050		VOH	1,3,6,8,9
High-speed input low voltage for BP_DAT (logic 0)	VIL-ACHS	V	0		Vref - 0.050	1,3,6,8,9
ODT mid-level (0.1*VDDQ) small-signal resistance	RODTss-M/RTT		0.75		1.05	4
ODT high-level (0.33*VDDQ) small-signal resistance	RODTss-H/RTT		0.85		1.05	4
ODT high-level (0.5*VDDQ) small-signal resistance	RODTss-H/RTT		0.95		1.75	4
Input crossing voltage for differential signals DQS*	VIX	mV	VDQSavg - 75	VDQSavg	VDQSavg + 75	5

 Notes:

- (1) This is the voltage at which the input receiver must be able to discriminate a high or a low input during a transient switching event.
- (2) The wider signal swing allowance causes a timing penalty due to the larger amount of cap charging and discharging in the receiver.
- (3) This narrower signal swing range incurs no timing penalty. Timing budgets are built using this signal swing.
- (4) Small-signal resistance is defined, for measured input current IIN as a function of applied voltage VIN, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

- (5) VDQSavg is defined as the average of 0.5*(DQS_t + DQS_c).
- (6) Low-speed refers to data rates less than or equal to 1600 Mbps.
- (7) High-speed refers to data rates greater than 1600 Mbps.
- (8) IO operation is supported only with ODT enabled.
- (9) The Vref value is dependent on pull-up/VOH and pull-down/ODT. PHY Vref is internally generated and programmed through local Vref generation at each receiver.

Table 5-32 LPDDR4X AC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Low-speed input high voltage for BP_DAT (logic 1)	VIH-ACLS	V	Vref + 0.050		VDDQ + 0.15	1,2,6,8
Low-speed input low voltage for BP_DAT (logic 0)	VIL-ACLS	V	-0.15		Vref - 0.050	1,2,6,8
High-speed input high	VIH-ACHS	V	Vref + 0.050		VOH	1,3,7,8

voltage for BP_DAT (logic 1)						
High-speed input low voltage for BP_DAT (logic 0)	VIL-ACHS	V	0		Vref – 0.050	1,3,7,8
ODT low-level (0.2*VDDQ) small-signal resistance	RODTss-L/RTT		0.75		1.05	4
ODT mid-level (0.5*VDDQ) small-signal resistance	RODTss-M/RTT		0.85		1.05	4
ODT high-level (0.75*VDDQ) small-signal resistance	RODTss-H/RTT		0.95		1.75	4
Input crossing voltage for differential signals DQS*	VIX	mV	VDQSavg – 75	VDQSavg	VDQSavg +75	5

 Notes:

- (1) This is the voltage at which the input receiver must be able to discriminate a high or a low input during a transient switching event.
- (2) The wider signal swing allowance causes a timing penalty due to the larger amount of cap charging and discharging in the receiver.
- (3) This narrower signal swing range incurs no timing penalty. Timing budgets are built using this signal swing.
- (4) Small-signal resistance is defined, for measured input current IIN as a function of applied voltage VIN, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

- (5) VDQSavg is defined as the average of 0.5*(DQS_t + DQS_c).
- (6) Low-speed refers to data rates less than or equal to 1600 Mbps.
- (7) High-speed refers to data rates greater than 1600 Mbps.
- (8) The Vref value is dependent on pull-up/VOH and pull-down/ODT. PHY Vref is internally generated and programmed through local Vref generation at each receiver.

Table 5-33 LPDDR5 AC Input Conditions

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
Low-speed input high voltage for BP_DAT (logic 1)	VIH-ACLS	V	Vref + 0.050		VDDQ + 0.15	1,2,6,8
Low-speed input low voltage for BP_DAT (logic 0)	VIL-ACLS	V	–0.15		Vref – 0.050	1,2,6,8
High-speed input high voltage for BP_DAT (logic 1)	VIH-ACHS	V	Vref + 0.050		VOH	1,3,7,8
High-speed input low voltage for BP_DAT (logic 0)	VIL-ACHS	V	0		Vref – 0.050	1,3,7,8

ODT low-level (0.2*VDDQ) small- signal resistance	RODTss-L/ RTT		0.75		1.05	4
ODT mid-level (0.5*VDDQ) small- signal resistance	RODTss-M/ RTT		0.85		1.05	4
ODT high-level (0.75*VDDQ) small- signal resistance	RODTss-H/ RTT		0.95		1.75	4
Input crossing voltage for differential signals DQS*	VIX	mV	VDQSavg – 75	VDQSavg g	VDQSavg +75	5

 Notes:

- (1) This is the voltage at which the input receiver must be able to discriminate a high or a low input during a transient switching event.
- (2) The wider signal swing allowance causes a timing penalty due to the larger amount of cap charging and discharging in the receiver.
- (3) This narrower signal swing range incurs no timing penalty. Timing budgets are built using this signal swing.
- (4) Small-signal resistance is defined, for measured input current IIN as a function of applied voltage VIN, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

- (5) VDQSavg is defined as the average of 0.5*(DQS_t + DQS_c).
- (6) Low-speed refers to data rates less than or equal to 1600 Mbps.
- (7) High-speed refers to data rates greater than 1600 Mbps.

The Vref value is dependent on pull-up/VOH and pull-down/ODT. PHY Vref is internally generated and programmed through local Vref generation at each receiver.

5.5.2.2 DDR Output I/O AC electrical characteristics

Table 5-34 Common AC Output Parameters

Parameter	Symbol	Unit	Mini	Typical	Max	Notes
Small-signal SEC pull-up driver variation at 0.5*VDD2H	RONssDQPu25 / RON		0.5		3	1
Small-signal SEC pull-down driver variation at 0.5*VDD2H	RONssDQPu05 / RON		0.5		3	1

 Notes:

- (1) Small-signal resistance is defined, for measured input current IIN as a function of applied voltage VIN, as the first derivative of the V-I plot:

$$R_{ODTss} = \frac{dV_{IN}}{dI_{IN}}$$

Table 5-35 Output Timing Characteristics

Parameter	Symbol	Unit	Minimum	Typical	Maximum	Notes
-----------	--------	------	---------	---------	---------	-------

M_DQS high pulse width	tDQSH	ps	0.46*tck		0.54*tck	1
M_DQS low pulse width	tDQSL	ps	0.46*tck		0.54*tck	1

 Notes:

(1) This timing parameter applies to all address, command, and control signals.

5.6 External peripheral interface parameters

5.6.1 JTAG timing parameters

The JTAG (IEEE Standard 1149.1-1990 Standard-Test-Access Port and Boundary Scan Architecture) interface is used for BSDL testing and emulation of the device. The trstn pin only needs to be released when it is necessary to use a JTAG controller to debug the device or exercise the device's boundary scan functionality. For maximum reliability, the device includes an Internal Pulldown (IPD) on the trstn pin to ensure that trstn is always asserted upon power up and the device's internal emulation logic is always properly initialized. JTAG controllers from Texas Instruments actively drive trstn high. However, some third-party JTAG controllers may not drive trstn high but expect the use of a Pullup resistor on trstn. When using this type of JTAG controller, assert trstn to initialize the device after powerup and externally drive trstn high before attempting any emulation or boundary-scan operations.

5.6.1.1 JTAG Electrical Data/Timing

Table 5-36 Timing Requirements for IEEE 1149.1 JTAG

NO.	PARAMETER	DESCRIPTION	MIN	MAX	UNIT
1	tc(TCK)	Cycle time, TCK	62.29		ns
1H	tw(TCKH)	Pulse duration, TCK high (40% of t_c)	24.92		ns
1L	tw(TCKL)	Pulse duration, TCK low (40% of t_c)	24.92		ns
3	tsu(TDI-TCK)	Input setup time, TDI valid to TCK high	6.23		ns
	tsu(TMS-TCK)	Input setup time, TMS valid to TCK high	6.23		ns
4	th(TCK-TDI)	Input hold time, TDI valid from TCK high	31.15		ns
	th(TCK-TMS)	Input hold time, TMS valid from TCK high	31.15		ns

Table 5-37 Switching Characteristics Over Recommended Operating Conditions for IEEE 1149.1 JTAG

NO.	PARAMETER	DESCRIPTION	MIN	MAX	UNIT
2	td(TCKL-TDOV)	Delay time, TCK low to TDO valid	0	30.5	ns

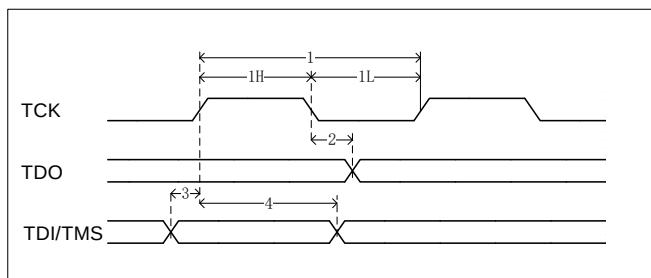


Figure 5-4 JTAG Timing

5.6.2 OSPI timing parameters

OSPI support SDR mode and DDR mode, the following describe the SDR mode timing parameter. Figure 5-5 is the SDR mode input timing and Figure 5-6 is the SDR mode output timing.

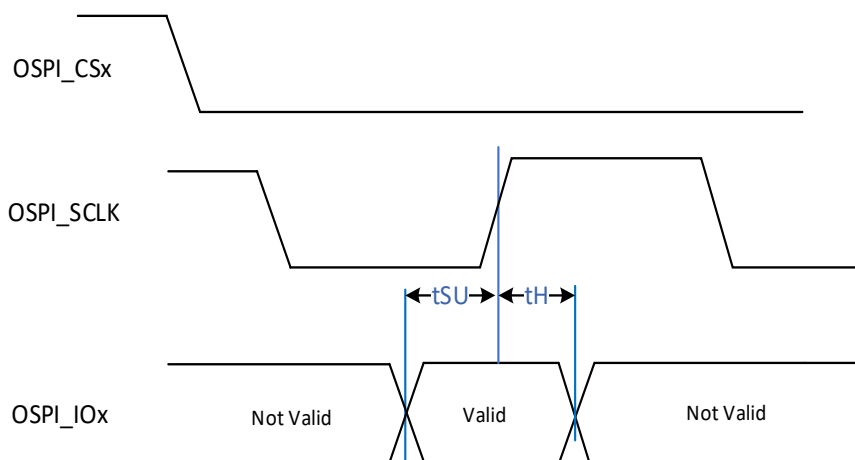


Figure 5-5 SDR mode input timing

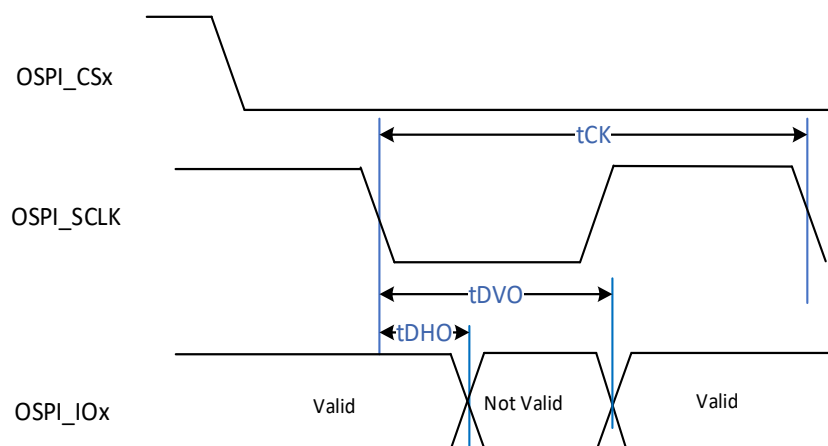


Figure 5-6 SDR mode output timing

The OSPI SDR mode timing parameter as the following table.

Table 5-38 OSPI SDR mode timing parameter

Parameter	Symbol	Min.	Max.	Unit	Remark
Clock					
Clock frequency	tCK	0	100	MHz	

Duty cycle ratio	-	45/55	55/45	%	
Inputs data (referenced to OSPI clock)					
Input setup time	tSU	5	-	ns	
Input hold time	tH	1	-	ns	
Outputs data (referenced to OSPI clock)					
Output data valid time	tDVO	-	5.7	ns	
Output data hold time	tDHO	2.0	-	ns	

The OSPI DDR mode timing as the following Figure.

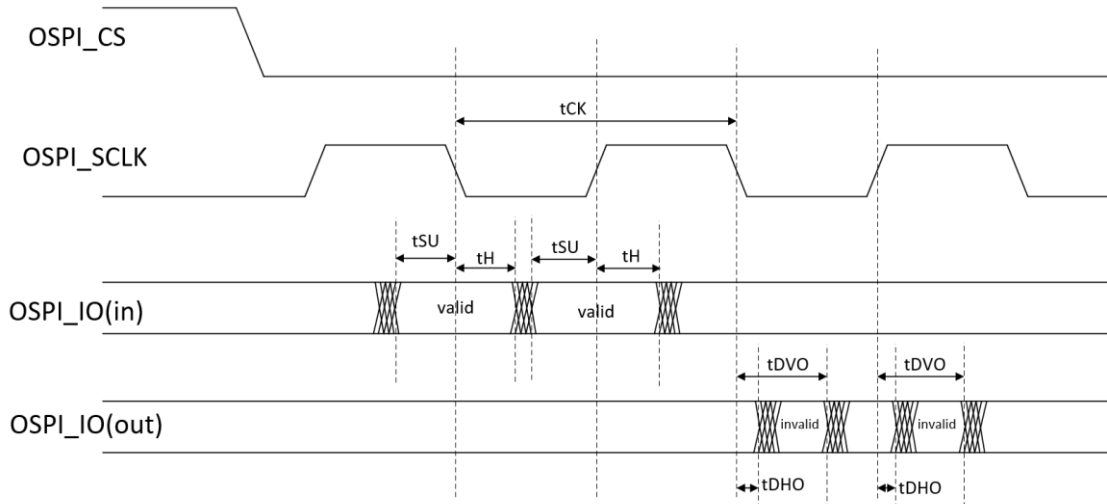


Figure 5-7 DDR mode timing

The OSPI DDR mode timing parameter as the following table.

Table 5-39 OSPI DDR mode timing parameter

Clock					
Clock frequency	tCK	0	50	MHz	
Duty cycle ratio	-	45/55	55/45	%	
Inputs data (referenced to OSPI clock)					
Input setup time	tSU	1.3	-	ns	
Input hold time	tH	1.3	-	ns	
Outputs data (referenced to OSPI clock)					
Output data valid time	tDVO	-	1.3	ns	
Output data hold time	tDHO	-2.0	-	ns	



Notes:

Timing requirement: the delay difference between data and clock should not exceed 200ps

5.6.3 CAN FD timing parameters

CAN FD is a pure asynchronous half duplex communication bus, and the protocol defines two of bit rates, one bit rates is a nominal bit rates, and the other is data bit rates. Nominal bit rates is a long time bit rate, and the time of data bit rates is less than nominal bit rates.

CAN FD protocol defines a sample point to sample data, and the detail of the sample point please reference the CAN FD protocol which release by BOSH.

The CAN FD main support 125kbps/250kbps/500kbps/1Mbps/2Mbps/4Mbps/5Mbps/8Mbps bit rate. The sample clock frequency support 20MHz/40MHz/80MHz, and the IO timing is shown in

follow figure.

Table 5-40 CANFD timing parameter

Parameter	min	typ	max	Unit
Serial Data Input Delay Time			25	ns
Serial Data Output Delay Time			40	ns

5.6.4 GMAC timing parameters

5.6.4.1 GMAC signals map

Table 5-41 GMAC signals map

Ball Name	Description	Mode	I/O	Notes
ST_RGMII_TX CLK	Transmit reference clock	RGMII	Output	The clock will be 125Mhz,25Mhz,or 2.5Mhz +/-50ppm depending on speed.
ST_RGMII_TX CTL	Transmit control signal.	RGMII/ RMII	Output	This signal is the control signal for the transmit data. RGMII mode:This signal is the control signal for the transmit data, and it is driven on both edges of TXCLK. RMII mode:When high, indicates that valid data is being transmitted on the TXD bus
ST_RGMII_TX D0	Bit0 of transmit data signal.	RGMII/ RMII	Output	RGMII mode:The data bus changes with both rising and falling edges of TXCLK. The data is valid only when TXCTL is high. RMII mode:The data is valid only when the TXCTL signal is high.
ST_RGMII_TX D1	Bit1 of transmit data signal.	RGMII/ RMII	Output	RGMII mode:The data bus changes with both rising and falling edges of TXCLK. The data is valid only when TXCTL is high. RMII mode:The data is valid only when the TXCTL signal is high.
ST_RGMII_TX D2	Bit2 of transmit data signal.	RGMII	Output	The data bus changes with both rising and falling edges of TXCLK. The data is valid only when TXCTL is high.
ST_RGMII_TX D3	Bit3 of transmit data signal.	RGMII	Output	The data bus changes with both rising and falling edges of TXCLK. The data is valid only when TXCTL is high.
ST_RGMII_RX CLK	Receive reference clock	RGMII/ RMII	Input	RGMII mode:The continuous receive reference clock will be 125Mhz,25Mhz,or 2.5Mhz+/-50ppm, and shall be derived from the receive data stream. RMII mode:The continuous receive reference clock will be 50Mhz+/-50ppm.
ST_RGMII_RX CTL	Receive control signal	RGMII/ RMII	Input	This is the receive control signal that is used to qualify the received data. RGMII mode:This signal is sampled both edges of RXCLK. RMII mode:Contains the crs and data valid information of the receive interface.
ST_RGMII_RX D0	Bit0 of receive data signal	RGMII/ RMII	Input	RGMII mode:The data bus is sampled with both rising and falling edges of RXCLK. The data is valid only when RXCTL signal is high. RMII mode:The data is valid only when the RXCTL signal is high.

ST_RGMII_RX D1	Bit1 of receive data signal	RGMII/ RMII	Input	RGMII mode: The data bus is sampled with both rising and falling edges of RXCLK. The data is valid only when RXCTL signal is high. RMII mode: The data is valid only when the RXCTL signal is high.
ST_RGMII_RX D2	Bit2 of receive data signal	RGMII	Input	The data bus is sampled with both rising and falling edges of RXCLK. The data is valid only when RXCTL signal is high.
ST_RGMII_RX D3	Bit3 of receive data signal	RGMII	Input	The data bus is sampled with both rising and falling edges of RXCLK. The data is valid only when RXCTL signal is high.
ST_RGMII_MD IO	Config data	RGMII/ RMII	Inout	MDIO is a bidirectional signal between the PHY and the MAC. It is used to transfer control information and status between the PHY and the MAC.
ST_RGMII_MD C	Config clock	RGMII/ RMII	Output	MDC is sourced by the MAC to the PHY as the timing reference for transfer of information on the MDIO signal.
ST_RGMII_TRI G_IN0	1588 trigger in	RGMII/ RMII	Input	When this signal goes high (rising edge), a snapshot of the System Time is taken.
ST_RGMII_TRI G_IN1	1588 trigger in	RGMII/ RMII	Input	When this signal goes high (rising edge), a snapshot of the System Time is taken
ST_RGMII_PPS O0	1588 pps output	RGMII/ RMII	Output	

5.6.4.2 RGMII timing parameters

Table 5-42 RGMII signal switching specifications

Symbol	Description	Min.	Max.	Unit
T_{skewT}^1	Data to Clock output Skew (at Transmitter)	-500	500	ps
T_{skewR}^1	Data to Clock input Skew (at Receiver)	1	2.6	ns
T_{cyc}^2	Clock Cycle Duration	7.2	8.8	ns
Duty G^3	Duty Cycle for Gigabit	45	55	%
Duty T^3	Duty Cycle for 10/100T	40	60	%
T_r / T_f	Rise / Fall Time (20-80%)		.75	ns



Notes:

- (1) For all versions of RGMII prior to 2.0; This implies that PC board design will require clocks to be routed such that an additional trace delay of greater than 1.5ns and less than 2.0ns will be added to the associated clock signal. For 10/100 the Max value is unspecified. There are +/-0.2ns skew margin on board to meet RGMII PHY standard timing.
- (2) For 10Mbps and 100Mbps, Tcyc will scale to 400ns+-40ns and 40ns+-4ns respectively.
- (3) Duty cycle may be stretched/shrunk during speed changes or while transitioning to a received packet's clock domain as long as minimum duty cycle is not violated and stretching occurs for no more than three Tcyc of the lowest speed transitioned between.

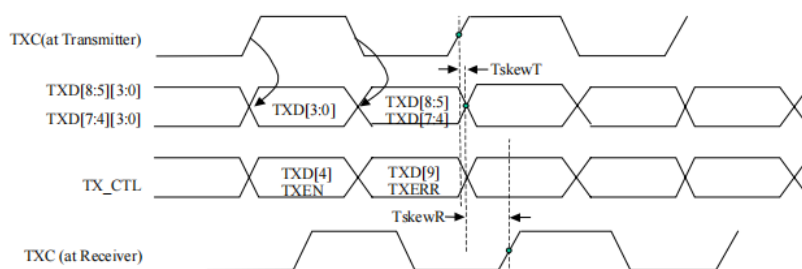


Figure 5-8 RGMII transmit signal timing diagram

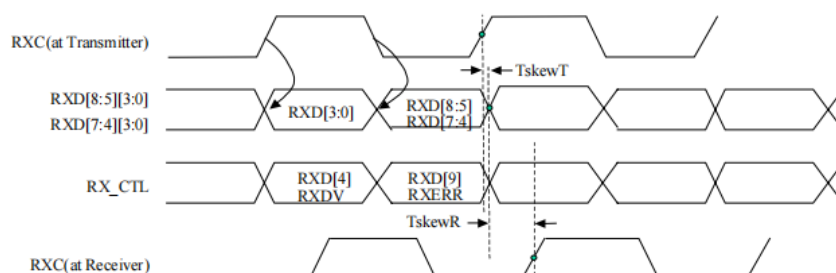


Figure 5-9 RGMII receive signal timing diagram

5.6.4.3 RMII timing parameters

Table 5-43 RMII signal switching specifications

Symbol	Description	Min.	Max.	Unit
f_{Freq}	REF_CLK Frequency	50 MHz -50 ppm	50MHz +50 ppm	MHz
Duty Cycle	REF_CLK Duty Cycle	35	65	%
t_{SETUP}	TXD[1:0],TX_EN,RXD[1:0],CRS_DV Setup to REF_CLK ring edge	4	-	ns
t_{HOLD}	TXD[1:0],TX_EN,RXD[1:0],CRS_DV hold from REF_CLK ring edge	2	-	ns

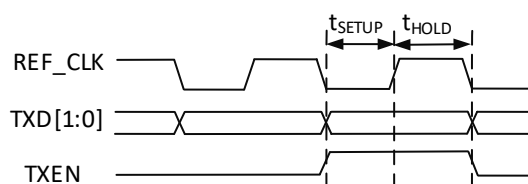


Figure 5-10 RMII transmit signal timing diagram

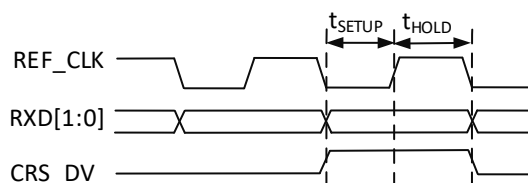


Figure 5-11 RMII receive signal timing diagram

5.6.4.4 MDIO timing parameters

Table 5-44 MDIO signal switching specifications

Symbol	Description	Min.	Max.	Unit
f_{FREQ}	MDC Frequency		10	MHz
t_{SETUP}	MDIO Setup to RX_CLK ring edge	10	-	ns
t_{HOLD}	RXD, RX_DV, RX_ER hold from RX_CLK ring edge	10	-	ns
T_{skew}	Data to Clock output Skew	15	45	ns



Notes: The frequency of the MDC should not exceed 10MHz.

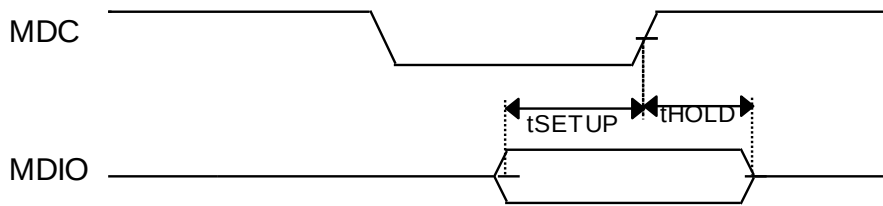


Figure 5-12 MDIO transmit signal timing diagram

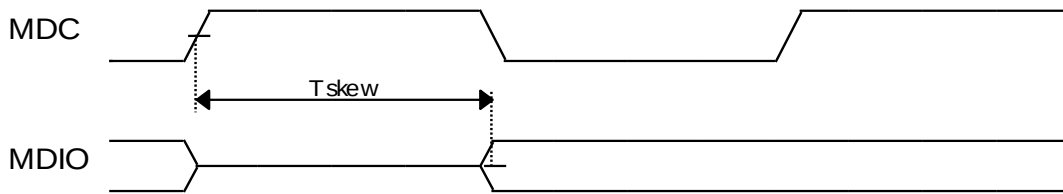


Figure 5-13 MDIO receive signal timing diagram

5.6.5 UART timing parameters

The serial uart is asynchronous, additional bits (start and stop) are added to the serial data to indicate the beginning and end. Utilizing these bits allows two devices to be synchronized. This structure of serial data- accompanied by start and stop bits-is referred to as a character, as shown in **Figure 5-14**.

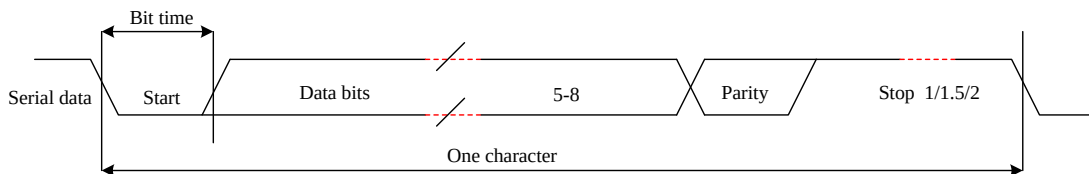


Figure 5-14 Serial data format

- (1) Fbaud_rate: Baud rate frequency. The maximum baud rate the UART can support is 10M.

5.6.6 I2C timing parameters

The Inter-Integrated Circuit (I2C) provides functionality of a standard I2C master and slave. The I2C is designed to be compatible with the I2C Bus Specification, version 3.0, dated April 2014.

Support three speeds, as following.

- (1) Standard mode (0 to 100 Kb/s) .
- (2) Fast mode (≤ 400 Kb/s) or fast mode plus (≤ 1000 kb/s)
- (3) high-speed mode (≤ 3.4 Mb/s)

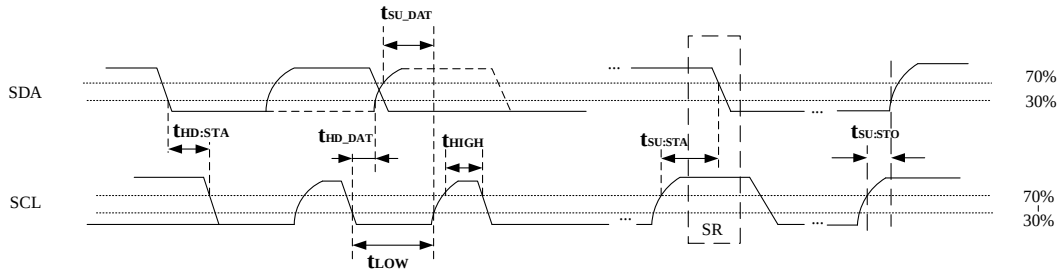


Figure 5-15 I2C timing

Table 5-45 I2C timing parameters

Parameter	Symbol	min	typ	max	Unit
Standard mode					
Clock period T	f_{CLK}	0		100	kHz
Clock Low Time	tLOW	4.7			μs
Clock High Time	tHIGH	4.0			μs
Setup Time for Repeated Start Condition	tSU:STA	4.7			μs
Start Condition Hold Time	tHD:STA	4.0			μs
SDA Hold Time from SCL Falling	tHD_DAT	0		3.45	μs
SDA Setup Time before SCL Rising	tSU_DAT	250			ns
Setup Time for Stop Condition	tSU:STO	4.0			μs
Fast Mode					
Clock period T	f_{CLK}	0		400	kHz
Clock Low Time	tLOW	1.3			μs
Clock High Time	tHIGH	0.6			μs
Setup Time for Repeated Start Condition	tSU:STA	0.6			μs
Start Condition Hold Time	tHD:STA	0.6			μs
SDA Hold Time from SCL Falling	tHD_DAT	0		0.9	μs
SDA Setup Time before SCL Rising	tSU_DAT	100			ns
Setup Time for Stop Condition	tSU:STO	0.6			μs
Fast Mode Plus					
Clock period T	f_{CLK}	0		1	MHz
Clock Low Time	tLOW	0.5			μs
Clock High Time	tHIGH	0.26			μs
Setup Time for Repeated Start Condition	tSU:STA	0.26			μs
Start Condition Hold Time	tHD:STA	0.26			μs
SDA Hold Time from SCL Falling	tHD_DAT	0			μs
SDA Setup Time before SCL Rising	tSU_DAT	50			ns
Setup Time for Stop Condition	tSU:STO	0.26			μs
High Speed Mode					
Clock period T	f_{CLK}	0		3.4	MHz
Clock Low Time	tLOW	160			ns

Clock High Time	tHIGH	60			ns
Setup Time for Repeated Start Condition	tSU:STA	160			ns
Start Condition Hold Time	tHD:STA	160			ns
SDA Hold Time from SCL Falling	tHD DAT	0		70	ns
SDA Setup Time before SCL Rising	tSU DAT	10			ns
Setup Time for Stop Condition	tSU:STO	160			ns

5.6.7 PWM_OUT timing parameters

The PWM can be programmed to select one of three clock signals as its source frequency. The selected clock signal is passed through a prescaler before being input to the counter. The output is available at the pulse-width modulator output (PWM_OUT) external pin.

Figure 5-16 depicts the timing of the PWM, and Table 5-46 lists the PWM timing parameters.

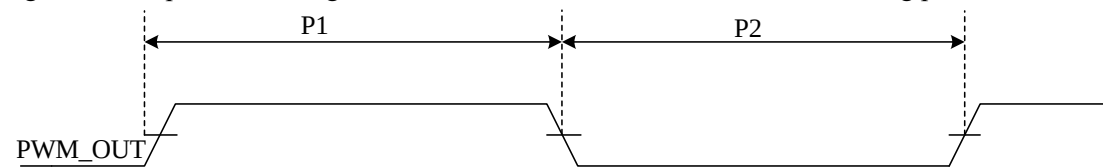


Figure 5-16 PWM timing

Table 5-46 PWM output timing parameters

ID	Parameter	Min	Max	Unit
	PWM output clock frequency		25MHz	MHz
P1	PWM output pulse width high	5	-	ns
P2	PWM output pulse width low	5	-	ns

(1)

5.6.8 ERR_RPT timing parameters

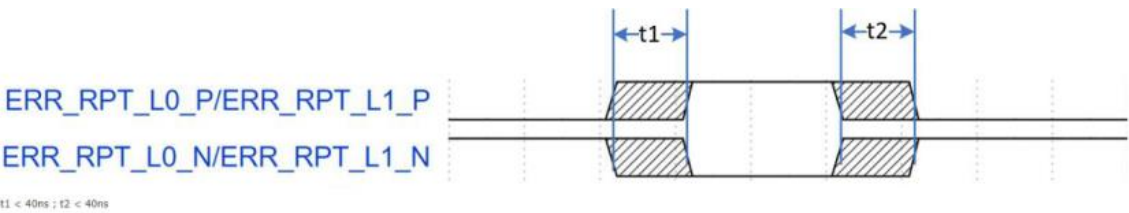


Figure 5-17 Error report pin timing

t1<40ns, t2<40ns

5.6.9 I3C timing parameters

The I3C provides functionality of a standard I3C master and slave. The I3C is designed to be compatible with the MIPI® Alliance Specification for I3C®, Version 1.1.

I3C support three speeds, as following.

- (1) High Data Rate Double Data Rate (HDR-DDR) mode.
- (2) HDR Bulk Transport (HDR-BT) mode.

(3) Multi-lane data transfers: Up to 4 SDA lanes supported for HDR-BT mode.

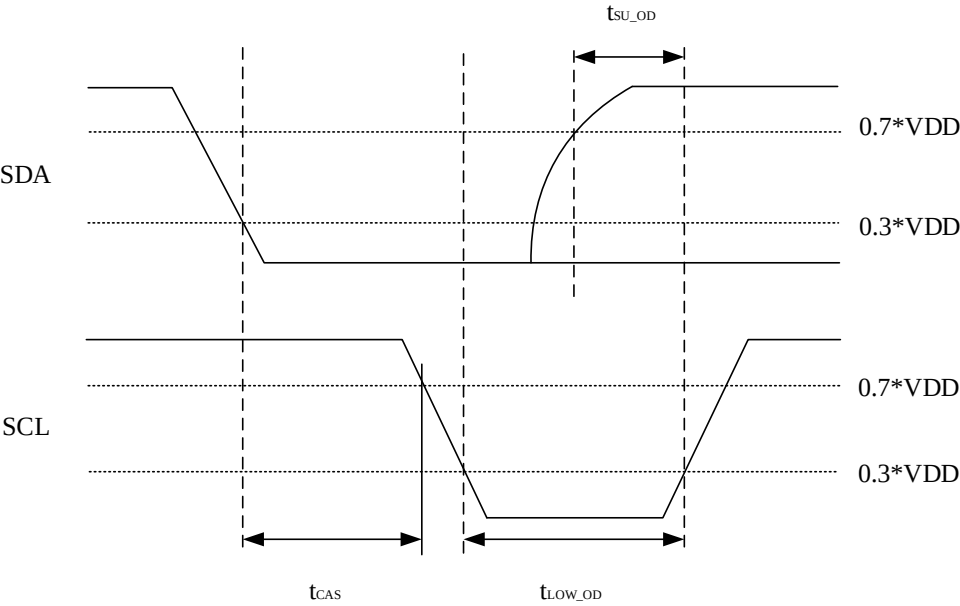


Figure 5-18 I3C START Timing

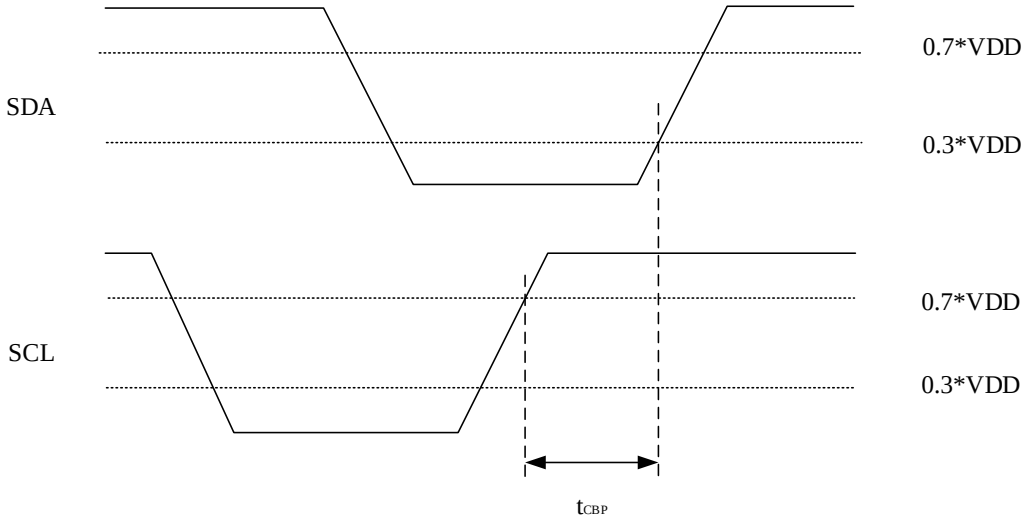


Figure 5-19 I3C STOP Timing

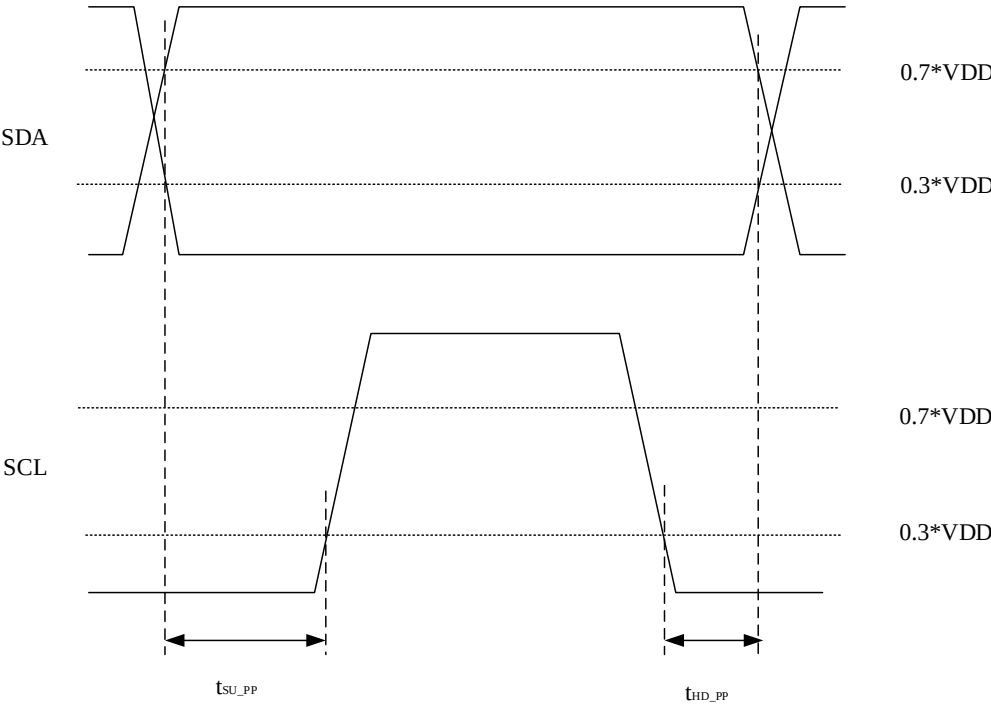


Figure 5-20 Controller SDR Timing

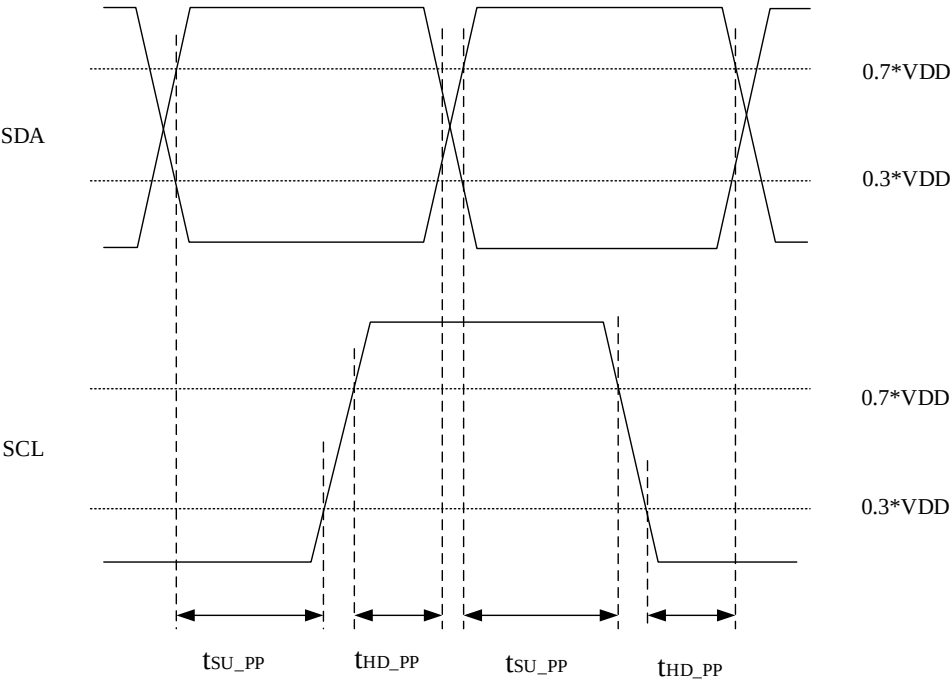


Figure 5-21 Controller DDR Timing

Table 5-47 I3C timing parameters

Parameter	Symbol	min	typ	max	Unit
Clock After START Condition	t_{CAS}	45			ns
Clock Before Stop Condition	t_{CBP}	50			ns
Clock Low Time in open drain mode	t_{LOW_OD}	200			ns
Clock High Time in open drain mode	t_{HIGH_OD}			40	ns
Clock Low Time in push-pull mode	t_{LOW_PP}	40			ns
Clock High Time in push-pull mode	t_{HIGH_PP}	40			ns
SDA Setup Time during open drain mode	t_{SU_OD}	160			ns
SDA setup time during push-pull mode	t_{SU_PP}	10			ns
SDA hold time during push-pull mode	t_{HD_PP}	10			ns

5.6.10 SPI timing parameters

The Inter-Integrated Circuit(SPI) provides functionality of a standard SPI transmitter and receive. The SPI is designed to be compatible with the SPI Bus Specification, Based on Motorola. The timing as the following diagram for spi.

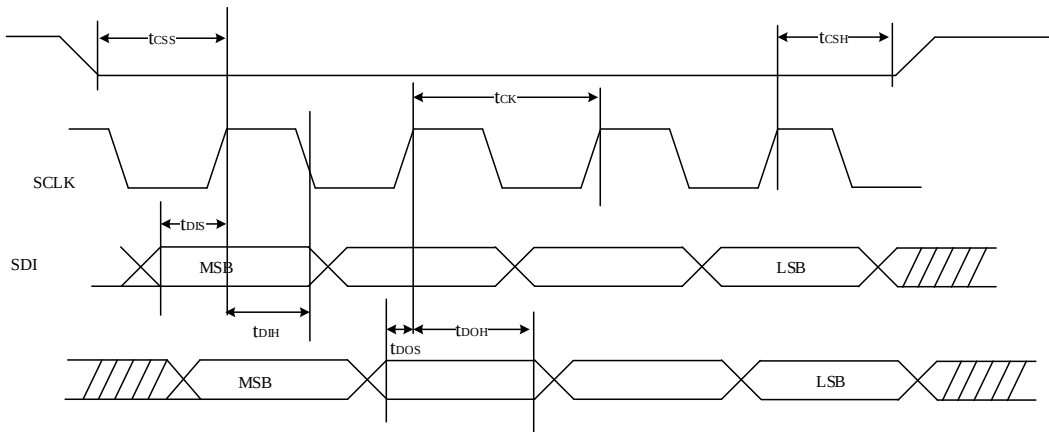


Figure 5-22 SPI timing

The timing parameter for SPI as the following table.

Table 5-48 SPI master timing parameter

	Value		Unit	Description
	min	max		
Clock period T/tCK	-	50	MHz	
Clock Duty cycle ratio	45/55	55/45	%	
MISO signal/input signal				
t_{DISU}	10.2	-	ns	Input setup time
t_{DIH}	2.4	-	ns	Input hold time
MOSI, CS signal/output signal				
t_{DOS}	14		ns	Output data valid time
t_{DOH}	-1.8	-	ns	Output data hold time
CS signal/output signal				
t_{CSS}	14	-	ns	CS setup time

tCSH	-1	-	ns	CS hold time
------	----	---	----	--------------

Table 5-49 SPI slave timing parameter

	Value		Unit	Description
	min	max		
Clock period T/tCK	-	50	MHz	
Clock Duty cycle ratio	45/55	55/45	%	
MISO signal/input signal				
tDISU	1.9	-	ns	Input setup time
tDIH	1	-	ns	Input hold time
MOSI, CS signal/output signal				
tDOS	2.8	-	ns	Output data valid time
tDOH	6.2	-	ns	Output data hold time
CS signal/input signal				
tCSS	3.1	-	ns	CS setup time
tCSH	1	-	ns	CS hold time

Note:The maximum transfer rate of spi in the soc domain is 12.5MHz.

5.6.11 I2S timing parameters

I2S bus consists of a serial data line (sd), a word selects line (ws), and a serial clock (sclk). The serial data line is time multiplexed to allow the transfer of two data streams (such as, left and right stereo data). The I2S bus stereo frame format as the following diagram.

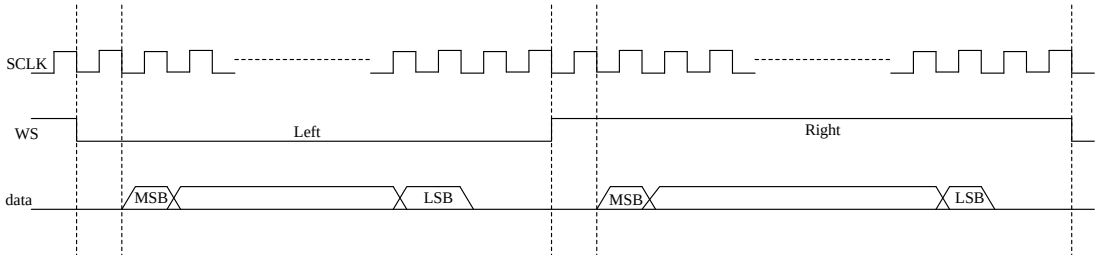


Figure 5-23 I2S bus stereo frame format

The bus for transmitter timing as the Figure 5-24.

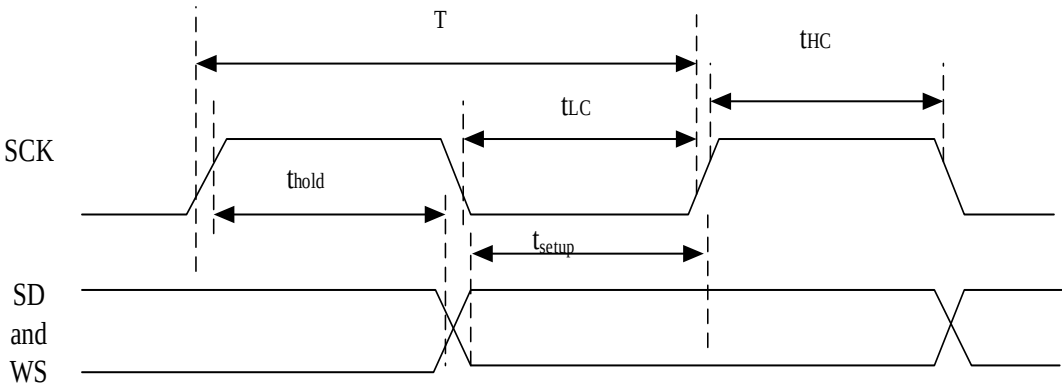
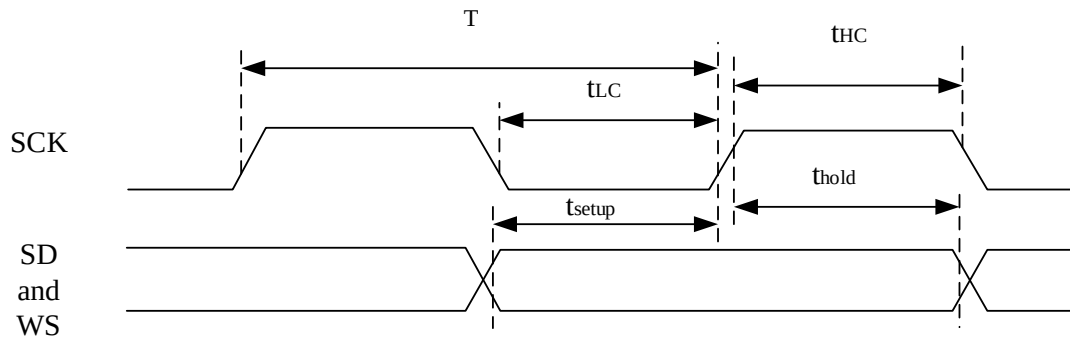


Figure 5-24 Timing for I2S transmitter

The bus for receiver timing as the Figure 5-25.

**Figure 5-25 Timing for I2S receiver****Table 5-50 I2S master timing parameter**

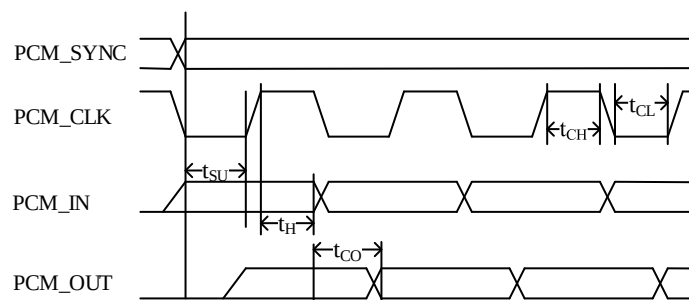
Parameter	Symbol	min	typ	max	Unit
Clock period T	f_{CLK}	-	-	12.288	MHz
Clock Duty cycle ratio		45	50	55	%
Serial Data Input Setup Time	t_{setup}	12	-	-	ns
Serial Data Input Hold Time	t_{hold}	-1	-	-	ns
Serial Data Output Setup Time	t_{setup}	-	-	24	ns
Serial Data Output Hold Time	t_{hold}	0	-	-	ns

Table 5-51 I2S slave timing parameter

Parameter	Symbol	min	typ	max	Unit
Clock period T	f_{CLK}	-	-	12.288	MHz
Clock Duty cycle ratio		45	50	55	%
Serial Data Input Setup Time	t_{setup}	20	-	-	ns
Serial Data Input Hold Time	t_{hold}	0	-	-	ns
Serial Data Output Setup Time	t_{setup}	-	-	15	ns
Serial Data Output Hold Time	t_{hold}	-24	-	-	ns

5.6.12 TDM timing parameters

Time Division Multiplexed (TDM) interface provides functionality of a standard TDM master, support speeds up to 25MHz.

**Figure 5-26 TDM interface timing****Table 5-52 TDM timing parameter**

Parameter	Symbol	min	typ	max	Unit
Clock period T	f_{CLK}	-	-	25	MHz
Clock Duty cycle ratio		45	50	55	%
Serial Data Input Setup Time	t_{SU}	14	-	-	ns
Serial Data Input Hold Time	t_H	9	-	-	ns
Serial Data Output Delay Time	t_{CO}	6	-	34	ns

5.6.13 VIN timing parameters

5.6.13.1 VIN signals map

Table 5-53 VIN signals map

Ball Name	VIN signal	Description	I/O
LVDS0 ODD TX0 P	VIN_R0	VIN pixel data0	input
LVDS0 ODD TX1 P	VIN_R1	VIN pixel data1	input
LVDS0 ODD TX2 P	VIN_R2	VIN pixel data2	input
LVDS0 ODD TX3 P	VIN_R3	VIN pixel data3	input
LVDS0 ODD TX4 P	VIN_R4	VIN pixel data4	input
LVDS0 ODD TX0 N	VIN_G0	VIN pixel data5	input
LVDS0 ODD TX1 N	VIN_G1	VIN pixel data6	input
LVDS0 ODD TX2 N	VIN_G2	VIN pixel data7	input
LVDS0 ODD TX3 N	VIN_G3	VIN pixel data8	input
LVDS0 ODD TX4 N	VIN_G4	VIN pixel data9	input
LVDS0 ODD CLKP	VIN_G5	VIN pixel data10	input
LVDS0 ODD CLKN	VIN_B0	VIN pixel data11	input
LVDS0 EVEN TX0 P	VIN_B1	VIN pixel data12	input
LVDS0 EVEN TX1 P	VIN_B2	VIN pixel data13	input
LVDS0 EVEN TX2 P	VIN_B3	VIN pixel data14	input
LVDS0 EVEN TX3 P	VIN_B4	VIN pixel data15	input
LVDS0 EVEN TX4 P	VIN_HS	VIN horizontal synchronization	input
LVDS0 EVEN TX0 N	VIN_VS	VIN vertical synchronization	input
LVDS0 EVEN TX1 N	VIN_DE	VIN data valid	input
LVDS0 EVEN TX2 N	VIN_LLC	Clock for VIN Pixel data	input

5.6.13.2 VIN timing parameters

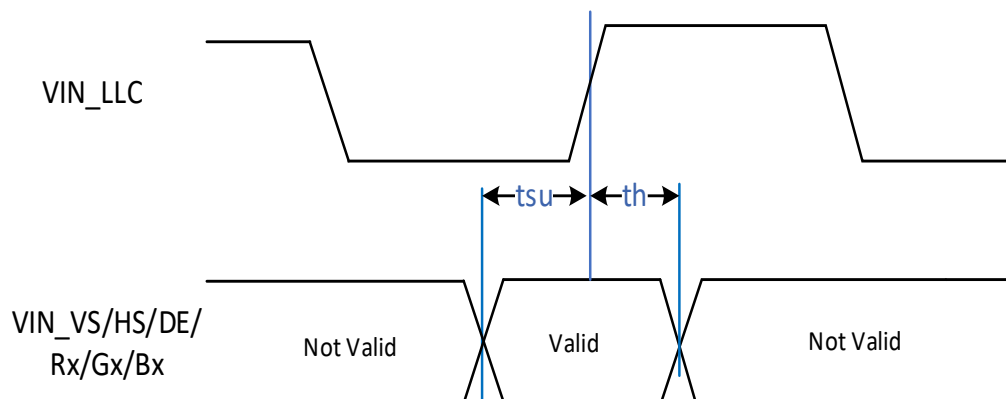


Figure 5-27 VIN timing diagram

Table 5-54 VIN timing parameters

Symbol	Parameter Description	MIN	Typical	Max	Unit
VIN_LLC	Input clock frequency	13.5		165	MHz
tsu	Input Data Setup time	2.2			ns
th	Input Data Hold time	0.3			ns

 Notes:

If select ADV7611 as the HDMI Receiver, the delay between clock input and data input must not exceed 200ps.

5.6.14 VOUT timing parameters

5.6.14.1 VOUT signals map

Table 5-55 VOUT signals map

Ball Name	VOUT Signal	Description	I/O
WK_GPIO0	SC_VOUT_PIXEL_CLK	Display output clock	output
WK_GPIO1	SC_VOUT_DATAEN	Display output horizontal valid	output
WK_GPIO2	SC_VOUT_HSYNC	Display output horizontal sync	output
WK_GPIO3	SC_VOUT_VSYNC	Display output vertical sync	output
WK_GPIO4	SC_VOUT_PXLDATA0	sc_vout_pxldata0_b[0]	output
WK_GPIO5	SC_VOUT_PXLDATA1	sc_vout_pxldata1_b[1]	output
WK_GPIO6	SC_VOUT_PXLDATA2	sc_vout_pxldata2_b[2]	output
WK_GPIO7	SC_VOUT_PXLDATA3	sc_vout_pxldata3_b[3]	output
WK_GPIO8	SC_VOUT_PXLDATA4	sc_vout_pxldata4_b[4]	output
WK_GPIO9	SC_VOUT_PXLDATA5	sc_vout_pxldata5_b[5]	output
WK_GPIO10	SC_VOUT_PXLDATA6	sc_vout_pxldata6_b[6]	output
WK_GPIO11	SC_VOUT_PXLDATA7	sc_vout_pxldata7_b[7]	output
WK_GPIO12	SC_VOUT_PXLDATA8	sc_vout_pxldata8_b[8]	output
WK_GPIO13	SC_VOUT_PXLDATA9	sc_vout_pxldata9_b[9]	output
WK_GPIO14	SC_VOUT_PXLDATA10	sc_vout_pxldata10_g[0]	output
WK_GPIO15	SC_VOUT_PXLDATA11	sc_vout_pxldata11_g[1]	output
WK_GPIO16	SC_VOUT_PXLDATA12	sc_vout_pxldata12_g[2]	output
WK_GPIO17	SC_VOUT_PXLDATA13	sc_vout_pxldata13_g[3]	output

WK_GPIO18	SC_VOUT_PXLDATA14	sc_vout_pxldata14_g[4]	output
WK_GPIO19	SC_VOUT_PXLDATA15	sc_vout_pxldata15_g[5]	output
WK_GPIO20	SC_VOUT_PXLDATA16	sc_vout_pxldata16_g[6]	output
WK_GPIO21	SC_VOUT_PXLDATA17	sc_vout_pxldata17_g[7]	output
WK_GPIO22	SC_VOUT_PXLDATA18	sc_vout_pxldata18_g[8]	output
WK_GPIO23	SC_VOUT_PXLDATA19	sc_vout_pxldata19_g[9]	output
WK_GPIO24	SC_VOUT_PXLDATA20	sc_vout_pxldata20_r[0]	output
WK_GPIO25	SC_VOUT_PXLDATA21	sc_vout_pxldata21_r[1]	output
WK_GPIO26	SC_VOUT_PXLDATA22	sc_vout_pxldata22_r[2]	output
WK_GPIO27	SC_VOUT_PXLDATA23	sc_vout_pxldata23_r[3]	output
WK_GPIO28	SC_VOUT_PXLDATA24	sc_vout_pxldata24_r[4]	output
WK_GPIO29	SC_VOUT_PXLDATA25	sc_vout_pxldata25_r[5]	output
WK_GPIO30	SC_VOUT_PXLDATA26	sc_vout_pxldata26_r[6]	output
WK_GPIO31	SC_VOUT_PXLDATA27	sc_vout_pxldata27_r[7]	output
WK_GPIO32	SC_VOUT_PXLDATA28	sc_vout_pxldata28_r[8]	output
WK_GPIO33	SC_VOUT_PXLDATA29	sc_vout_pxldata29_r[9]	output
WK_GPIO34	SC_VOUT_TRANSFACTIVE	Image transmission start	output

5.6.14.2 VOUT timing parameters

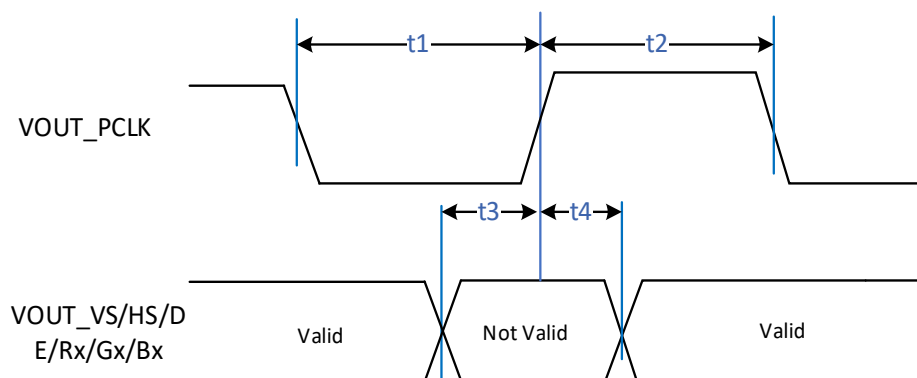


Figure 5-28 VOUT timing diagram

Table 5-56 VOUT timing parameters

Parameter	Symbol	Description	Min	Typ	Max	Unit
VOUT_PCLK Frequency Range			27		165	MHz
VOUT_PCLK Duty Cycle	t1:t2		45:55		55:45	
Data Output Transition Time	t3	End of valid data to negative clock edge			1.2	ns
	t4	Negative clock edge to start of valid data			0.8	ns

Notes:

If the bit width of external devices is less than 10 bits, should be truncating for hardware design.

If select ADV7513 as the HDMI Transmitter, the delay between clock input and data input must not exceed 200ps.

5.6.15 SDEMMC timing parameters

5.6.15.1 SD controller timing

Supports five modes: DS, HS, SDR12, SDR25, SDR50, SDR104. The SD controller input timing diagram as following.

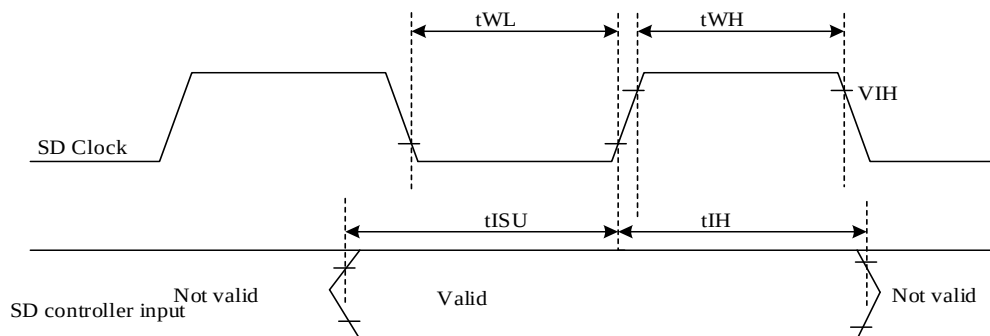


Figure 5-29 SD input timing

The SD controller output timing diagram as following.

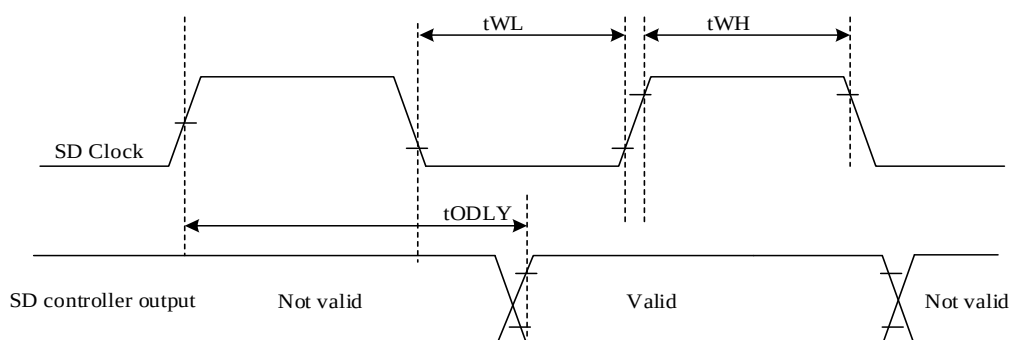


Figure 5-30 SD output timing

The SD timing parameter as the following table.

Table 5-57 DS/HS/SDR12/SDR25/SDR50 mode timing parameter

Parameter	Symbol	Min.	Max.	Unit	Remark
Max clock frequency			100	Mhz	SDR50
Input setup time	tISU	2.2	-	ns	
Input hold time	tIH	1.1	-	ns	
Clock duty cycle ratio	tWL/tWH	45/55	55/45	%	
Output delay	tODLY	1.2	6.6	ns	

 Notes:

Timing requirement: the delay difference between data and clock should not exceed 200ps

5.6.15.2 EMMC timing parameters

The eMMC defines two bus speed modes. Following table summarizes the various modes.

Table 5-58 EMMC bus speed modes summarizes

Mode name	Data rate	I/O voltage	Bus width	Frequency	Max data transfer
-----------	-----------	-------------	-----------	-----------	-------------------

Backwards Compatibility with legacy MMC card	Single	1.8/3.0	8	0-25MHz	25MB/s
High speed SDR	Single	1.8/3.0	8	0-50MHz	50MB/s
High speed DDR	Dual	1.8	8	0-52MHz	104MB/s
HS200	Single	1.8	8	0-200MHz	200MB/s

Figure 5-31 shows the data input/output timing and note that data must always be sampled on the rising edge of the clock.

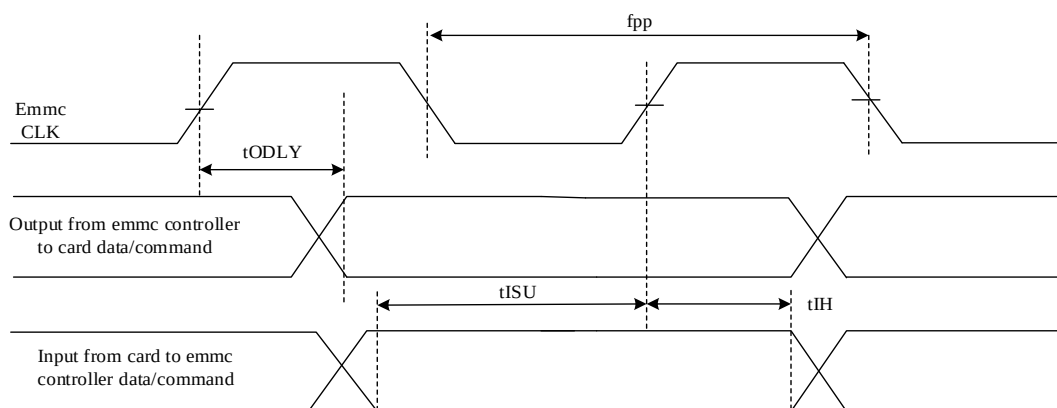


Figure 5-31 Data input/output timing

Table 5-59 Timing parameters of EMMC(SDR Mode)

Parameter	Symbol	Min.	Max.	Unit	Remark
Max clock frequency	fpp	-	50	Mhz	
Input setup time	tISU	5.7	-	ns	
Input hold time	tIH	1.9	-	ns	
Clock duty cycle ratio		45/55	55/45	%	
Output delay	tODLY	3.6	16.4	ns	

Table 5-60 Timing parameters of EMMC(HS200 Mode)

Parameter	Symbol	Min.	Max.	Unit	Remark
Max clock frequency	fpp	-	200	Mhz	
Input setup time	tISU	1.4	-	ns	
Input hold time	tIH	0.8	-	ns	
Clock duty cycle ratio		45/55	55/45	%	
Output delay	tODLY	0	5	ns	



Notes:

- (1) In identification mode(SD), card clock must be lower than 400 kHz, voltage ranges from 2.7 to 3.6 V.
- (2) To satisfy timing requirement, the delay difference between clock and cmd/data must not exceed 200ps.

5.6.16 LIN timing parameters

Local Interconnect Network(LIN) interface provides functionality of a standard LIN master and

slave. The LIN is designed to be compatible with the Local Interconnect Network, version 1.3, 2.0, 2.1, and 2.2.

Support bit rates up to 20 Kbit/s (LIN protocol).

Table 5-61 LIN timing parameter

Parameter	min	typ	max	Unit
Serial Data Input Delay Time			25	ns
Serial Data Output Delay Time			40	ns

5.6.17 FlexRay timing parameters

The FlexRay support FlexRay Communications System Protocol Specification Version 2.1, Revision A compliant protocol implementation and FlexRay Communications System Electrical Physical Layer Specification, Version 2.1, Revision A compliant bus driver interface.

Support for FlexRay bus data rates of 10 Mbit/s, 8 Mbit/s, 5 Mbit/s, and 2.5 Mbit/s.

Table 5-62 Flexray timing parameter

Parameter	min	typ	max	Unit
Serial Data Input Delay Time			30	ns
Serial Data Output Delay Time			45	ns

5.6.18 ADC parameters

The ADC provides functionality of translating analog electrical signals for data processing purposes. The timing is as the following diagram for ADC.

Table 5-63 Timing parameter of ADC

Parameter		min	typ	max	unit
bit width		10			bit
input channel		12 Single-ended			-
input signal range		VSS		AVDD(1.8)	V
LSB _{GAIN-ERROR}	Gain Error	± 1.0% with External Cal.			%
LSB _{OFFSET-ERROR}	Offset Error		± 2		LSB
DNL	Differential Non-Linearity		± 1		LSB
INL	Integral Non-Linearity		± 1.5		LSB
SNR @Fin100KHz, -1 DBFS output	Signal-to-Noise Ratio		56		dBFS
SNDR @Fin100KHz, -1 DBFS output	Signal-to-Noise Dynamic Ratio		55		dBFS
THD @Fin100KHz, -1 DBFS output	Total Harmonic Distortion		-60		dBc
SFDR @Fin100KHz, -1 DBFS output	Spurious Free Dynamic Range		62		dBc
T _R	Sampling Rate		2	4	MSPS
F _{SMPL_CLK}	SMPL_CLK Frequency			64	MHz
Tr	Clock rise time	-	-	50	ps
Tf	Clock fall time	-	-	50	ps

DCclk	Clock duty clock	47	50	53	%
tdc	Clock output delay	0	-	1.5	ns
tdd	Clock output to Data Output delay	0	-	0.5	ns
tsclk	Clock to input signal	-	-	5	ns
ttds	Control signal setup time	0.6	-	-	ns
ttdn	Control signal hold time	0.6	-	-	ns
	Input to data latency		32		cycles
Tstup	Start-up time of ADC circuit			256	cycles

 Notes:

- (1) The ADC input clock frequency is 16 times of the sampling rate.
- (2) The specified cycle number refers to the input clock signal, cycle period time = $1/F_{clk}$

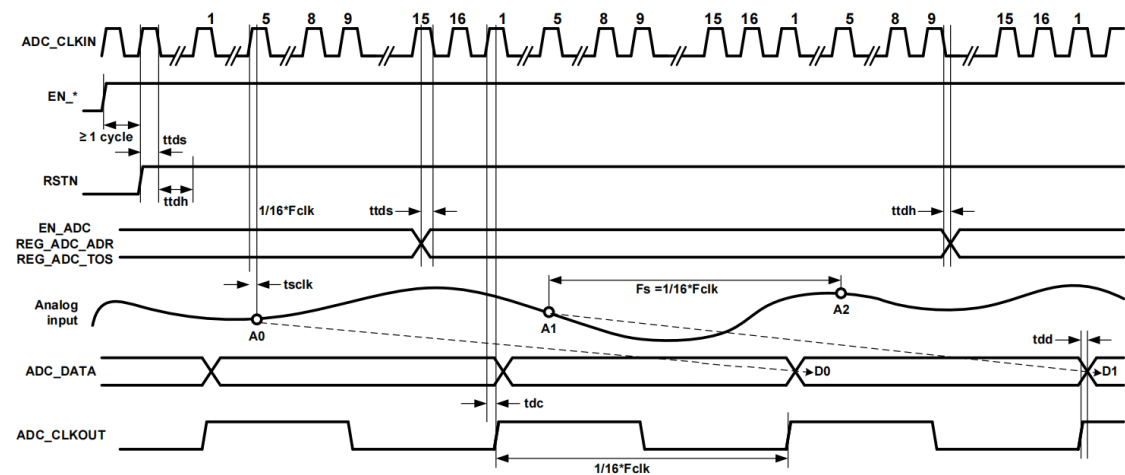


Figure 5-32 Timing Waveform of ADC

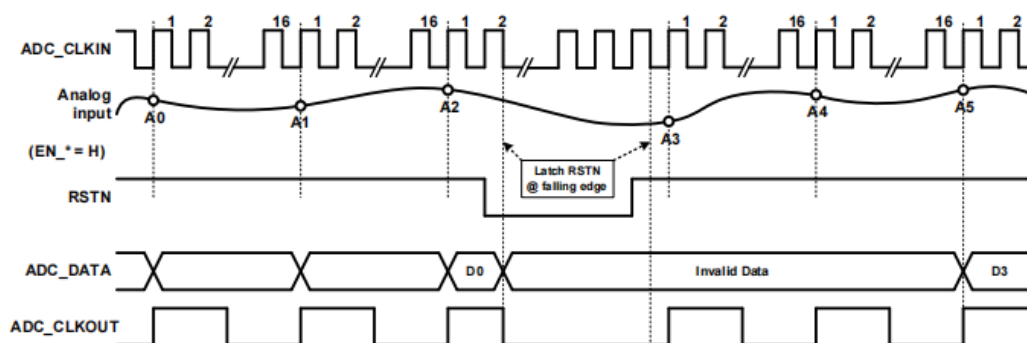


Figure 5-33 Timing Waveform of RSTN Function

5.6.19 10G Serdes PHY parameters

5.6.19.1 Reference Clock: Input Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

The table below lists differential reference clock requirements for protocols other than PCIe are provided for data rates less than and greater than 8 Gbaud.

Table 5-64 Reference Clock Input Requirements

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	312.5	MHz	Not continuous; per protocol as noted in the databook
FREF_OFFSET	Reference clock frequency offset	-100	100	ppm	All protocols except for SATA
RMSJREF_CLK ≤ 8 Gbps	Reference clock random Jitter		1.6	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			1.2		Integrated RJ from 2 MHz to 20 MHz
DJREF_CLK ≤ 8 Gbps	Reference clock deterministic jitter		2.8 5.6	ps, pp	0.75–10MHz (offset) ^a 0.2–50MHz (offset) ^a
RMSJREF_CLK > 8 Gbps	Reference clock random jitter		1.0	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			0.8		Integrated RJ from 2 MHz to 20 MHz
DJREF_CLK > 8 Gbps	Reference clock deterministic jitter		1.7 3.4	ps, pp	0.75–10MHz (offset) ^a 0.2–50MHz (offset) ^a
DCREF_CLK	Duty cycle	40	60	%	
VCMREF_CLK	Common mode input level	0	VP	V	Differential inputs
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDSEREF_CLK_DC	Input Voltage Level for ref_pad_clk_{p,m}	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_DC.
VDSEREF_CLK_AC ^b	Absolute voltage level at the pin including the overshoot for ref_pad_clk_{p,m}	-0.3	1.15	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_AC.
VDREF_CLK_ALT	Differential input swing for ref_alt_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_ALT.
VDSEREF_CLK_ALT	Input Voltage Level for ref_alt_clk_{p,m}	0	VP	V	Differential inputs must meet both VDREF_CLK_ALT and VDSEREF_CLK_ALT.
SKEWDIF_CLK	Differential clock input skew		100	ps	
SWREF_CLK	Differential clock input slew rate	0.6		V/ns	20–80%

a. DJ noise limits are for noise within an offset from the divided reference clock and harmonics of the divided reference clock.

b. Allowed overshoot time: 10% of the FREF_CLK period

In Figure 5-34, VDREF_CLK is the peak-peak differential swing for ref_pad_clk_{p,m}.

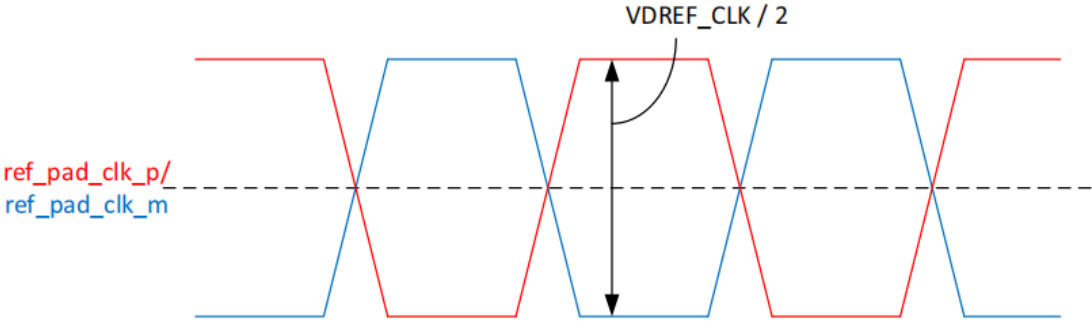


Figure 5-34 VDREF_CLK

In **Figure 5-35**, VDREF_CLK_ALT is the peak-peak differential swing for ref_alt_clk_{p,m}.

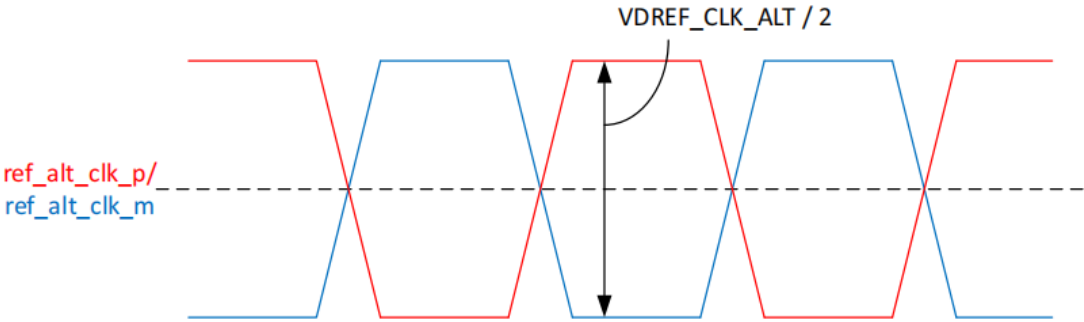


Figure 5-35 VDREF_CLK_ALT

5.6.19.2 Reference Clock External Pad Input Parasitics

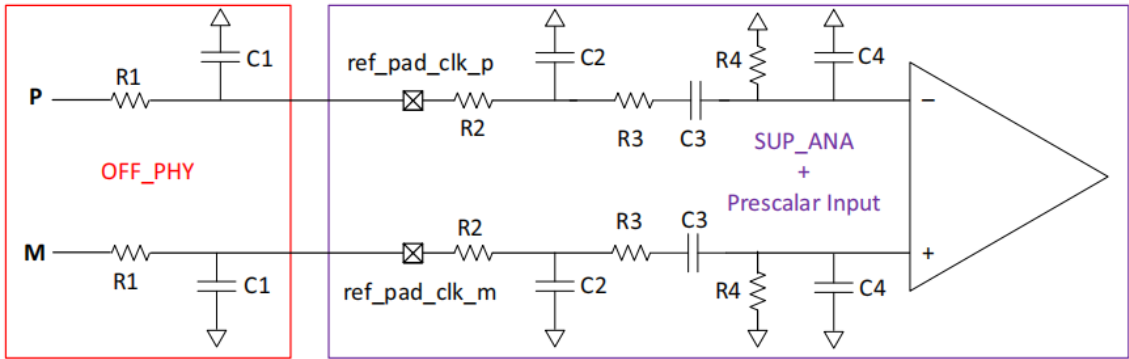


Figure 5-36 Model of Reference Clock Pad Input Load

Table 5-65 Reference Clock External Pad Input Parasitics

Location	Parasitics Device	Value	Unit	Notes
OFF PHY	R1	≤ 50	Ohm	
OFF PHY	C1	≤ 3	pF	
Within PHY: Bump to prescaler input	R2	≤ 50	Ohm	

Within PHY: Bump to prescaler input	C2	≤ 1	pF	
Within prescaler: 2nd ESD Protection resistor	R3	1.5	kOhm	Required for 6A peak discharge current ESD rating.
Within prescaler: AC capacitor	C3	920	fF	
Within prescaler: DC Bias resistor	R4	468	kOhm	
Within prescaler: Input capacitor of prescaler input RX	C4	≤ 50	fF	

5.6.19.3 Reference Resistor Requirements

Reference resistor requirements are outlined in Table 5-66.

Table 5-66 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2mA	None
External Reference Resistor parasitic capacitance C_{load}	$< 20\text{pF}$	-The maximum parasitic capacitance includes external loading of package trace and PCB, as well as, PHY loading. -Each PHY connected to the resref pin adds an additional 1.5 pF loading. -For example, a single PMA can support up to 18.5 pF (20 pF–1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 14 pF (20 pF–4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor must be as close as possible to the PMA ground voltage. Noise adjacent signals (for example, clocks) must be spaced away from resref or must be shielded with ground, if possible.

Package and board connectivity for a single PMA implementation is shown in **Figure 5-37**.

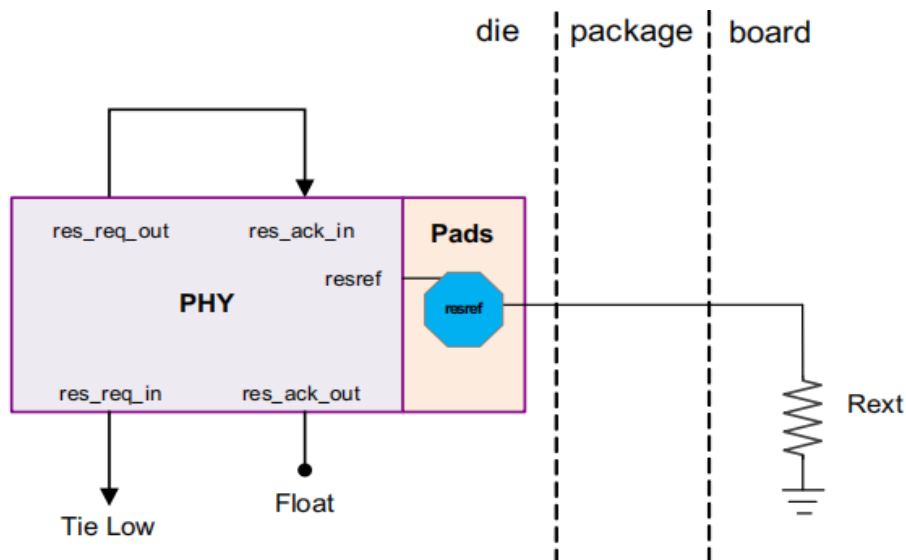


Figure 5-37 External Reference Resistor Connectivity for a Single PMA

The external reference resistor may be shared among a maximum of four instances of the Multi-Protocol 16G LP AP PHY x2 for TSMC 7FF PMA by connecting the resref signals together in the package substrate. Sharing an external reference resistor by tying the resref bus together in the package is shown in Figure 5-38.

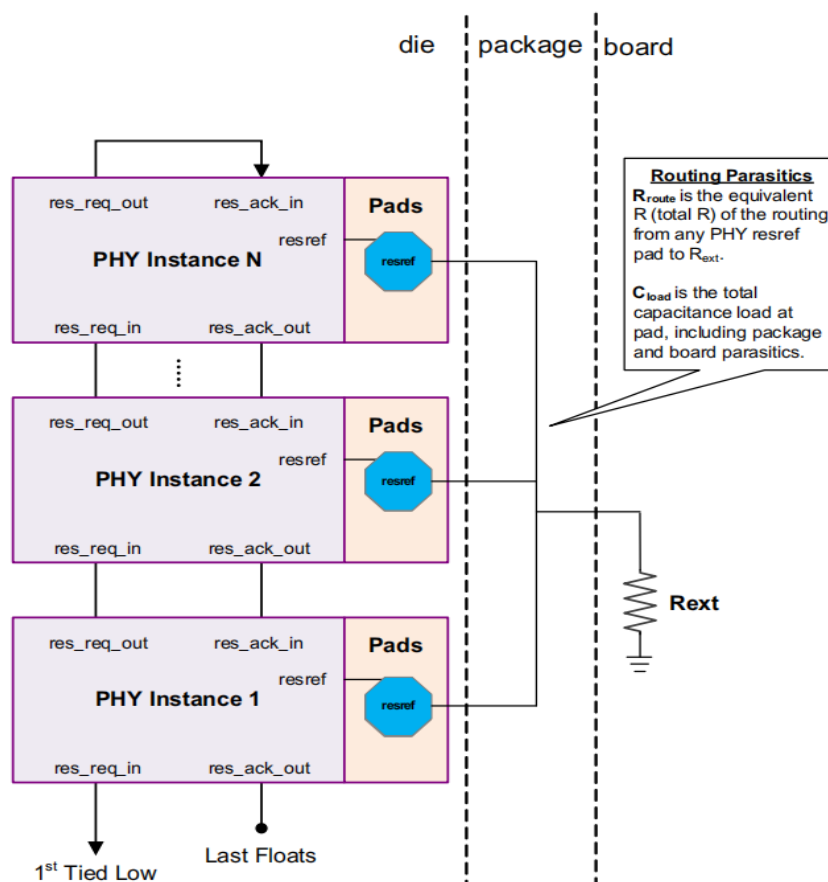


Figure 5-38 External Resistor Sharing

5.6.19.4 External Components

The PHY requires the following external components:

- AC coupling capacitors as specified by the standard, for reference, see Table 5-67.
- Supply bypass capacitors selected to minimize supply noise.
- Precision 200-Ohm ($\pm 1\%$) resistor used for biasing and termination impedance calibration; the precision resistor requirement is shown in Table 5-68.

Table 5-67 AC Coupling Capacitors - Target Protocol

Protocol	AC-Coupling Capacitor	Units
1000BASE-KX	4.7	nF
5GBase-KR	100	nF
10GBase-KR	100	nF
SGMII	Specification does not provide for a specific value.	nF
5G-SXGMII	100	nF
10G-SXGMII	100	nF

Table 5-68 External Component Requirements

Component	Value
resref	200 ohm (+/- 1%)

5.6.19.5 RX and TX Differential Signal Guidelines

You must design and simulate each of the RX and TX differential routes to ensure that signal integrity requirements are met.

Table 5-69 Package Guidelines for RX and TX Differential Routes

Parameter	Target	Unit	Note
Nominal differential impedance	85-100	Ohm	Trace nominal impedance should be selected to minimize the return loss and to maximize the system performance considering specific features of the protocols, other interconnect components in the system channel and the TX and RX termination via SI simulation.
Skew within differential pair	< 1	ps	Refer to the applicable standard for the specific SerDes protocol.
Skew between differential pairs			Refer to the applicable standard for the specific SerDes protocol
Differential insertion loss	Better than 1.5	dB	Up to Nyquist frequency for data rate lower than 25 Gbps
	Better than 2	dB	Up to Nyquist frequency for data rate at 25 or 32 Gbps

			Up to Nyquist frequency for data rate at 25 or 32 Gbps
Differential return loss	Better than -15	dB	Below Nyquist frequency
	Better than -10	dB	Below Baud rate
Common mode return loss	-6	dB	Up to Baud rate
Common mode to differential mode conversion loss	-20	dB	Up to Baud rate
Differential to common mode conversion loss	-20	dB	Up to Baud rate
Differential crosstalk coupling (Near-end crosstalk between TX and RX at die side, sum of all attackers)	-50	dB	Up to Nyquist frequency
Differential crosstalk coupling (Far-end crosstalk between TX and TX, or RX and RX, sum of all attackers)	-40	dB	Up to Nyquist frequency

5.6.19.6 SGMII LVDS AC DC Specification

The basis of the LVDS and termination scheme can be found in IEEE1596.3-1996. Some parameters have been modified to accommodate the 1.25Gb/s requirements. SGMII consists of the most lenient DC parameters between the general purpose and reduced range LVDS. Both the data and clock signals are DC balanced; therefore, implementations that meet the AC parameters but fail to meet the DC parameters may be AC coupled.

Figure 5-39 shows the DDR circuit at the source of the LVDS. The circuit passes data and clock with a 90 degree phase difference. The receiver samples data on both edges of the clock.

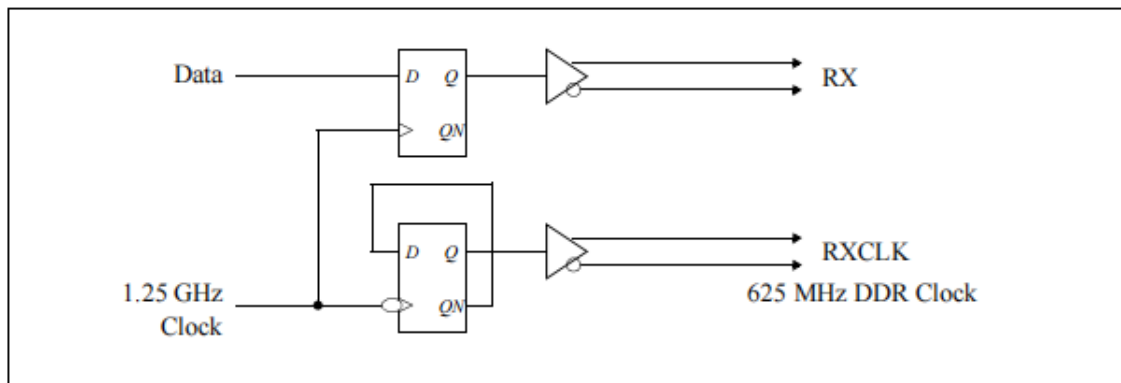


Figure 5-39 Reference data and clock circuit

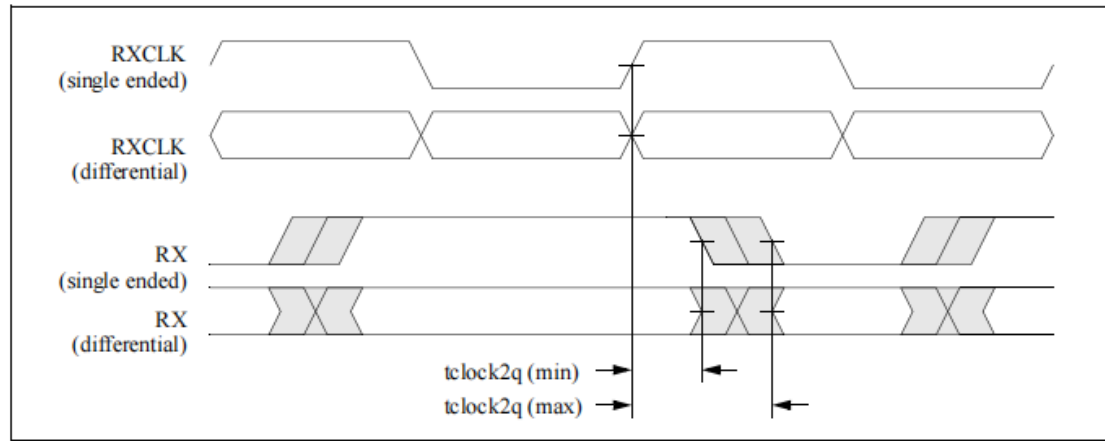


Figure 5-40 Driver Clock and Data Alignment

Table 5-70 Driver DC specification

Symbol ^a	Parameter ^b	Min	Max	Units
Voh	Output voltage high,		1525	mV
Vol	Output voltage low	875		mV
Vring	Output ringing		10	%
Vod	Output Differential Voltage	150	400	mV
Ro	Output impedance (single ended)	40	140	ohms
ΔRo	Mismatch in a pair		10	%
$\Delta Vod $	Change in Vod between “0” and “1”		25	mV
ΔVos	Change in Vos between “0” and “1”		25	mV
Isa, Isb	Output current on Short to GND		40	mA
Isab	Output current when a, b are shorted		12	mA
Ixa, Ixb	Power off leakage current		10	mA

a. For a detailed description of the symbols please refer to the IEEE1596.3-1996 standard.

b. All parameters measured at $R_{load} = 100\text{ohms} \pm 1\%$ load

Table 5-71 Receiver DC specification

Symbol	Parameter	Min	Max	Units
Vi	Input Voltage range a or b	675	1725	mV
Vidth	Input differential threshold	-50	+50	mV
Vhyst	Input differential hysteresis	25		mV
Rin	Receiver differential input impedance	80	120	ohms

Table 5-72 Driver AC specification

Symbol ^a	Parameter	Min	Max	Units
clock	Clock signal duty cycle @ 625MHz	48	52	%
t_{fall}	Vod fall time (20%-80%)	100	200	pSec
t_{rise}	Vod rise time (20%-80%)	100	200	pSec
t_{skew1}^b	Skew between two members of a differential pair - $ t_{PHLA} - t_{PLHB} $ or $ t_{PLHA} - t_{PHLB} $		20	pSec
$t_{clock2q}^c$	Clock to Data relationship: from either edges of the clock to valid data	250	550	pSec

- a. For a detailed description of the symbols please refer to the IEEE1596.3-1996 standard
- b. Skew measured at 50% of the transition
- c. Skew measured at 0v differential

Table 5-73 Receiver AC specification

Symbol	Parameter	Min	Max	Units
t_{setup}^a	setup time	100		pSec
t_{hold}	hold time	100		pSec

- a. Measured at 50% of the transition

5.6.19.7 10GBASE-KR electrical characteristics

1. Transmitter characteristics

Table 5-74 Transmitter characteristics for 10GBASE-KR

Parameter	Value	Units
Signaling speed	10.3125 ± 100 ppm	GBd
Differential peak-to-peak output voltage (max.)	1200	mV
Differential peak-to-peak output voltage (max.) with TX disabled	30	mV
Common-mode voltage limits	0 to 1.9	V
Common-mode voltage deviation (max) during LPI	150	mV
Differential output return loss (min.)	See equation (for $50 \text{ MHz} \leq f < 2500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9$) and (for $2500 \text{ MHz} \leq f \leq 7500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9 - 12 \log_{10}(f/2500, \text{Hz})$)	dB
Common-mode output return loss (min.)	See equation (for $50 \text{ MHz} \leq f < 2500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 6$) and (for $2500 \text{ MHz} \leq f \leq 7500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 6 - 12 \log_{10}(f/2500, \text{Hz})$)	dB
Transition time (20%–80%)	24 to 47	ps
Max output jitter (peak-to-peak)		
Random jitter ^a	0.15	UI
Deterministic jitter	0.15	UI
Duty Cycle Distortion ^b	0.035	UI
Total jitter	0.28	UI

- a. Jitter is specified at BER 10^{−12}.

- a. Duty Cycle Distortion is considered part of the deterministic jitter distribution.

(1) Test fixture

The test fixture of Figure 4-38 or its functional equivalent, is required for measuring the transmitter specifications, with the exception of return loss.

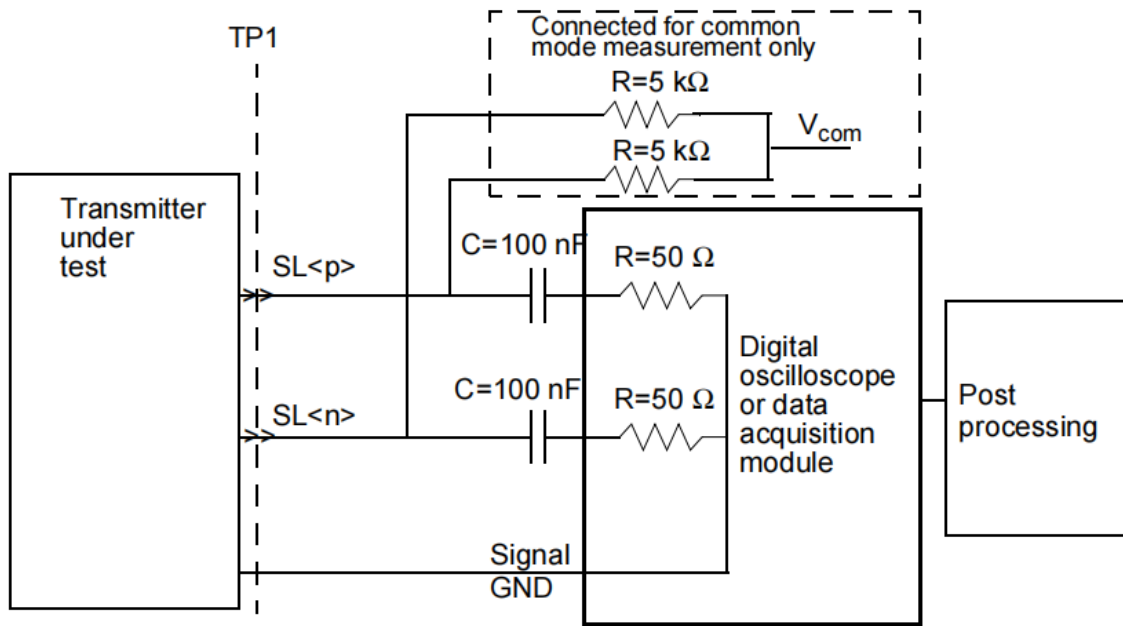


Figure 5-41 Transmit test fixture for 10GBASE-KR

(2) Test fixture impedance

The differential load impedance applied to the transmitter output by the test fixture depicted in Figure 5-41 shall be 100 Ω. The differential return loss, in dB with f in MHz, of the test fixture shall meet the requirements of Equation (for $100 \text{ MHz} \leq f < 5000 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 15$) and Equation (for $5000 \text{ MHz} \leq f \leq 10000 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 15 - 26.57 \log_{10}(f/500 \text{ MHz})$).

(3) Signaling speed

The 10GBASE-KR signaling speed shall be $10.3125 \text{ GBd} \pm 100 \text{ ppm}$.

(4) Output amplitude

The differential output voltage is constrained via the transmitter output waveform requirements. For a 1010 pattern, the peak-to-peak differential output voltage shall be less than 1200mV, regardless of equalization setting. The transmitter output voltage shall be less than 30mV peak-to-peak when disabled. The differential output voltage test pattern shall consist of no fewer than eight symbols of alternating polarity.

NOTE 1—The required test patterns may be found in the training pattern field of the training frames.

NOTE 2—See for an illustration of the definition of differential peak-to-peak output voltage.

DC-referenced voltage levels are not defined since the receiver is AC-coupled. The common-mode voltage of SL<p> and SL<n> shall be between 0 V and 1.9 V with respect to signal ground as measured at Vcom in Figure 5-42.

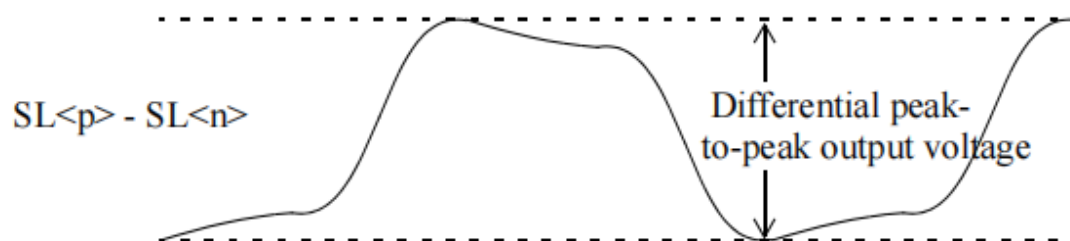


Figure 5-42 Transmitter differential peak-to-peak output voltage definition

NOTE—SL<p> and SL<n> are the positive and negative sides of the differential signal pair. For EEE capability, the transmitter's differential peak-to-peak output voltage shall be less than 30 mV within 500 ns of tx_mode being set to QUIET and remain so while tx_mode is set to QUIET. Furthermore, the transmitter's differential peak-to-peak output voltage shall be greater than 720 mV within 500 ns of tx_mode being set to ALERT. The transmitter output shall be fully compliant within 5 μ s after tx_mode is set to DATA. During LPI mode, the common mode shall be maintained to within ± 150 mV of the pre-LPI value.

(5) Differential output return loss

For frequencies from 50 MHz to 7500 MHz, the differential return loss, in dB with f in MHz, of the transmitter shall meet the requirements of Equation (for $50 \text{ MHz} \leq f < 2500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9$) and Equation (for $2500 \text{ MHz} \leq f \leq 7500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9 - 12\log_{10}(f/2500\text{MHz})$). This output impedance requirement applies to all valid output levels. The reference impedance for differential return loss measurements shall be 100 Ω .

The minimum differential output return loss is shown in Figure 5-43.

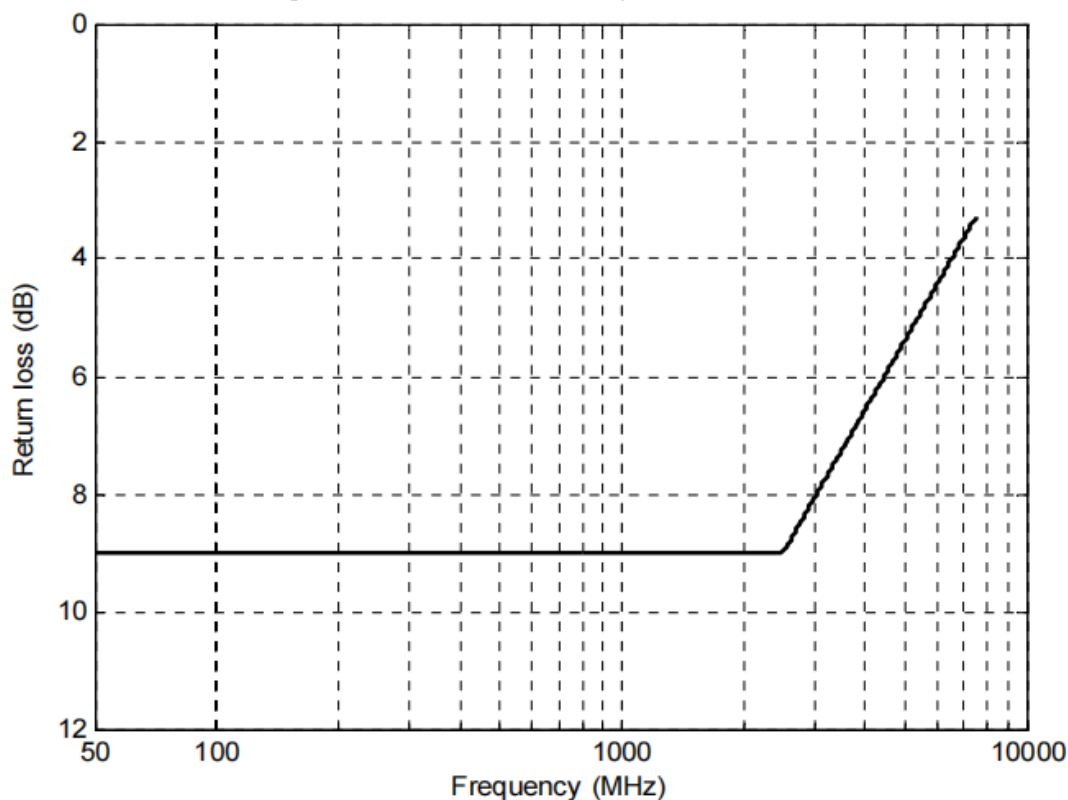
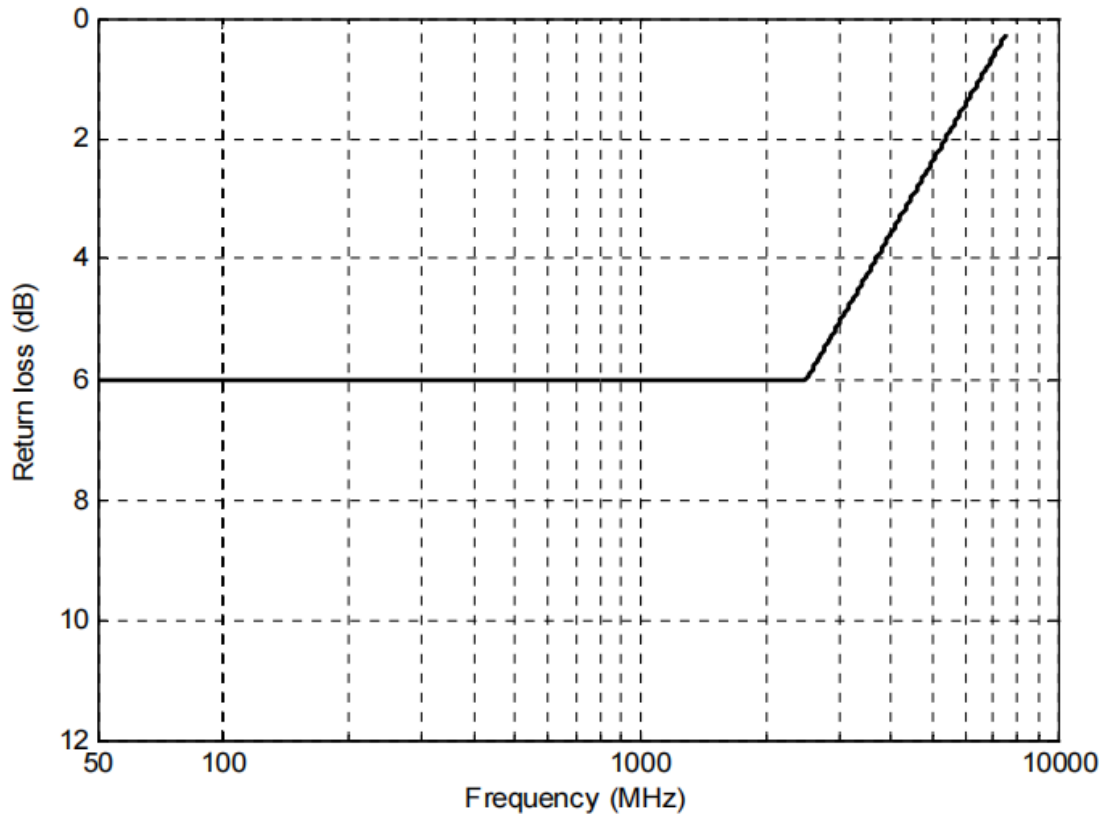


Figure 5-43 Minimum differential output return loss**(6) Common-mode output return loss**

The transmitter common-mode return loss shall meet the requirements of Equation (for $50 \text{ MHz} \leq f < 2500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 6$) and Equation (for $2500 \text{ MHz} \leq f \leq 7500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 6 - 12 \log_{10}(f/2500 \text{ MHz})$). The reference impedance for common-mode return loss measurements is 25Ω .

**Figure 5-44 Minimum common-mode output return loss****(7) Transition time**

The rising and falling edge transition times shall be between 24 ps and 47 ps as measured at the 20% and 80% levels referenced to v2 and v5. Measurement is done using the square wave test pattern, with no equalization and a run of at least eight consecutive ones. Transmit equalization may be disabled by asserting the preset control.

(8) Transmit jitter test requirements

Transmit jitter is defined with respect to a test procedure resulting in a BER bathtub curve. For the purpose of jitter measurement, the effect of a single-pole high-pass filter with a 3 dB point at 4 MHz is applied to the jitter. The data pattern for jitter measurements shall be test patterns 2 or 3. Crossing times are defined with respect to the mid-point (0 V) of the AC-coupled differential signal.

Equalization shall be off during jitter testing. Transmit equalization may be disabled by asserting the preset control. The duty cycle distortion test pattern shall consist of no fewer than eight symbols of alternating polarity.

(9) Transmit jitter

The transmitter shall have a maximum total jitter of 0.28 UI peak-to-peak, composed of a maximum deterministic component of 0.15 UI peak-to-peak and a maximum random component of 0.15 UI peak-to-peak. Duty cycle distortion (DCD) is considered a component of deterministic jitter and shall not exceed 0.035 UI peak-to-peak. The peak-to-peak duty cycle distortion is defined as the absolute value of the difference in the mean pulse width of a 1 pulse or the mean pulse width of a 0 pulse (as measured at the mean of the high- and low-voltage levels in a clock-like repeating 0101 bit sequence) and the nominal pulse width.

(10) Transmitter output waveform

The 10GBASE-KR transmitter includes programmable equalization to compensate for frequency-dependent loss in the backplane channel and facilitate data recovery at the receiver. This equalization may be accomplished with a three-tap finite impulse response (FIR) structure as shown in Figure 5-45. The actual implementation of the transmit equalizer, including the incorporation of additional taps, is beyond the scope of this standard.

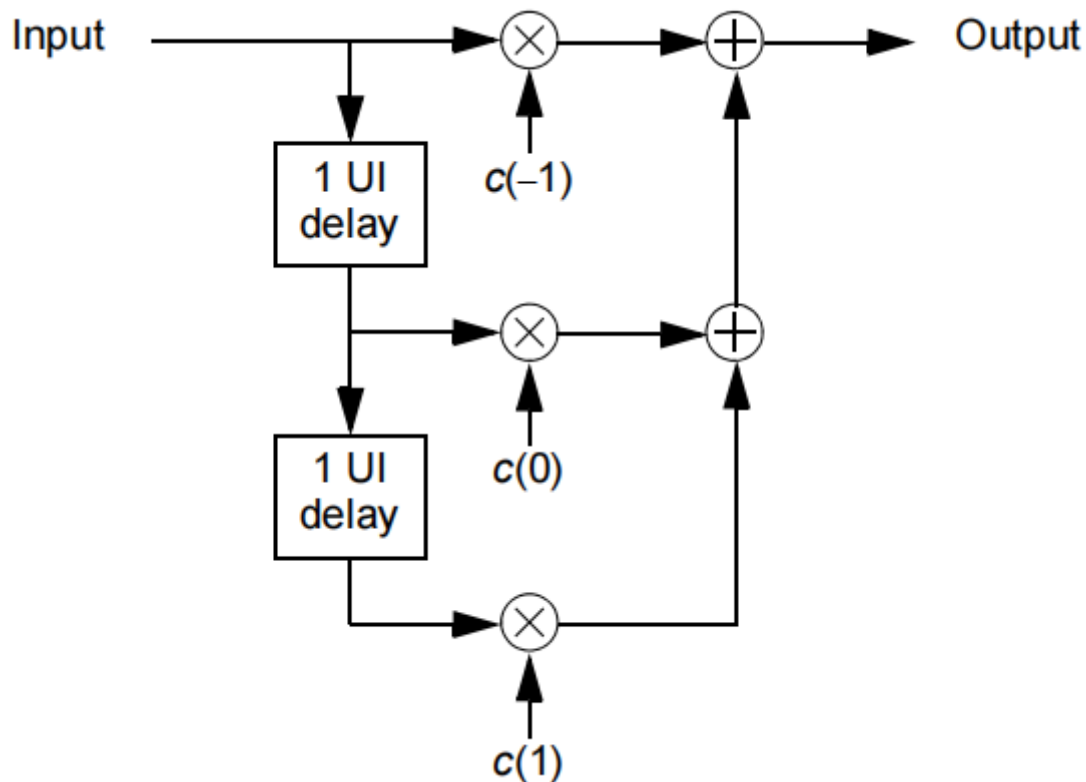


Figure 5-45 Transmit equalizer example

It should be noted that the valid ranges of the $c(1)$ and $c(-1)$ coefficients may include positive and negative values. A value of zero is used to turn off equalization for the tap.

(11) Transmitter output waveform requirements

The transmitter output waveform test is based on the voltages v_1 through v_6 , Δv_2 , and Δv_5 , which shall be measured as shown in Figure 5-46 and described below.

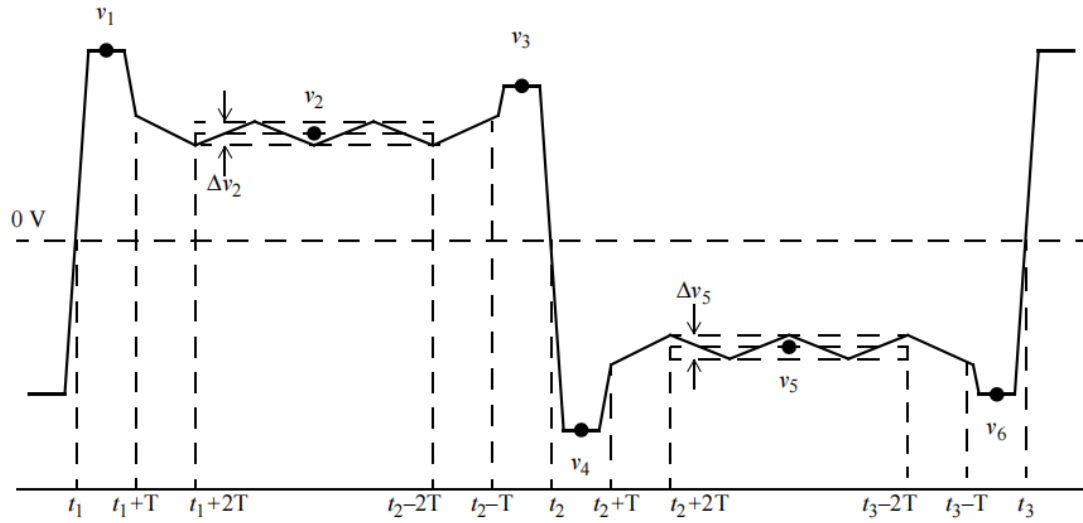


Figure 5-46 Transmitter output waveform

T = symbol period

t_1 = zero-crossing point of the first rising edge of the AC-coupled signal

t_2 = zero-crossing point of the falling edge of the AC-coupled signal

t_3 = zero-crossing point of the second rising edge of the AC-coupled signal

v_1 = maximum voltage measured in the interval t_1 to $t_1 + T$

v_2 = positive steady-state voltage measured as the average voltage in the interval $t_1 + 2T$ to $t_2 - 2T$

v_3 = maximum voltage measured in the interval $t_2 - T$ to t_2

v_4 = minimum voltage measured in the interval t_2 to $t_2 + T$

v_5 = negative steady-state voltage measured as the average voltage in the interval $t_2 + 2T$ to $t_3 - 2T$

v_6 = minimum voltage measured in the interval $t_3 - T$ to t_3

Δv_2 = positive voltage ripple measured as the peak-to-peak value of the difference between the voltage in the range $t_1 + 2T$ to $t_2 - 2T$ and v_2

Δv_5 = negative voltage ripple measured as the peak-to-peak value of the difference between the voltage in the range $t_2 + 2T$ to $t_3 - 2T$ and v_5

From these voltages, the pre- and post-cursor equalization ratios R_{pre} and R_{pst} are derived from Equation ($R_{pre} = v_3/v_2$) and Equation ($R_{pst} = v_1/v_2$).

The state of the transmitter equalizer and hence the transmitter output waveform is manipulated via the protocol defined in (10) or via management. The changes in the transmitter output waveform resulting from coefficient update requests shall meet the requirements stated in **Table 5-75**. The coefficient update requests in **Table 5-75** are to be followed by a coefficient update equal to hold for all taps. The results shall be verified after the coefficient status for all taps is reported as not_updated.

Table 5-75 Transmitter output waveform requirements related to coefficient update

Coefficient update ^a			Requirements ^b		
$c(1)$	$c(0)$	$c(-1)$	$v_1(k) - v_1(k - 1)$	$v_2(k) - v_2(k - 1)$	$v_3(k) - v_3(k - 1)$

			(mV)	(mV)	(mV)
increment	hold	hold	-20 to -5	5 to 20	5 to 20
decrement	hold	hold	5 to 20	-20 to -5	-20 to -5
hold	increment	hold	5 to 20	5 to 20	5 to 20
hold	decrement	hold	-20 to -5	-20 to -5	-20 to -5
hold	hold	increment	5 to 20	5 to 20	-20 to -5
hold	hold	decrement	-20 to -5	-20 to -5	5 to 20

a Step size requirements for the tap under test apply regardless of the current value of the other taps.

b This difference is measured relative to the voltage prior to the assertion coefficient update k equal to hold.

For any coefficient update, the magnitudes of the changes in v_1 , v_2 , and v_3 shall be within 5 mV of each other. When sufficient increment or decrement updates have been applied to a given tap, it will reach a maximum or minimum limit governed by the coefficient range or by restrictions placed on minimum steady-state or maximum peak voltage, and the coefficient status is reported accordingly.

The transmitter output waveform shall meet the requirements of **Table 5-76** for all of the limiting cases represented in the table. Implementation of $c(-1)$ or $c(1)$ coefficient values greater than zero or less than the minimum defined by $R_{pre}(\min)$ and $R_{pst}(\min)$ is optional. A coefficient may be disabled by first asserting the preset control, then manipulating the other coefficients as required by the test.

Table 5-76 Transmitter output waveform requirements related to coefficient status

Coefficient status			Requirements		
$c(1)$	$c(0)$	$c(-1)$	R_{pre}	R_{pst}	v_2 (mV)
disabled	minimum	disabled	0.90 to 1.10	0.90 to 1.10	220 to 330
disabled	maximum	disabled	0.95 to 1.05	0.95 to 1.05	400 to 600
minimum	minimum	disabled	—	4.00 (min)	—
disabled	minimum	minimum	1.54 (min)	—	—

In addition:

- The quantities Δv_2 and Δv_5 shall not exceed 40 mV peak-to-peak.
- The positive and negative voltages shall match such that each of the quantities $(v_1 + v_4)/v_1$, $(v_2 + v_5)/v_2$, and $(v_3 + v_6)/v_3$ does not exceed 0.05.
- The quantity v_2 shall be greater than or equal to 40 mV.
- Any coefficient update equal to decrement applied to any tap that would result in v_2 less than 40 mV shall return a coefficient status value minimum.
- Any coefficient update equal to decrement that would result in a violation of (4) shall return a coefficient status value minimum for that coefficient.
- Any coefficient update equal to increment that would result in a violation of (4) shall return a coefficient status value maximum for that coefficient.

2. Receiver characteristics

Table 5-77 Receiver characteristics for 10GBASE-KR

Parameter	Value	Units
Bit error ratio	10 ⁻¹²	
Signaling speed	10.3125 ± 100 ppm	GBd
Receiver coupling	AC	
Differential input peak-to-peak amplitude (maximum)	1200a	mV
Differential input return loss (minimum)b	See equation	dB

	(for 50 MHz $\leq f < 2500$ MHz, ReturnLoss(f) ≥ 9) and (for 2500 MHz $\leq f \leq 7500$ MHz, ReturnLoss(f) $\geq 9 - 12 \log_{10}(f/2500, \text{Hz})$)	
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a The receiver shall tolerate amplitudes up to 1600 mV without permanent damage.

b Relative to 100 Ω differential.

(1) Receiver interference tolerance

The receiver interference tolerance shall consist of two separate tests as described in Annex 69A with the parameters specified in **Table 5-78**. The data pattern for the interference tolerance test shall be the test patterns 2 or 3. The receiver shall satisfy the requirements for interference tolerance specified in Annex 69A for both tests.

Table 5-78 10GBASE-KR interference tolerance parameters

Parameter	Test 1 values	Test 2 values	Units
Target BER	10^{-12}	10^{-12}	
m_{TC} (min.) ^a	1.0	0.5	
Amplitude of broadband noise (min. RMS)	5.2	12	mV
Applied transition time (20%–80%, min.)	47	47	ps
Applied Sinusoidal jitter (min. peak-to-peak)	0.115	0.115	UI
Applied random jitter (min. peak-to-peak) ^b	0.130	0.130	UI
Applied Duty Cycle Distortion (min. peak-to-peak)	0.035	0.035	UI

a m_{TC} is defined in Equation (69A–6) of Annex 69A.

b Applied random jitter is specified at a BER of 10^{-12} .

(2) Signaling speed range

A 10GBASE-KR receiver shall comply with the requirements of **Table 5-77** for any signaling speed in the range 10.3125 GBd $\pm$ 100 ppm.

(3) AC-coupling

The 10GBASE-KR receiver shall be AC-coupled to the backplane to allow for maximum interoperability between various 10 Gb/s components. AC-coupling is considered to be part of the receiver for the purposes of this specification unless explicitly stated otherwise. It should be noted that there may be various methods for AC-coupling in actual implementations.

NOTE—It is recommended that the maximum value of the coupling capacitors be limited to 100 nF. This will limit the inrush currents to the receiver that could damage the receiver circuits when repeatedly connected to transmit modules with a higher voltage level.

(4) Input signal amplitude

10GBASE-KR receivers shall accept differential input signal peak-to-peak amplitudes produced by compliant transmitters connected without attenuation to the receiver, and still meet the BER requirement. Note that this may be larger than the 1200 mV differential maximum due to the actual transmitter output and receiver input impedances. The input impedance of a receiver can cause the minimum signal into a receiver to differ from that measured when the receiver is replaced with a 100 Ω test load. Since the channel is AC-coupled, the absolute voltage levels with respect to the receiver ground are dependent on the receiver implementation.

(5) Differential input return loss

For frequencies from 100 MHz to 7500 MHz, the differential return loss, in dB with f in MHz, of the

receiver shall be greater than or equal to Equation (for $50 \text{ MHz} \leq f < 2500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9$) and Equation (for $2500 \text{ MHz} \leq f \leq 7500 \text{ MHz}$, $\text{ReturnLoss}(f) \geq 9 - 12 \log_{10}(f/2500, \text{Hz})$). This return loss requirement applies at all valid input levels. The reference impedance for differential return loss measurements is 100Ω .

5.6.20 UFS MIPI MPHY parameters

The MIPI MPHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

5.6.20.1 Reference Clock: Jitter Requirements

The tables below list reference clock jitter for applicable reference clock architectures.

Table 5-79 Reference clock Random Jitter (RJ) Characteristics

Protocol	REFCLK_RJ1 12k-20M(ps, RMS)	REFCLK_RJ2 2M-20M (ps, RMS)
MPHY G1 @ BER 1e-10	4.20	4.00
MPHY G2 @ BER 1e-10	3.20	2.20
MPHY G3 @ BER 1e-10	2.90	1.90
PHY G4 @ BER 1e-10	1.40	0.90
MPHY G1 @ BER 1e-12	3.80	3.60
MPHY G2 @ BER 1e-12	2.90	2.00
MPHY G3 @ BER 1e-12	2.50	1.70
MPHY G4 @ BER 1e-12	1.30	0.80

5.6.20.2 Reference Clock: Electrical Characteristics

Reference clock electrical characteristics are presented in Table.

Table 5-80 Reference Clock Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Conditions
FREF_OFFSET	Reference clock frequency offset	-300	300	ppm	
DCREF_CLK	Duty cycle	40	60	%	
	Maximum input voltage level		VP	V	Applies to ref_alt_clk_p only (CMOS level)
	Minimum input voltage level	0		V	
	Rise/Fall times (20%-80%)		100	ps	
	R load	200	300	Ohm	
	C load	20	40	fF	
	Maximum input voltage level		0.88	V	Applies to ref_pad_clk_p / ref_pad_clk_m only
	Minimum input voltage level	-0.15		V	
	Absolute maximum voltage at pin including overshoot		1.15 ^b	V	
	Absolute minimum voltage	-0.3 ^b		V	

	at pin including overshoot				
	Single-ended input swing	0.3	0.88	V _{pp}	
	Differential input swing	0.3	1.76	V _{ppdif}	
SWREF_CLK	Differential ^c input edge rate	0.6 ^d	4.0 ^d	V/ns	
REF_CLK_SKEW	Reference clock skew (±)		200	ps	

b. Allowed overshoot time: 10% of period.

c. In case ref_pad_clk is used in single mode, input edge rate is required to be fulfilled on the single ended ref_pad_clk_p/m signal.

d. To be verified from -150mV to +150mV (centered on the differential zero crossing) on the differential waveform. During this 300mV region, the signal must be monotonic through the both rise and fall times.

- ref_pad_clk_p/ref_pad_clk_m inputs can be low-voltage swing, or CMOS level
- ref_alt_clk inputs are CMOS-level on

Note: Only one of the on-chip and off-chip clock inputs may be used at a time.

■ If an on-chip clock source is used, then you must either tie the off-chip source to ground or leave it floating

■ If an off-chip clock source is used, then you must tie the on-chip source to ground

5.6.20.3 Reference Clock - External Off-Chip Clock Sources

When routing reference clocks in from an off-chip, a signal integrity (SI) analysis should be performed using a model of the clock input, package, board traces, and clock source. This analysis should confirm that the reference clock input specification is being met across the external on-chip reference clock input (ref_pad_clk_p/ref_pad_clk_m).

Table 5-81 Model of Reference Clock Pad Input Load

Location	Parasitic Device	Value	Units
OFF PHY	R1	<= 50	Ohm
OFF PHY	C1	<= 1	pF
Within PHY: Bump to prescaler input	R2	<= 50	Ohm
Within PHY: Bump to prescaler input	C2	<= 1	pF
Within prescaler: 2nd ESD Prot resistor	R3	1.5k	Ohm
Within prescaler: AC cap	C3	920	fF
Within prescaler: DC Bias resistor	R4	468k	Ohm
Within prescaler: Input cap of prescaler input RX	C4	<= 50	fF

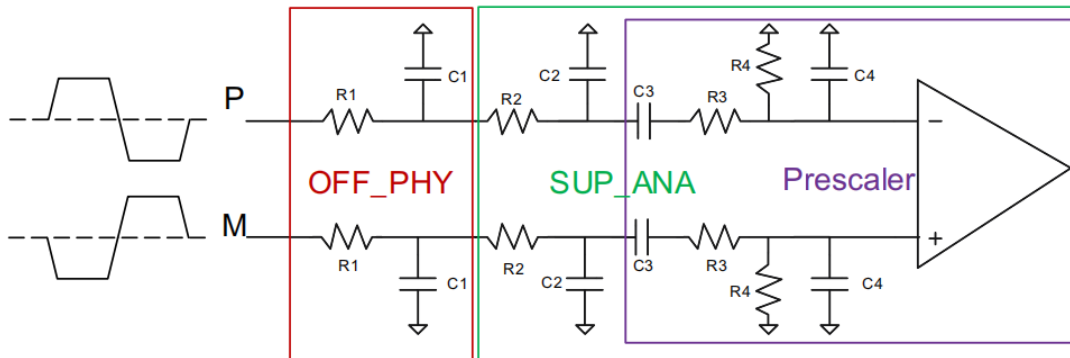


Figure 5-47 Model of Reference Clock Pad Input Load

The reference clock inputs are not terminated. This is to allow for connection of reference clocks to multiple PHYs at the same time. If termination is used, it should be placed on your board as close to the chip as possible.

5.6.20.4 Reference Resistor Requirements

Table 5-82 Table Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 Ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2mA	None
External Reference Resistor parasitic capacitance C_{load}	<10pF	It is important to meet the total capacitance at low frequency (in other words, the total capacitance of pad, package trace, and PCB). The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA can support up to 8.5 pF (10 pF- 1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 4 pF (10 pF -4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 Ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals, for example clocks, should be spaced away from resref or shielded with ground.

5.6.20.5 External Components

The PHY requires the following external components:

- Supply bypass capacitors selected to minimize supply noise
- Precision 200 Ohm ($\pm 1\%$) resistor used for biasing and termination impedance calibration

Table 5-83 Example Supply Bypass Capacitors

Component	Value
Power Supply Bypass Capacitors	0.01 uF, 0.1 uF, 4.7 uF, 10.0 uF

a. Smaller-value capacitors must be placed between the ferrite bead and the package.

Table 5-84 External Component Requirements

Component	Value
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resref	200 Ohm (+/- 1%)
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5.6.21 LVDS PHY parameters

5.6.21.1 LVDS timing parameters

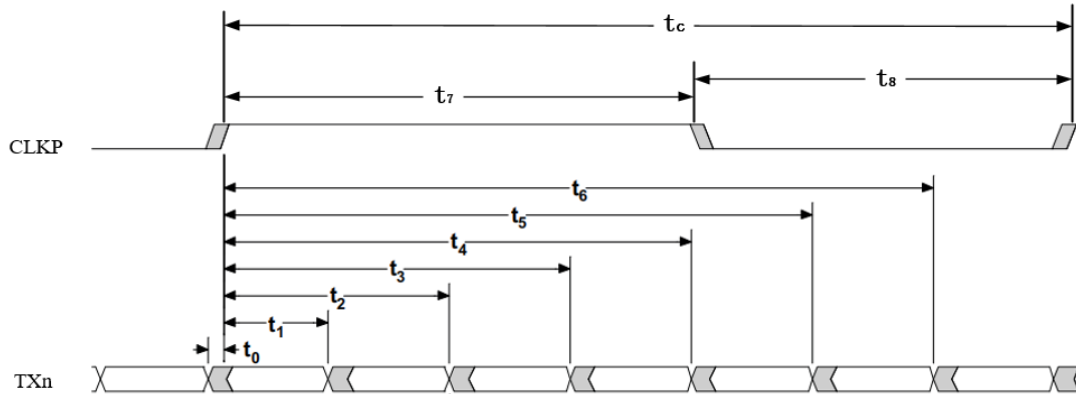


Figure 4-48 LVDS timing diagram

Table 5-85 LVDS timing parameters

Parameter	Symbol	Description	Min	Typ	Max	Unit
CLKP Frequency Range	tc		27		148.5	MHz
CLKP Duty Cycle	t7:t8			4:3		
Data Output Transition Time	t0	Delay time, CLKOUT↑ after TX0 valid	-0.1	0	0.1	ns
	t1	Delay time, CLKOUT↑ after TX1 valid	1/7tc-0.1	0	1/7tc+0.1	ns
	t2	Delay time, CLKOUT↑ after TX2 valid	2/7tc-0.1	0	2/7tc+0.1	ns
	t3	Delay time, CLKOUT↑ after TX3 valid	3/7tc-0.1	0	3/7tc+0.1	ns
	t4	Delay time, CLKOUT↑ after TX4 valid	4/7tc-0.1	0	4/7tc+0.1	ns
	t5	Delay time, CLKOUT↑ after TX5 valid	5/7tc-0.1	0	5/7tc+0.1	ns
	t6	Delay time, CLKOUT↑ after TX6 valid	6/7tc-0.1	0	6/7tc+0.1	ns

5.6.22 DP/eDP PHY parameters

The PHY is compatible with all the features and datarates defined in the following specifications:

- VESA DisplayPort (DP) Standard, Version 1.4, 23 February, 2016
- VESA DisplayPort Alt Mode on USB Type-C Standard, Version 1.0b, 03 November, 2017
- VESA DisplayPort Alt Mode on USB Type-C 1.0a Compliance Test Specification, Revision 1.0, 24 January, 2017

The PHY does not support DisplayPort Dual Mode specification over Type-C connector as stated in DisplayPort Alternate Mode v1.0a specification.

DP 1.4 compliant with support for RBR, HBR, HBR2 and HBR3.

DP BR_GEN2 is not supported.

Support for eDP 1.4b excluding the recommended low voltage swing.

Display port Dual Mode can be supported over DP connector with an external passive DP to HDMI

translator or DP++ cable adapter.

The PHY may impose PCB routing limitations at higher data rates due to the additional loading in the pads. Should the customer require relaxation on these PCB routing limitations, repeaters may be needed, or an alternative DP/HDMI specific metal option can be evaluated.

The PHY is designed to handle a wide range of input clock frequencies to support a wide range of applications.

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

5.6.22.1 Reference Clock: Jitter Requirements

The tables below list reference clock jitter for applicable reference clock architectures.

Table 5-86 Reference Clock Requirements: Random and Deterministic Jitter

Protocol	Parameter	Max	Units	Conditions
RJ _{REFCLK} DP HBR2	Reference clock Random jitter	3.5	ps, rms	Integrated Rj from 12kHz to 20MHz
		2.4		Integrated Rj from 2MHz to 20MHz
DJ _{REFCLK} DP HBR2	Reference clock Deterministic Jitter (Dj)	14.9	ps, pp	Integrated Dj from 800KHz to 70MHz
		7.4		Integrated Dj from 2MHz to 14MHz
RJ _{REFCLK} DP HBR3	Reference clock Random jitter	3.5	ps, rms	Integrated Rj from 12kHz to 20MHz
		2.4		Integrated Rj from 2MHz to 20MHz
DJ _{REFCLK} DP HBR3	Reference clock Deterministic Jitter (Dj)	14.9	ps, pp	Integrated Dj from 1MHz to 100MHz
		7.4		Integrated Dj from 3MHz to 20MHz

5.6.22.2 Reference Clock: Electrical Characteristics

Table 5-87 Reference Clock Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Conditions
FREF_OFFSET	Reference clock frequency offset	-300	300	ppm	
DCREF_CLK	Duty cycle	40	60	%	
	Maximum input voltage level		VP	V	Applies to ref_alt_clk_p only (CMOS level)
	Minimum input voltage level	0		V	
	Rise/Fall times (20%-80%)		100	ps	
	R load	200	300	Ohm	
	C load	20	40	fF	
	Maximum input voltage level		0.88	V	Applies to ref_pad_clk_p / ref_pad_clk_m only
	Minimum input voltage level	-0.15		V	
	Absolute maximum voltage at pin including overshoot		1.15 ^b	V	
	Absolute minimum voltage at pin including overshoot	-0.3 ^b		V	
	Single-ended input swing	0.3	0.88	V _{pp}	
	Differential input swing	0.3	1.76	V _{ppdif}	
	Differential ^c input edge rate	0.6 ^d	4.0 ^d	V/ns	
SWREF_CLK	Differential ^c input edge rate	0.6 ^d	4.0 ^d	V/ns	
REF_CLK_SKEW	Reference clock skew (±)		200	ps	

b. Allowed overshoot time: 10% of period.

c. In case ref_pad_clk is used in single mode, input edge rate is required to be fulfilled on the single ended ref_pad_clk_p/m signal.

d. To be verified from -150mV to +150mV (centered on the differential zero crossing) on the differential waveform. During this 300mV region, the signal must be monotonic through the both rise and fall times.

- ref_pad_clk_p/ref_pad_clk_m inputs can be low-voltage swing, or CMOS level
- ref_alt_clk inputs are CMOS-level only

Note: Only one of the on-chip and off-chip clock inputs may be used at a time.

■ If an on-chip clock source is used, then you must either tie the off-chip source to ground or leave it floating

■ If an off-chip clock source is used, then you must tie the on-chip source to ground

5.6.22.3 Reference Clock Implementation: External Off-Chip Clock Sources

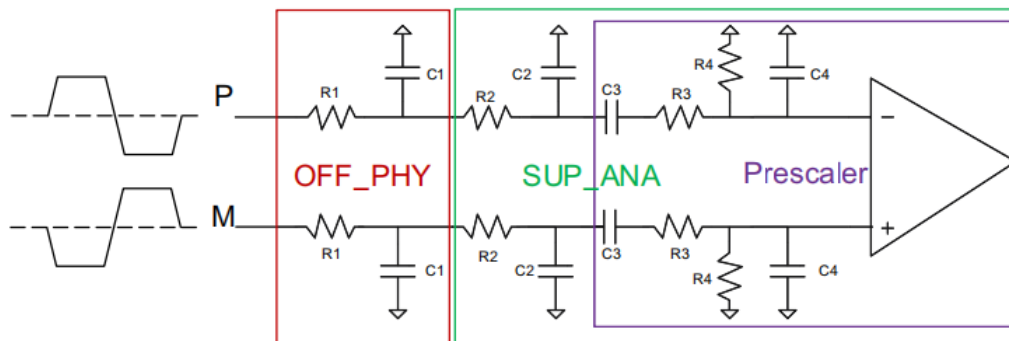


Figure 5-49 Model of Reference Clock Pad Input Load

Table 5-88 Model of Reference Clock Pad Input Load

Location	Parasitic Device	Value	Units
OFF PHY	R1	≤ 50	Ohm
OFF PHY	C1	≤ 1	pF

5.6.22.4 Reference Resistor Requirements

Table 5-89 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 Ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2mA	None
External Reference Resistor parasitic capacitance C_{load}	$<10\text{pF}$	It is important to meet the total capacitance at low frequency (in other words, the total capacitance of pad, package trace, and PCB). The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA can support up to 8.5 pF (10 pF- 1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 4 pF (10 pF -4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 Ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals, for example clocks, should be spaced away from resref or shielded with ground.

5.6.22.5 External Components

The PHY requires the following external components:

- Supply bypass capacitors selected to minimize supply noise.
- Precision 200 Ohm ($\pm 1\%$) resistor used for biasing and termination impedance calibration.

Table 5-90 Example Supply Bypass Capacitors

Component	Value ^a
Power Supply Bypass Capacitors	0.01 uF, 0.1 uF, 4.7 uF, 10.0 uF

a. Smaller-value capacitors must be placed between the ferrite bead and the package.

Table 5-91 External Component Requirements

Component	Value
resref	200 Ohm (+/- 1%)

5.6.22.6 DP AUX-I2C PHY

Table 5-92 Maximum Pad/Package/Board Requirements

Signal	Signal Description	Series Resistance ^a (Ω)	Series Inductance (nH)	Capacitance (pF)	Worst-Case Interconnect Metal Requirements (mA)
DVDD	Digital-core supply	1	2	-	1
DGND	Macro	<0.1	2	-	1
PADP PADN	Analog signal	0.2	2	1	8
VPH GD	Analog signal	0.2	2		8

Table 5-93 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Typical	Max	Units
DVDD	High Voltage IO supply	0.75 - 7%	0.75	0.75 + 10%	V
VPH	Analog supply	1.8 - 5%	1.8	1.8 + 10%	V

5.6.23 MIPI DSI/CSITX DPHY parameters

Table 5-94 TX DPHY High Speed Clock Timings

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
-	Maximum Serial Data rate (forward direction)	On DATAP/N outputs. $80\Omega \leq RL \leq 125\Omega$	80		2500	Mbps	
FDDRCLK	DDR CLK frequency	On CLKP/N outputs	40		1250	MHz	
TDDRCLK	DDR CLK period	$80\Omega \leq RL \leq 125\Omega$	0.8		25	ns	
UIINST	UI instantaneous		0.4		12.5	ns	This value corresponds to

							a minimum Mbps data rate.
Δ UI	UI variation		-10%		10%	UI	When UI $\geq$ 1ns, within a single burst.
			-5%		5%	UI	When 0.667ns $<$ UI $<$ 1ns, within a single burst.
tCDC	DDR CLK duty cycle	tCDC = tCPH / TDDRCLK		50		%	
tCPH	DDR CLK high time			1		UI	
tCPL	DDR CLK low time			1		UI	
-	DDR CLK / DATA Jitter			75		ps pk-pk	When UI $<$ 1ns, within a single burst.
TSKEW[TX]	Data to Clock Skew (TX)		-0.15		0.15	UI	For PHY operating at or below 1.5Gbps

5.6.24 MIPI RX CDPHY parameters

Table 5-95 MIPI RX DPHY High Speed Clock Timings

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
-	Maximum Serial Data rate (forward direction)	On DATAP/N outputs. $80\Omega \leq RL \leq 125\Omega$	80		4500	Mbps	
FDDRCLK	DDR CLK frequency	On CLKP/N outputs	40		1250	MHz	
TDDRCLK	DDR CLK period	$80\Omega \leq RL \leq 125\Omega$	4		25	ns	
UIINST	UI instantaneous		0.22		12.5	ns	This value corresponds to a minimum Mbps data rate.
Δ UI	UI variation		-10%		10%	UI	
Period Jitter			-5%		5%	UI	Applies when the DDR clock period within a single burst, that is clock rising edge to next rising edge is ≥ 0.444 ns and < 0.8 ns
tCDC	DDR CLK duty cycle	tCDC = tCPH / TDDRCLK		50		%	
tCPH	DDR CLK high time			1		UI	
tCPL	DDR CLK low time			1		UI	
-	DDR CLK / DATA Jitter			75		ps pk-pk	When UI $<$ 1ns, within a

							single burst.
TSKEW[TX]	Data to Clock Skew (TX)		-0.15		0.15	UI	For PHY operating at or below 1 Gbps
			-0.2		0.2	UI	For PHY operating above 1 Gbps and below or at 1.5Gbps
TSETUP[RX]	Data to Clock Setup Time (RX)		0.15			UI	For PHY operating at or below 1Gbps
			0.2			UI	For PHY operating above 1Gbps and below or at 1.5Gbps
TSKEW[LIS] static	Channel Data to Clock Skew		-0.2		0.2	UI	For PHY operating above 1 Gbps
THOLD[RX]	Data to Clock Hold Time (RX)		0.15			UI	For PHY operating at or below 1 Gbps
			0.20			UI	For PHY operating above 1 Gbps and below or at 1.5Gbps
tSKEW[TX] static	Static Data to Clock Skew (TX)		-0.2		0.2	UI	For PHY operating above 1.5Gbps
tSKEW[TLIS] static	Channel Static Data to Clock Skew		-0.10		0.1	UI	For PHY operating above 1 Gbps
TSKEW[RX] static	Static Data to Clock Skew RX Tolerance		-0.3		0.3	UI	For PHY operating above 1.5Gbps
BER	Target Bit Error Rate			10 ⁻¹²			
Q Factor for BER	QBER			7.0345			

Table 5-96 MIPI RX DPHY Calibration and Global Timing Parameters

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
TSKEWCA L initial	Time that the transmitter drives the skew-calibration pattern in the initial skew-calibration mode				100	us	For PHY Operating above 1.5Gbps
			2 ⁻¹⁵			UI	

TSKEWCA L periodic	Time that the transmitter drives the deskew calibration pattern in the periodic skew-calibration mode				10	us	For PHY Operating above 1.5Gbps
			2^{10}			UI	
TALTCAL	Time that the transmitter drives the Alternate Calibration Pattern				100	us	For PHY Operating above 2.5Gbps
			2^{15}			UI	
TLP-EXIT	Time that the transmitter drives the LP-11 between any LP sequences, or between an LP sequence and a HS burst.		2			tLPX	
TCLK-POST (8 bits)	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. The interval is defined as the period from the end of THS-TRAIL to the beginning of TCLK-TRAIL		$60\text{ns} + 52 \cdot \text{UI}$			ns	DPHY 8 bits
TCLK-POST (16 bits)	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. The interval is defined as the period from the end of THS-TRAIL to the beginning of TCLK-TRAIL		$60\text{ns} + 84 \cdot \text{UI}$			ns	DPHY 16 bits

Table 5-97 MIPI RX CPHY Calibration and Global Timing Parameters

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
t3-PREAMBLE	t3-PREBEGIN+t3-PROGSEQ+t3-PREEND		469			UI	t3-PREBEGIN should be adjustable at the transmitter from 7 UI minimum to 448 UI maximum in increments of 7 UI.
t3-CALPRE AMBLE_RX	Time for which the Receiver accepts the sequence of all “1” symbols		469		1792	UI	
t3-Post	Sequence of 4 symbols after end of Packet Data		224			UI	
TLP-EXIT	Time that the transmitter drives the LP-111 between any LP sequences, or between an LP sequence and a HS burst.		2			tLPX	

Table 5-98 MIPI RX DPHY Receiver Eye Diagram Details

Bit Error Rate	Teye_rx(UI)	Vdif_rx(mv)
10 ⁻¹²	0.5	40
10 ⁻⁶ (Prorated for validation)	0.53	47

Table 5-99 MIPI RX CPHY Receiver Eye Diagram Details

Symbol	Parameter	Condition	Mini.	Typ.	Max.	Unit	Notes
tEYE_RAM P_RX	Eye ramp time at the receiver		0.25			UI	
tEYE_WID TH_RX	Eye width at the receiver		0.5			UI	
tUI_AVER	UI average			UIIN			

AGE				ST			
UICHAN(n) _RX	Minimum Rx UI at the receiver		0.6			UI	
UI_JitterPE AK_RX	Maximum Rx UI Jitter at the receiver				0.4	UI	

5.6.25 USB3 PHY parameters

The PHY complies with all of the required features of the Universal Serial Bus 3.1 Specification, Revision 1.0, July 26 2013, USB Implementers Forum, Inc.

The PHY is complaint with all the features and datarates defined in the USB Type-C Standard, Version 1.0, 22 September, 2014.

The PHY is also backwards compatible to the USB3.0 SS (5 Gbps) protocol and data rate. Add-on USB 2.0 PHY is required to support USB 2.0 HS (480Mbps), FS (12Mbps) and LS (1.5Mbps) data rates.

The PHY is designed to handle a wide range of input clock frequencies to support a wide range of applications.

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

5.6.25.1 Reference Clock: Jitter Requirements

The tables below list reference clock jitter for applicable reference clock architectures.

Table 5-100 Reference Clock Requirements: Random and Deterministic Jitter

Protocol	Parameter	Max	Units	Conditions
Rj _{REFCLK} USB 3.0 - 5.0Gbps	Reference clock Random jitter (Rj)	4.3	ps, rms	Integrated Rj from 12kHz to 20MHz
		3.0		Integrated Rj from 2MHz to 20MHz
Dj _{REFCLK} USB 3.0 - 5Gbps	Reference clock Deterministic Jitter (Dj)	12.0	ps, pp	Integrated Dj from 800kHz to 70MHz
		6.0		Integrated Dj from 2MHz to 14MHz
Rj _{REFCLK} USB 3.1 - 10.0Gbps	Reference clock Random Jitter	2.3	ps, rms	Integrated Rj from 12kHz to 20MHz
		1.7		Integrated Rj from 2MHz to 20Mhz
Dj _{REFCLK} USB 3.1 - 10.0Gbps	Reference clock Deterministic Jitter (Dj)	7.4	ps, pp	Integrated Dj from 1MHz to 100MHz
		3.7		Integrated Dj from 3MHz to 20MHz

5.6.25.2 Reference Clock: Electrical Characteristics

Table 5-101 Reference Clock Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Conditions
FREF_OFFSET	Reference clock frequency offset	-300	300	ppm	
DCREF_CLK	Duty cycle	40	60	%	
	Maximum input voltage level		VP	V	Applies to ref_alt_clk_p only (CMOS level)
	Minimum input voltage level	0		V	
	Rise/Fall times (20%-80%)		100	ps	
	R load	200	300	Ohm	
	C load	20	40	fF	
	Maximum input voltage level		0.88	V	Applies to ref_pad_clk_p / ref_pad_clk_m only
	Minimum input voltage level	-0.15		V	
	Absolute maximum voltage at pin including overshoot		1.15 ^b	V	
	Absolute minimum voltage at pin including overshoot	-0.3 ^b		V	
	Single-ended input swing	0.3	0.88	V _{pp}	
	Differential input swing	0.3	1.76	V _{ppdif}	
	Differential ^c input edge rate	0.6 ^d	4.0 ^d	V/ns	
SWREF_CLK	Differential ^c input edge rate	0.6 ^d	4.0 ^d	V/ns	
REF_CLK_SKEW	Reference clock skew (±)		200	ps	

b. Allowed overshoot time: 10% of period.

c. In case ref_pad_clk is used in single mode, input edge rate is required to be fulfilled on the single ended ref_pad_clk_p/m signal.

d. To be verified from -150mV to +150mV (centered on the differential zero crossing) on the differential waveform. During this 300mV region, the signal must be monotonic through the both rise and fall times.

- ref_pad_clk_p/ref_pad_clk_m inputs can be low-voltage swing, or CMOS level
- ref_alt_clk inputs are CMOS-level only

Note: Only one of the on-chip and off-chip clock inputs may be used at a time.

■ If an on-chip clock source is used, then you must either tie the off-chip source to ground or leave it floating

■ If an off-chip clock source is used, then you must tie the on-chip source to ground

5.6.25.3 Reference Clock Implementation: External Off-Chip Clock Sources

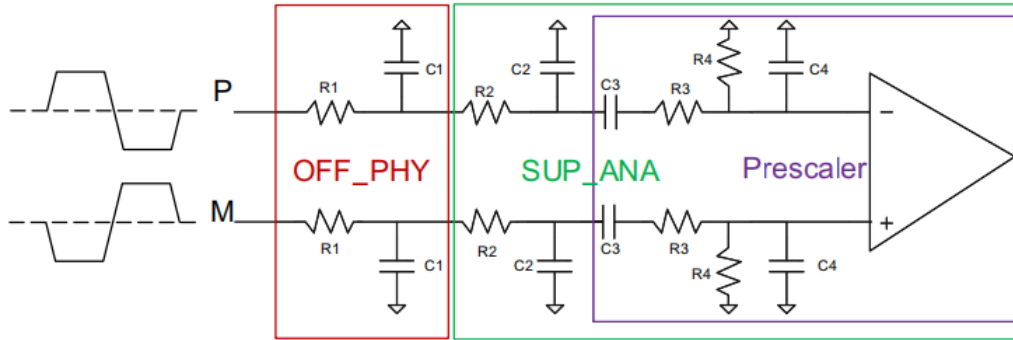


Figure 5-50 Model of Reference Clock Pad Input Load

Table 5-102 Model of Reference Clock Pad Input Load

Location	Parasitic Device	Value	Units
OFF PHY	R1	≤ 50	Ohm
OFF PHY	C1	≤ 1	pF

5.6.25.4 Reference Resistor Requirements

Table 5-103 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 Ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2mA	None
External Reference Resistor parasitic capacitance C_{load}	$<10\text{pF}$	It is important to meet the total capacitance at low frequency (in other words, the total capacitance of pad, package trace, and PCB). The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA can support up to 8.5 pF (10 pF- 1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 4 pF (10 pF -4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 Ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals, for example clocks, should be spaced away from resref or shielded with ground.

5.6.25.5 External Components

The PHY requires the following external components:

- AC coupling capacitors as specified by the standard.
- Supply bypass capacitors selected to minimize supply noise.
- Precision 200 Ohm ($\pm 1\%$) resistor used for biasing and termination impedance calibration.

Table 5-104 AC Coupling Capacitors

Protocol	Min	Typ	Max	Units
USB	75	-	265	nF

Table 5-105 Example Supply Bypass Capacitors

Component	Value ^a
Power Supply Bypass Capacitors	0.01 uF, 0.1 uF, 4.7 uF, 10.0 uF

a. Smaller-value capacitors must be placed between the ferrite bead and the package.

Table 5-106 External Component Requirements

Component	Value
resref	200 Ohm (+/- 1%)

5.6.26 USB2 PHY parameters

The USB 2.0 femtoPHY supports the USB 2.0 480-Mbps protocol and data rate, and is backward compatible with the USB 1.1 1.5-Mbps and 12-Mbps protocol and data rates.

The USB 2.0 femtoPHY complies with the following standards:

- Universal Serial Bus Specification, Revision 2.0, with a High-Speed (HS) Transmit to Receive InterPacket Delay requirement of 38 bit-times provided by the UTMI specification version 1.05.
- Battery Charging Specification, Revision 1.2, USB Implementers Forum, Inc. ACA (Accessory Charger Adaptor) functions are not supported.
- On-The-Go and Embedded Host Supplement to the USB Revision 2.0 Specification, Revision 2.0, USB Implementers Forum, Inc. ADP (Attach Detach Protocol) is not supported.
- UTMI+ Specification, Revision 1.0 (Level 3)

The USB 2.0 femtoPHY supports the following reference clock source:

On-chip clock connected to the CLKCORE pin:

The clock must have a fundamental frequency of 19.2/20/24/50/100 MHz, with a frequency tolerance of ± 400 ppm, peak jitter of ± 100 ps, duty cycle between 40/60 and 60/40 percent, and signal swing equal to the digital power supply voltage. The PHY supports an optional TESTBURNIN mode, and for this mode only, a 50-MHz input clock is required.

Table 5-107 Top-Level Pin Settings for Supported Reference Clocks (in functional mode)

Reference Clock	REFCLKSEL[1:0]	Reference Clock Frequency	FSEL[2:0]	PLLBTUNE
On-chip clock connected to CLKCORE	10	19.2 MHz	000	1

		20 MHz	001	1
		24 MHz	010	1
		100 MHz	111	1

Table 5-108 Top-Level Pin Settings for Supported Reference Clocks in TESTBURNIN Mode

Reference Clock	REFCLKSEL[1:0]	Reference Clock Frequency	FSEL[2:0]	PLLBTUNE
On-chip clock connected to CLKCORE	10	50 MHz ^a	111	0

a. If TESTBURNIN mode is used, a 50-MHz clock must be provided before setting TESTBURNIN to 1'b1. The 50-MHz clock is only valid for TESTBURNIN mode and not for functional mode.

5.6.27 PCIE PHY parameters

5.6.27.1 Reference Clock: Jitter Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

The table below lists reference clock jitter requirements for all supported PCIe clock architectures.

Table 5-109 Reference Clock Jitter Requirements for PCIe 1.1,2.1,3.1 and 4.0.

Application	Reference Clock Jitter Requirements
PCIe 1.1 – 2.5 Gbps	Refer to the PCI Express 3.0 Card Electromechanical specification.
PCIe 2.1 – 5.0 Gbps	Refer to the PCI Express 3.1 base specification.
PCIe 3.1 – 8.0 Gbps	Refer to the PCI Express 3.1 base specification.
PCIe 4.0 – 16.0 Gbps	Refer to the PCI Express 4.0 base specification.

Table 5-110 Reference Clock Amplitude Requirements

Symbol	Parameter	Min	Max	Units	Conditions
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDSEREF_CLK_DC	Input Voltage Level for ref_pad_clk_{p,m}	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_DC.
VDSEREF_CLK_AC*	Absolute voltage level at the pin including the overshoot for ref_pad_clk_{p,m}	-0.3	1.15	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_AC.
VDREF_CLK_ALT	Differential input swing for ref_alt_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_ALT.

VDSEREF_CLK_ALT	Input Voltage Level for ref_alt_clk_{p,m}	0	VP	V	Differential inputs must meet both VDREF_CLK_ALT and VDSEREF_CLK_ALT.
*Allowed overshoot time: 10% of the reference clock period.					

5.6.27.2 Reference Clock Input Requirements

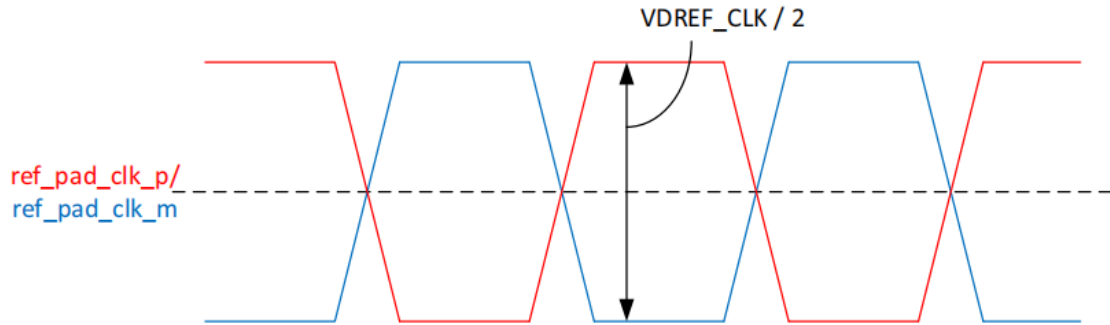
Differential reference clock requirements for protocols other than PCIe are provided in Table 4-53 for data rates less than and greater than 8 Gbaud.

Table 5-111 Reference Clock Electrical Characteristics

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	312.5	Mhz	Not continuous; per protocol as noted in the databook
FREF_OFFSET	Reference clock frequency offset	-100	100	ppm	All protocols except for SATA
RMSJREF_CLK ≤ 8 Gbps	Reference clock random Jitter		1.6	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			1.2		Integrated RJ from 2 MHz to 20 MHz
DJREF_CLK ≤ 8 Gbps	Reference clock deterministic jitter		2.8 5.6	ps,pp	0.75–10MHz (offset)a 0.2–50MHz (offset)a
RMSJREF_CLK > 8 Gbps	Reference clock random jitter		1.0	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			0.8		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK > 8 Gbps	Reference clock deterministic jitter		1.7 3.4	ps,pp	0.75–10MHz (offset)a 0.2–50MHz (offset)a
DCREF_CLK	Duty cycle	40	60	%	
VCMREF_CLK	Common mode input level	0	VP	V	Differential inputs
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDSEREF_CLK_DC	Input Voltage Level for ref_pad_clk_{p,m}	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_DC.
VDSEREF_CLK_Acb	Absolute voltage level at the pin including the overshoot for ref_pad_clk_{p,m}	-0.3	1.15	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_AC.
VDREF_CLK_ALT	Differential input swing for ref_alt_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_ALT.
VDSEREF_CLK_ALT	Input Voltage Level for ref_alt_clk_{p,m}	0	VP	V	Differential inputs must meet both VDREF_CLK_ALT and VDSEREF_CLK_ALT.
SKEWDIF_CLK	Differential clock input skew		100	ps	
SWREF_CLK	Differential clock input slew rate	0.6		V/ns	20–80%

- DJ noise limits are for noise within an offset from the divided reference clock and harmonics of the divided reference clock.
- Allowed overshoot time: 10% of the FREF_CLK period

In below figure, VDREF_CLK is the peak-peak differential swing for ref_pad_clk_{p,m}.



5.6.27.3 Model of Reference Clock Pad Input Load

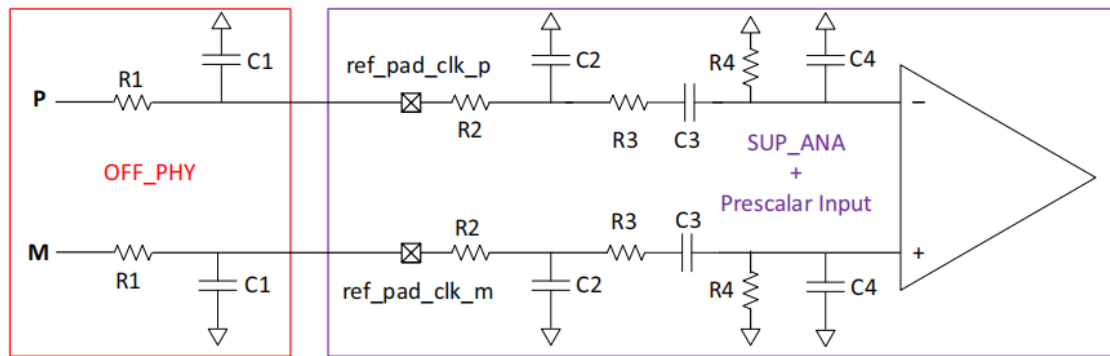


Figure 5-51 Model of Reference Clock Pad Input Load

Table 5-112 Reference Clock External Pad Input Parasitics

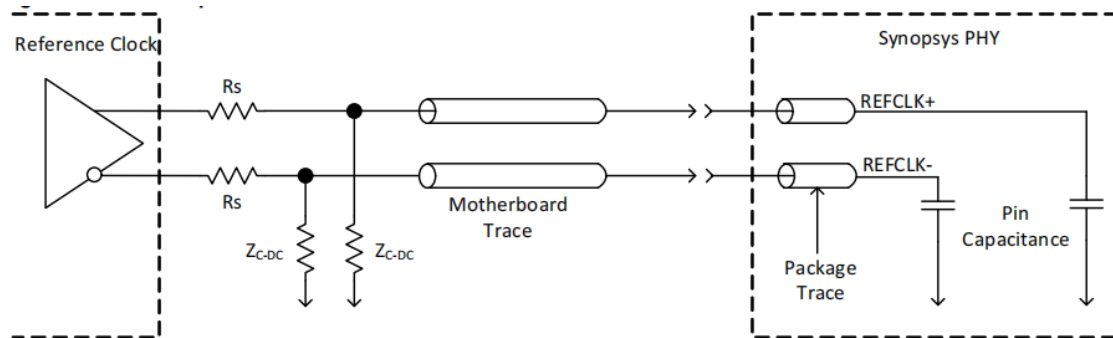
Location	Parasitics Device	Value	Unit	Notes
OFF PHY	R1	≤ 50	Ohm	
OFF PHY	C1	≤ 3	pF	
Within PHY: Bump to prescaler input	R2	≤ 50	Ohm	
Within PHY: Bump to prescaler input	C2	≤ 1	pF	
Within prescaler: 2nd ESD Protection resistor	R3	1.5	kOhm	Required for 6A peak discharge current ESD rating
Within prescaler: AC capacitor	C3	920	fF	
Within prescaler: DC Bias resistor	R4	468	kOhm	
Within prescaler: Input capacitor of prescaler input RX	C4	≤ 50	fF	

Note: To improve Common-Mode Rejection Ratio (CMRR), differential traces must be as well matched as possible

To enable connecting reference clocks to multiple PHYs at the same time, the reference clock inputs are not terminated. If termination is used, place it on your board as close to the chip as possible. For PCIe, see the PCIe CEM specification for details on how to implement the on-board termination.

If you use off-chip reference clock (that is, `ref_pad_clk_p` or `ref_pad_clk_m`), you must terminate the off-chip clock source to suppress reflection. Also, make sure that the clock swing at the Synopsys PHY input is within reference clock specification. An example of this implementation is shown in Figure 4-30.

Figure 5-30 Off-chip Clock Source Termination



5.6.27.4 Reference Resistor Requirements

Reference resistor requirements are outlined in Table 5-113.

Table 5-113 Reference Resistor Requirements

Item	Value	Design Guidelines
External reference resistor resistance	200-Ohm precision resistor ($\pm 1\%$)	None
External reference resistor maximum current	2 mA	None
External reference resistor parasitic capacitance C_{load}	<20 pF	■ The maximum parasitic capacitance includes external loading of package trace and PCB, as well as, PHY loading. ■ Each PHY connected to the resref pin adds an additional 1.5 pF loading. ■ For example, a single PMA can support up to 18.5 pF ($20 \text{ pF} - 1 \times 1.5 \text{ pF}$) of external load. Sharing a single reference resistor between 4 PMAs can support up to 14 pF ($20 \text{ pF} - 4 \times 1.5 \text{ pF}$) of external load.
External reference resistor parasitic resistance through package and board R_{route}	1% of 200 Ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board external resistor. The ground used on the board to terminate the external resistor

		must be as close as possible to the PMA ground voltage. Noise adjacent signals (for example, clocks) must be spaced away from resref or must be shielded with ground, if possible.
--	--	---

5.6.27.5 External Components

The PHY requires the following external components:

- AC coupling capacitors as specified by the standard, for reference, see Table 4-90
- Supply bypass capacitors selected to minimize supply noise.
- Precision 200-Ohm ($\pm 1\%$) resistor used for biasing and termination impedance calibration; the precision resistor requirement is shown in Table 4-58.

Table 5-114 AC Coupling Capacitors - - Target Protocols

Protocol	AC-Coupling Capacitor	Units
PCIe 1.1 and PCIe 2.1	200	nF
PCIe 3.1 and PCIe 4.0	176-265	nF

Table 5-115 External Component Requirements

Component	Value
resref	200 ohm (+/- 1%)

5.6.28 USB3 PHY DC electrical characteristics

5.6.28.1 Power Supply Limits

Table 5-116 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Nominal	Max	Units
vph	High Voltage IO supply	io_nominal -5%	1.2, 1.8	io_nominal +10%	V
vp/vptxX/vpdig	Analog Core supply	core_nominal -7%	0.75	core_nominal +10%	V

5.6.28.2 vp/vptxX/vpdig Supply Requirements

The total amount of tolerated noise in this band depends on the applied DC level, being maximized when DC level is centered within vp/vptxX/vpdig DC range limits.

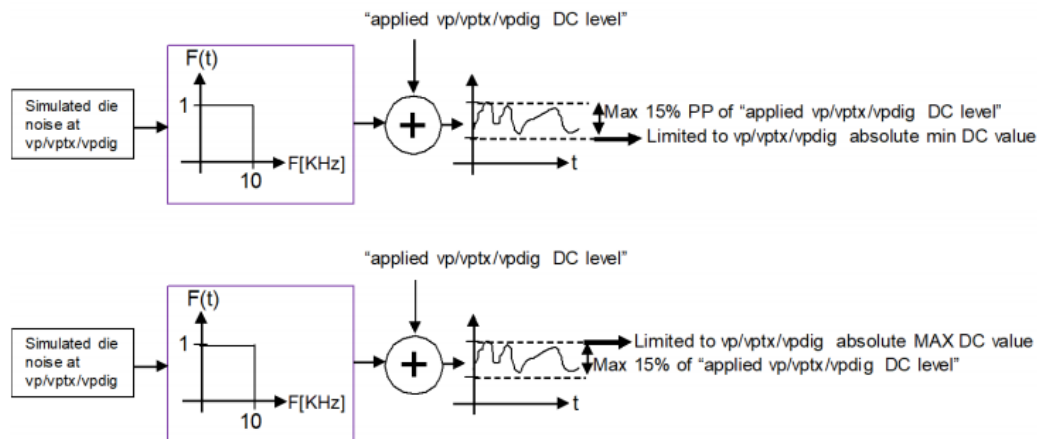


Figure 5-52 Specified DC Limits

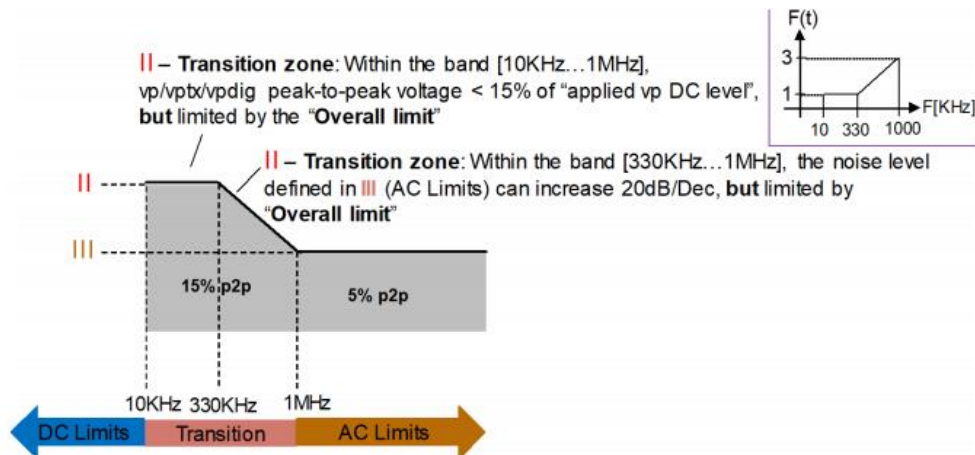
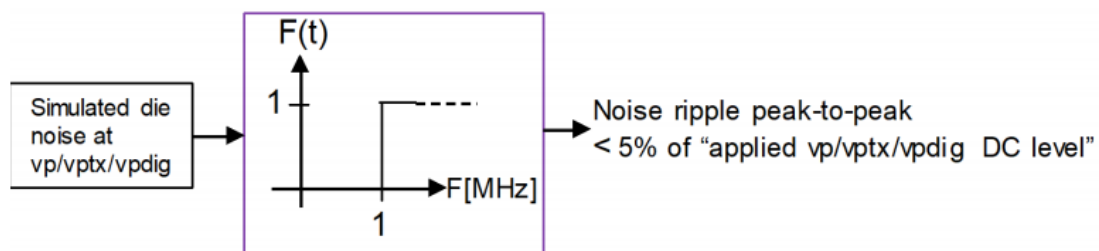
Figure 5-53 $v_p/v_{ptxX}/v_{pdig}$ AC Limits at the Die

Figure 5-54 Specified AC Limits

Overall limits:

In addition to the previously defined DC/Transition/AC limits, there is also a limit to the minimum allowed instantaneous simulated $v_p/v_{ptx}/v_{pdig}$ voltage, that must be always above “Absolute min $v_p/v_{ptx}/v_{pdig}$ DC value” – 2.5%.

5.6.28.3 vph Supply Requirements

The total amount of tolerated noise in this band depends on the applied DC level, being maximized when DC level is centered within vph DC range limits.

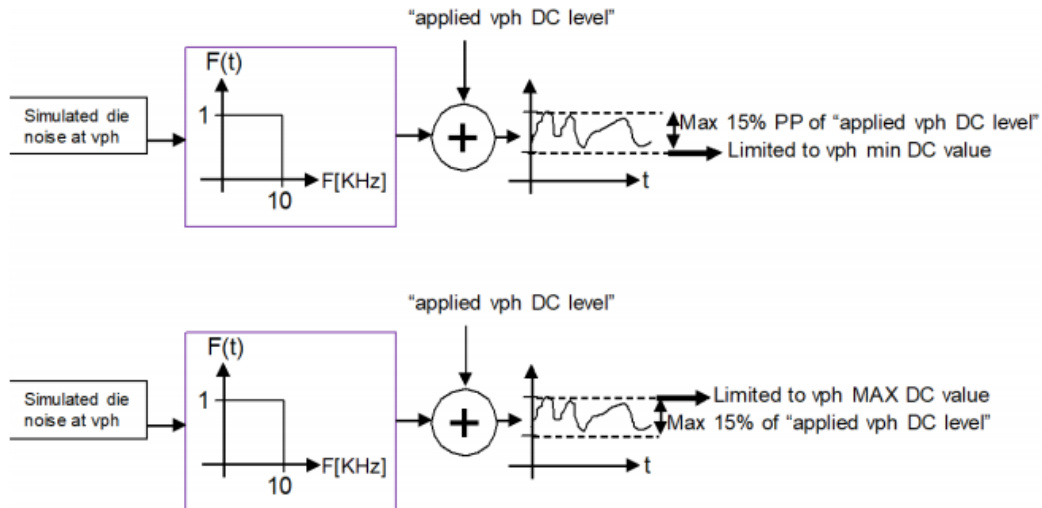


Figure 5-55 Specified DC Limits

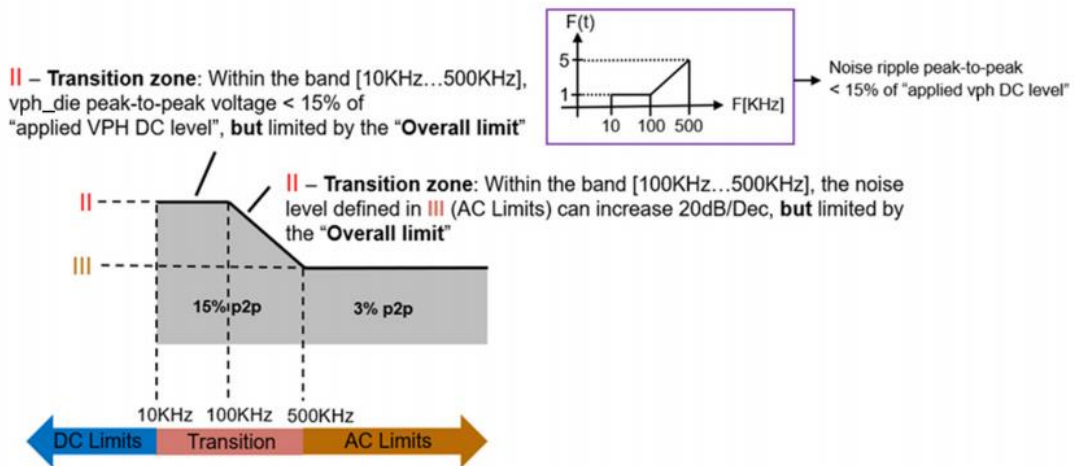


Figure 5-56 vph AC Limits at the Die

The maximum peak-to-peak voltage is relative to the applied vph DC level.



Figure 5-57 Specified AC Limits

Overall limits:

- In addition to the previously defined DC/Transition/AC limits, there is also a limit to the minimum allowed instantaneous simulated vph voltage, that must be always above “Absolute min vph DC value” – 1.5%
- vph slew rate < “applied vph DC level”/ 10us. As an example, for a vph of 1.2V, this corresponds to a required maximum slew rate < 120mV/us.

5.6.29 USB2 PHY DC electrical characteristics

Table 5-117 Maximum Power Supply Slew Rates

VDD33<#>/VDDH<#>	DVDD	Unit
0.1	0.1	V/μs

Table 5-118 Maximum Pad/Package/Board Requirements

Signal	Signal Description	A Series Resistance (Ω)	Series Inductance (nH)	Capacitance (pF)	5-V Tolerance Needed	Worst-Case Interconnect Metal DC Current Handling Requirements (mA)
ID0	USB mini-receptacle identifier	1	4	5	N	1
VBUS0	USB mini-receptacle VBUS	1	4	5	N	8
VDD33<#>	Macro supply	1	4	-	N	35
VSSAb	Macro supply	1	4	-	N	50
VDDH<#>	Macro supply	1	4	-	N	35
DP<#>	Analog signal	1	4	5	Y	30
DM<#>	Analog signal	1	4	5	Y	30
TXRTUNE	Analog signal	1	4	5	N	15
DVDD	Digital-core supply	1	4	-	N	15

5.6.30 PCIe PHY DC electrical characteristics

The PCIe PHY complies with the required features specified in: Peripheral Component Interconnect Express 4.0 specification, Revision 1.0.

- PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG
- PCI Express Card Electromechanical Specification, Revision 3.0, July 2013, PCI-SIG
- PCI Express Mini Card Electromechanical Specification, Revision 2.0, April 2012, PCI-SIG
- Energy Efficient Ethernet (EEE) Physical Layer electrical specifications
- IEEE Standard 1149.1-2001, IEEE
- IEEE Standard 1149.6-2003, IEEE

The PHY supports the power supply voltages in Table 4-95.

The values are supported across a temperature range of -40°C to 125°C.

Table 5-119 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Typical	Max	Units
vph	High Voltage IO supply	1.2-5%	1.2/1.8V	1.8V+10%	V
vp/vptx X/vpdig	Analog supply	0.75+10%	0.75V	0.75V-5%	V

The following sections provide simulated power consumption values for the PHY.

- The typical (Typ) values are for TT process corner, 25°C junction temperature and typical voltage supplies for specified data rate.
- The Worst-Case (WC) values are for FF process corner, 125°C junction temperature and maximum voltage supplies for specified data rate.

5.6.31 MIPI RX CDPHY DC electrical characteristics

The MIPI DPHY is conformant with the MIPI D-PHY interface specification, revision v2.1

The MIPI CPHY is conformant with the MIPI C-PHY interface specification, revision v1.2

Table 5-120 MIPI RX DPHY and CPHY Input DC Specifications

Apply to CLKP/N and DATAP/N Inputs.

Symbol	Parameter	Minimum	Maximum	Unit	Notes
VPIN	Pin signal voltage range	-50	1350	mV	
ILEAK	Pin leakage current	-100	100	μA	When the Lane Module is in LP receive mode and the pad voltage is in the signal voltage range VPIN for LP mode or in the signal range VPIN_LVL for LVLV operation, ILEAK should be well within the limits in LVLV operation
VGNDSH	Ground shift	-50	50	mV	Ground shift operating in Half Swing Mode
		-5	5	mv	
VPIN(absmax)	Maximum transient output voltage level	-0.15	1.45	V	
VPIN(absmax)	Maximum transient time above VOH(absmax)		20	ns	The voltage overshoot and undershoot beyond the VPIN is only allowed during a single 20 ns window after any LP-0 to LP-1 transition or vice versa. For all other situations the voltage overshoot and undershoot must stay within the VPIN range

Table 5-121 MIPI RX DPHY HS Line Receiver DC Specifications

Symbol	Parameter	Mini	Typical	Max	Unit	Notes
VCMRXDC	Input common mode voltage	70		330	mV	Excluding possible additional RF interference of 100 mV peak sine wave beyond 450 MHz. This table value includes a ground difference of 50 mV between the transmitter

						and the receiver, the static common-mode level tolerance and variations below 450 MHz
ZID	Differential input impedance	80	100	125	ohm	
ZID_OPEN	ZID_OPEN	10k			ohm	This mode can only be used when a transmitter is in Half Swing mode. ZID_OPEN is defined for a differential voltage with maximum amplitude of VOD_Halfswing and within the common voltage range of VCMTX_Halfswing

Table 5-122 MIPI RX CPHY HS Line Receiver DC Specifications

Symbol	Parameter	Mini	Typical	Max	Unit	Notes
VCPRX	Common-Point voltage HS receive mode	95		390	mV	Excluding possible additional RF interference of 100 mV peak sine wave beyond 450 MHz. This table value includes a ground difference of 50 mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450 MHz
		95		325	mV	HS unterminated mode
ZID_AB ZID_BC ZID_CA	Differential input impedance	80	100	120	ohm	
		10k			ohm	HS unterminated mode
Delta ZID	Differential input impedance mismatch			10	%	

Table 5-123 MIPI RX DPHY and CPHY LP Line Receiver DC Specifications

Symbol	Parameter	Minimum	Typical	Maximum	Unit
VIL	Input low voltage			550	mV
VIH	Input high voltage	740			mV
VIL-ULPS	Logic 0 input voltage, ULPS			300	mV
VHYST	Input hysteresis	25			mV

Table 5-124 MIPI RX DPHY and CPHY Contention Line Receiver DC Specifications

Symbol	Parameter	Minimum	Typical	Maximum	Unit
VILF	Input low fault voltage			200	mV
VIHF	Input high fault voltage	450			mV

5.6.32 MIPI MPHY electrical characteristics

- IEEE Standard 1149.1-2001, IEEE
- IEEE Standard 1149.6-2003, IEEE
- MIPI Alliance Specification for M-PHY, Version 4.1 Revision 03, 01 December 2016
- MIPI M-PHY Physical Layer Conformance Test Suite, Version 1.00 Revision Candidate 2, 21 March 2013
- MIPI Alliance Specification for Unified Protocol (UniProSM), Version 1.6 Revision 9, 6 August 2013
- JEDEC STANDARD Universal Flash Storage (UFS), JESD220-1, September 2013

The PHY supports the power supply voltages in Table 4-65.

The values are supported across a temperature range of -40°C to 125°C.

Table 5-65 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Typical	Max	Units
vph	High Voltage IO supply	io_nominal -7%	1.2, 1.8	io_nominal +10%	V
vp/vptxX/vpdig	Analog Core supply	core_nominal -7%	0.75	core_nominal +10%	V

5.6.33 DDR

5.6.33.1 Operating Conditions

All voltages are referenced to VSS.

5.6.33.2 Recommended Operating Conditions

The following table shows the DDR operating conditions, including supply voltages and temperature.

Table 5-125 Recommended Operating Conditions¹

Symbol	Description	Minimum	Typical	Maximum	Units	Note
VDD	Core Voltage - Supports data rates of 6400 Mbps and below	VDD(Typ) -7%	0.75	VDD(Typ) +10%	V	1, 2, 3, 4
VAC_VDD (pk-pk)	Core power/ground supply rail peak-to-peak AC noise			4	%VDD (Type)	1,5
VDDQ_DRAM	SDRAM VDDQ supply: LPDDR4 mode	1.06	1.1	1.17	V	1, 6
VDDQ	PHY VDDQ supply: LPDDR4 mode, DC average	VDD(Min)	VDD(Typ)	VDD(Max)	V	1, 2, 7, 8
VDDQ	PHY VDDQ supply: LPDDR4X mode, DC average	0.57	0.6	0.65	V	1, 2, 7, 8
VDDQ	PHY VDDQ supply: LPDDR5 mode, DC average	0.47	0.5	0.57	V	1, 2, 7,8, 9

VDDQ	PHY VDDQ supply: LPDDR5 mode, DC average	0.27	0.3	0.37	V	1, 2, 7, 8, 10
VAC_VDDQ (pk-pk)	I/O power/ground supply rail peak-to-peak AC noise: LPDDR4 mode			10	% VDDQ(Typ)	1, 5
VAC_VDDQ (pk-pk)	I/O power/ground supply rail peak-to-peak AC noise: LPDDR4X mode			10	% VDDQ(Typ)	1, 5
VAC_VDDQ (pk-pk)	I/O power/ground supply rail peak-to-peak AC noise: LPDDR5 mode			10	% VDDQ(Typ)	1, 5
VRANGE_VDDQ	PHY VDDQ supply LPDDR4 mode, allowed range	VDDQ (LPDDR4 Typ) -10%		VDDQ (LPDDR4 Typ) +10%	V	1, 2, 11
VRANGE_VDDQ	PHY VDDQ supply LPDDR4X mode, allowed range	VDDQ (LPDDR4X Typ) -10%		VDDQ (LPDDR4X Typ) +10%	V	1, 2, 11
VRANGE_VDDQ	PHY VDDQ supply LPDDR5 mode, allowed range	VDDQ (LPDDR5 Typ) -10%		0.57 or 0.37	V	1, 2, 9, 10, 11
VDD2H	VDD2H supply: LPDDR4 mode	1.06	1.1	1.17	V	1, 2, 7, 12
VDD2H	V supply: LPDDR4X mode	1.06	1.1	1.17	V	1, 2, 7
VDD2H	VDD2H supply: LPDDR5 mode	1.01	1.05	1.12	V	1, 2, 7
VAC_VDD2H(pk-pk)	VDD2H power/ground supply rail peak-to-peak AC noise: LPDDR4 mode			10	% VDD2H(Typ)	1, 5
VAC_VDD2H(pk-pk)	VDD2H power/ground supply rail peak-to-peak AC noise: LPDDR4X mode			10	% VDD2H(Typ)	1, 5
VAC_VDD2H(pk-pk)	VDD2H power/ground supply rail peak-to-peak AC noise: LPDDR5 mode			10	% VDD2H(Typ)	1, 5
VRANGE_VDD2H	PHY VDD2H supply LPDDR4 mode, allowed range	VDD2H (LPDDR4 Typ) -10%		VDD2H (LPDDR4 Typ) +10%	V	1, 2, 11
VRANGE_VDD2H	PHY VDD2H supply LPDDR4X mode, allowed range	VDD2H (LPDDR4X Typ) -10%		VDD2H (LPDDR4X Typ) +10%	V	1, 2, 11
VRANGE_VDD2H	PHY VDD2H supply LPDDR5 mode, allowed range	VDD2H (LPDDR5 Typ) -10%		VDD2H (LPDDR5 Typ) +10%	V	1, 2, 11
VAA_VDD2H	PLL supply voltage VAA_VDD2H connected to independent 1.8 V supply	VAA_VDD2H(Typ) -7%	1.8	VAA_VDD2H(Typ) +10%	V	1, 2
VAC_VAA_VDD2H(pk-pk)	PLL power/ground supply rail peak-to-peak AC noise when connected to VAA_VDD2H = 1.8 V			5	%VAA_VDD2H(Typ)	1, 5
VAA_VDD2H(VDD2H)	PLL supply voltage VAA_VDD2H connected to VDD2H	See Minimum VDD2H (per mode above)	See Typical VDD2H (per mode above)	See Maximum VDD2H (per mode above)	V	1, 2, 13
VAC_VAA_VDD2H(VDD2H)(pk-pk)	PLL power/ground supply rail peak-to-peak AC noise when connected VDD2H			2	%VAA_VDD2H(VDD2H)	1, 5

)					D2 H(Typ))	
RVDDQ RVDD2H RVAA_VD D2H	Supply power-up ramp rate for VDDQ, VDD2H, VAA_VDD2H			5	mV/us	1, 14
RVDD	Supply power-up ramp rate for VDD			150	mV/us	1, 15
TJ	Junction Temperature	-40		125	oC	1

 Note:

- Correct operation of the DDR PHY is not guaranteed if the recommended operating conditions for VDD, VDDQ, VDD2H, and VAA_VDD2H are exceeded.
- Voltages VDD, VDDQ, VDD2H, and VAA_VDD2H are measured at the pins of the hard macros.
- Core voltage range (VDD) includes both DC variation and AC noise.
- Power, jitter, and aging analysis are performed at DC average VDD levels of +/- 5%.
- AC noise is noise of high enough frequency to erode the timing margins of source-synchronous DDR interfaces. Anything greater than 20 MHz will be considered AC noise.
- This supply is referenced to the DRAM supply for IO spec purposes. It is not a supply used in the PHY.
- The "VDDQ" and "VDD2H" voltage specifications are for DC voltage only. DC is inclusive of all noise up to 20 MHz.
- Power and jitter analysis are performed at DC average of VDDQ(min) and VDDQ(max).
- Intended for IO operation with both ODT enabled and disabled.
- Intended for IO operation with ODT disabled.
- The "VRANGE_*" specifications include both DC variation and AC noise and are the absolute voltage limit for the supply.
- Connect VDD2H supply to DRAM's VDDQ(VDDQ_DRAM) for receiver VREF tracking purposes.
- PLL power supply VAA_VDD2H is allowed to be connected to either VAA_VDD2H or VDD2H voltage levels. When VAA_VDD2H of PLL is connected to VDD2H, VDD2H needs to be filtered on the package or board in order to provide clean supply for PLL. Noise limit for VDD2H filtered is same as when it is connected to VAA_VDD2H, which is VAC_VAA_VDD2H in "Operating Conditions".
- The VDDQ, VDD2H and VAA_VDD2H power supplies are limited to a maximum supply ramp rate listed in Table 9-1 above. This limit prevents triggering of the internal clamps on those power supplies. All IP designs must make a trade-off around protecting from ESD events. A super-fast power rail ramp rate during normal operation looks significantly like an ESD discharge that has the potential to damage the chip. Our power rail requirements are linked to our trade-offs between allowing power rails to be driven by a wide variety of sources with different characteristics, versus having ESD clamps which trigger at conservative rates to ensure the chip is not damaged.
- VDD power supplies are limited to a maximum supply ramp rate of 150mV/us. This limit is required for proper operation of the Power-On-Reset custom circuit. A super-fast power rail

ramp rate during normal operation could create glitching on reset signals.

5.6.33.3 Core Voltage Tolerance

The following figures show the core voltage tolerance, allowable jitter, and relationship to PVT points for both minimum and maximum voltages.

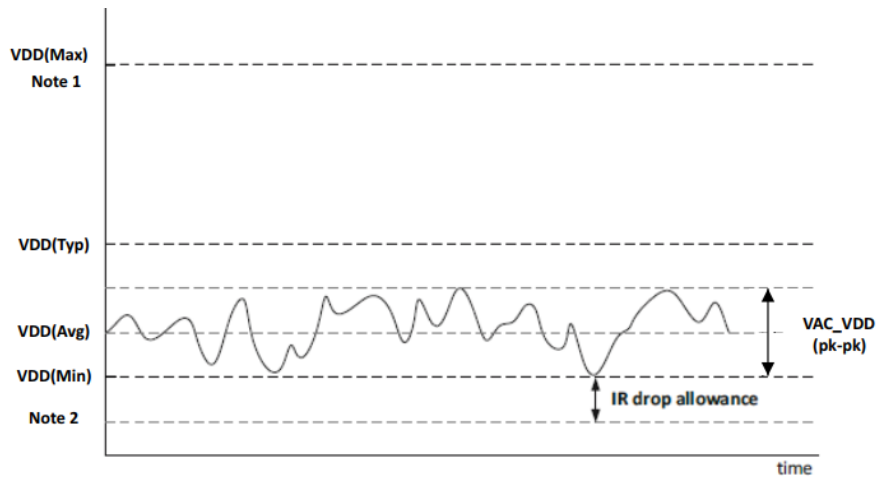


Figure 5-58 Core Voltage Tolerance for Minimum VDD Voltage

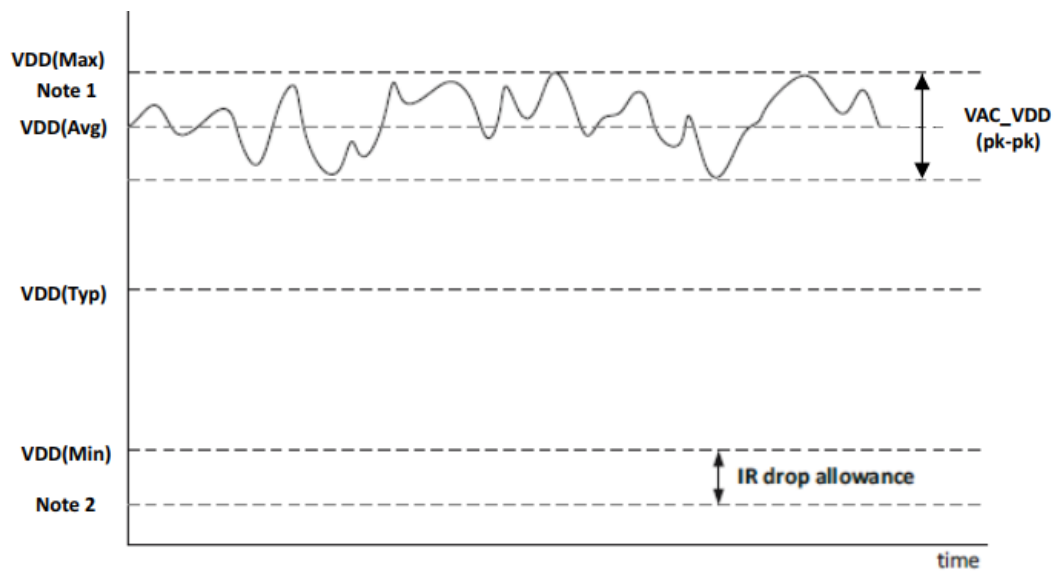


Figure 5-59 Core Voltage Tolerance for Maximum VDD Voltage

 Note:

- (1) VDDMax is the maximum voltage.
- (2) As shown in the figures, there is an IR drop allowance for the supply voltage to the individual transistors in die between VDDMin and the minimum voltage

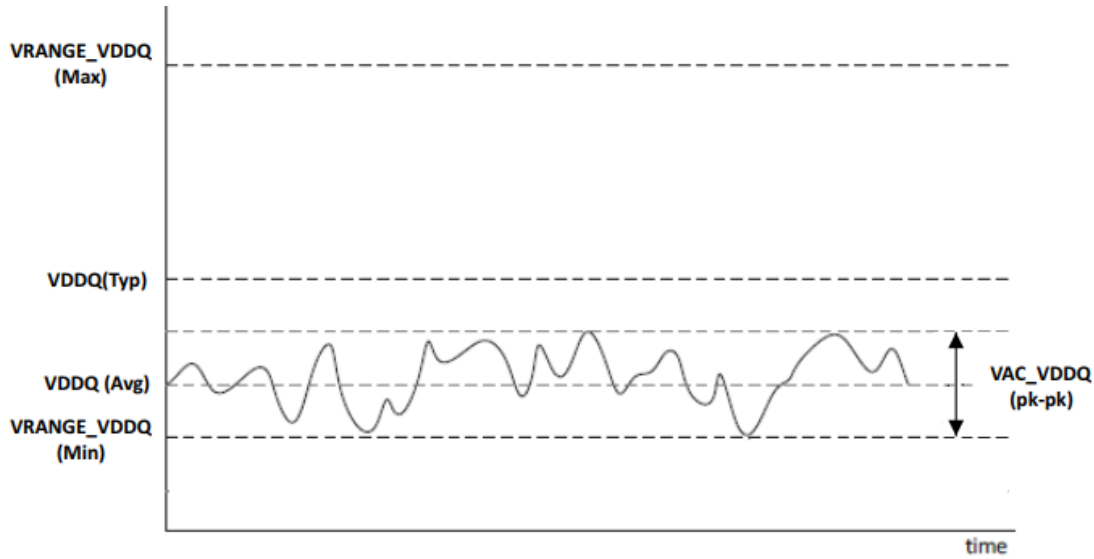


Figure 5-60 Core Voltage Tolerance for Minimum VDDQ Voltage

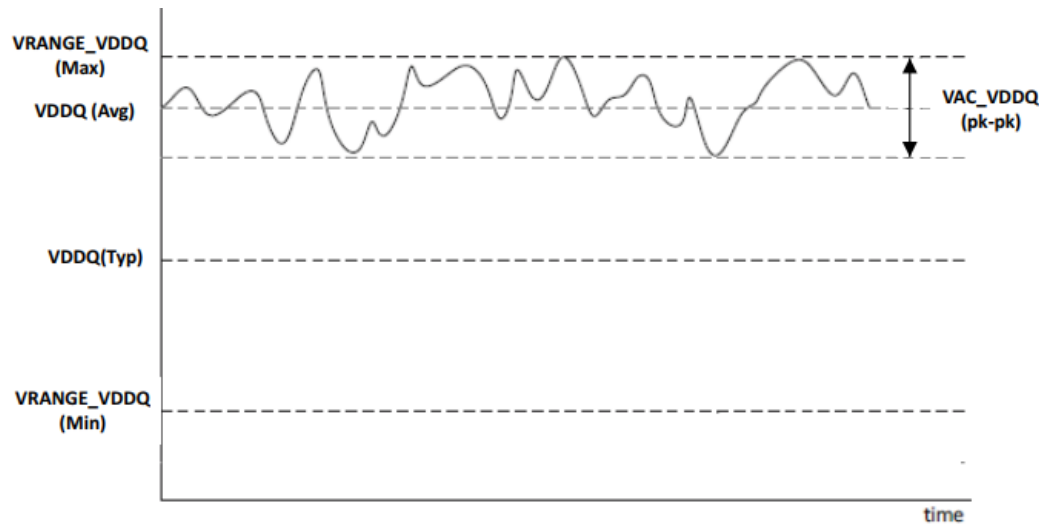


Figure 5-33 Core Voltage Tolerance for Maximum VDDQ Voltage

6 Boot configuration

6.1 Boot device interface allocation

The following table lists the interfaces that can be used by the boot process in accordance with the specific Boot mode configuration.

Table 6-1 Interface allocation during boot

Interface	IP Instance	Allocated Pads During Boot	Comment
OSPI Flash	OSPI0/OSPI1	ST_OSPI0_CS0~1, ST_OSPI0_SCLK, ST_OSPI0_IO0~7/ ST_OSPI1_CS0~1, ST_OSPI1_SCLK, ST_OSPI1_IO0~7	will boot from OSPI1 if the OSPI0 boot failed

6.2 Boot strap pin configuration

The following table lists the strap pin that can be used by the boot process to configure system mode.

Table 6-2 Boot strap pin configuration

PAD Name	Strap pin name	Input/ Output	PU/PD	Description
ST_GPIO0	bootsel0	input	pull down	boot_sel[1:0] select boot mode : 2'b00: secure boot,secure r5 boot from bootrom first; 2'b01: safe boot,safety r5 boot from OSPI first;Can be disabled by OTP; 2'b1x: reserved
ST_GPIO1	bootsel1	input	pull down	
ST_RGMII_TXD0	bootsel2	input	pull down	boot_sel[4:2] select boot storage mode : 3'b1xx: UART/CAN/USB intercept mode ; 3'b000: QSPI/OSPI 3'b001: eMMC 3'b010: UFS 3'b011: QSPI/OSPI
ST_RGMII_TXD1	bootsel3	input	pull down	
ST_RGMII_PPS00	bootsel4	input	pull down	
ST_GPIO2	st_r5_lockstep	input	pull down	boot R5 work on lock-step mode 1: enable 0: disable
ST_GPIO4	ist_post_bypass	input	pull down	System IST bypass control 1: bypass 0: no bypass
ST_GPIO5	mem_repair_bypass	input	pull down	System Memory Repair bypass control 1: bypass 0: no bypass
ST_GPIO6	st_parity_en	input	pull down	secure&safety subsystem parity check control 1: enable 0: disable
ST_GPIO7	st_ecc_en	input	pull down	secure&safety subsystem ECC check control 1: enable 0: disable
ST_RGMII_TXD2	ospi_mode0	input	pull down	ospi_mode[1:0]: 2'b00: xip default 2bit mode ; 2'b01: xip default 4bit mode ; 2'b1x: xip default 8bit mode ;
ST_RGMII_TXD3	ospi_mode1	input	pull down	

7 PCB Guideline or Requirement

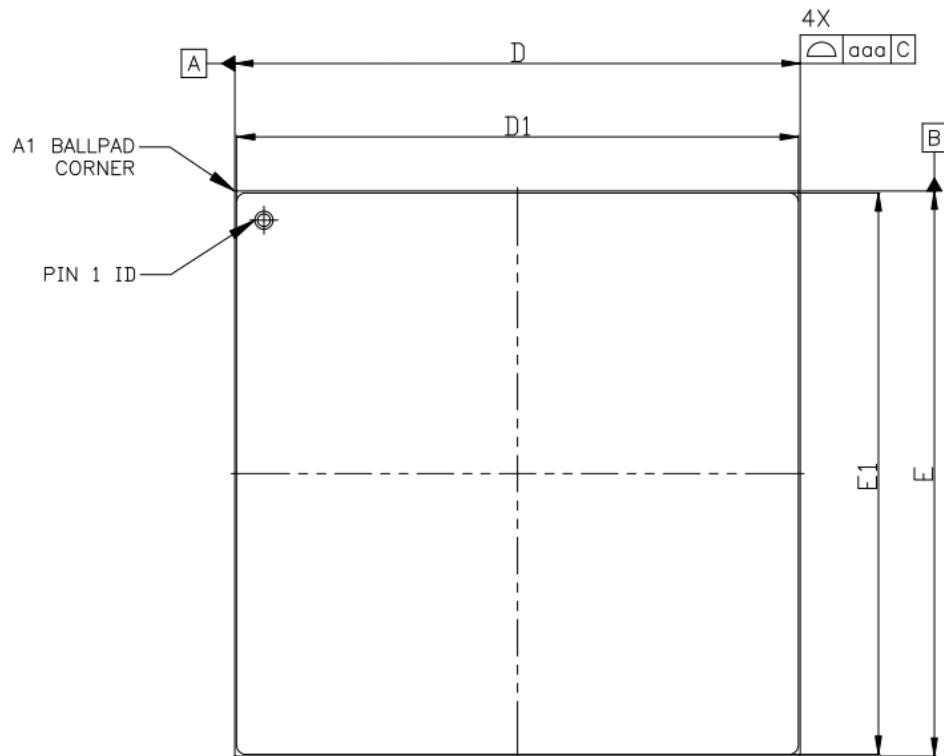
7.1 PCB Guideline

The PCB Guideline or Requirement please refer to *BST C1200 Hardware Design Guidelines.pdf*.

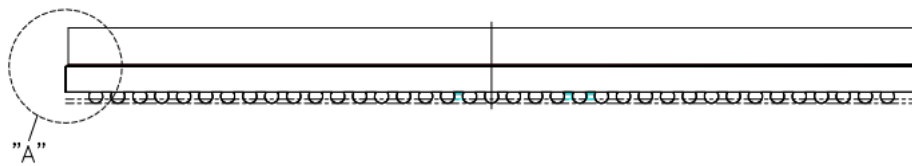
8 Package information and contact assignments

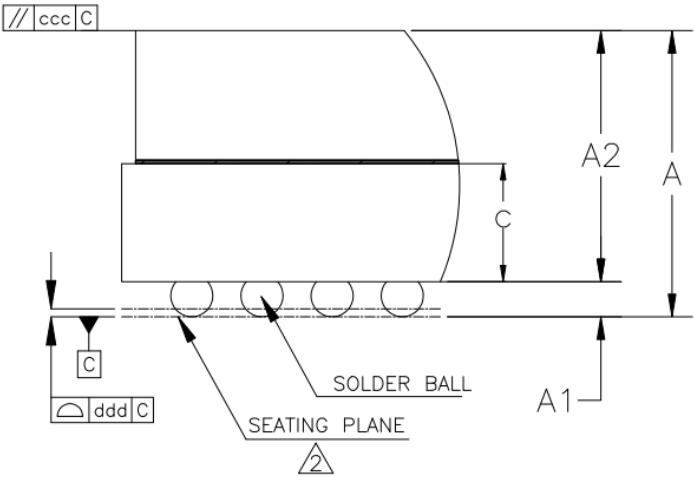
8.1 Package outline Dimension

Figure 8-1 shows the outline dimension of the 31×31 mm, 0.8 mm pitch BGA package.

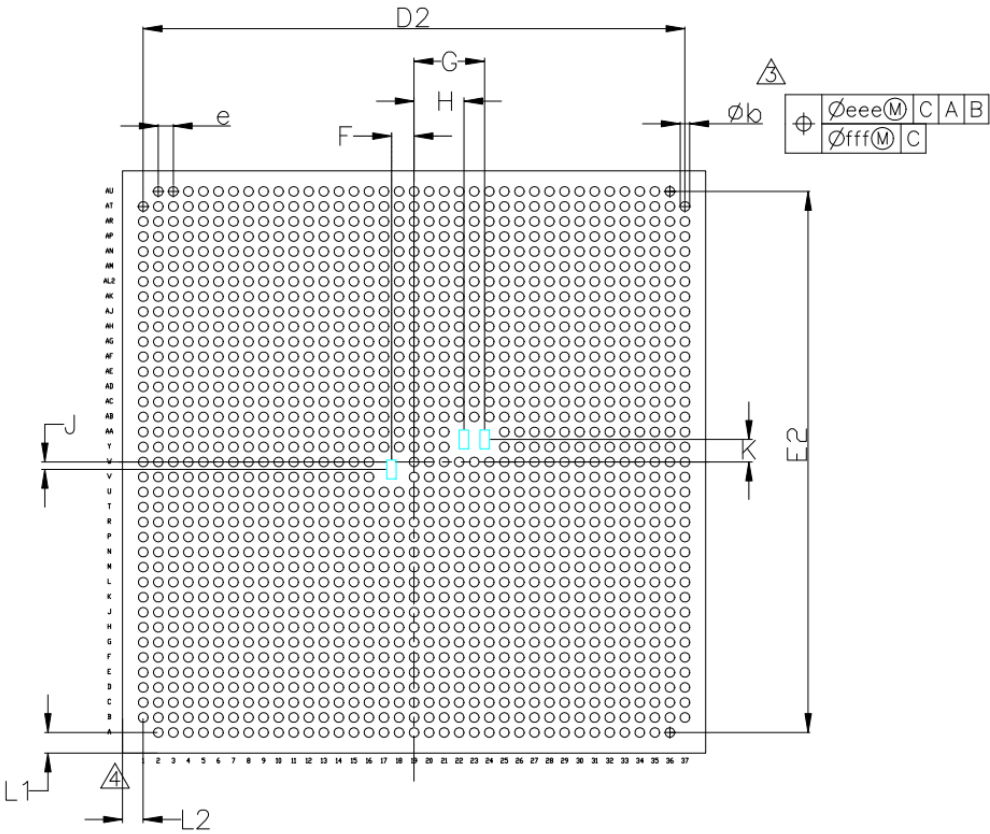


TOP VIEW





DETAIL : "A"



BOTTOM VIEW

1355 SOLDER BALLS

ITEM	DIMENSION IN MM		
	MIN	NOM	MAX
D/E	30.900	31.000	31.100
D1/E1	---	30.800	---
D2/E2	---	28.800	---
F	---	1.200	---
G	---	3.750	---
H	---	2.650	---
J	---	0.400	---
K	---	1.200	
A	2.589	2.742	2.895
c	---	0.942	---
A1	0.350	0.400	0.450
A2	---	2.342	---
L1/L2	---	1.100	---
b	0.460	0.510	0.560
e	---	0.800	---
aaa		0.200	
ccc		0.350	
ddd		0.152	
eee		0.200	
fff		0.080	

NOTE :

1. CONTROLLING DIMENSION : MILLIMETER.

2. PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.

3. DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO PRIMARY DATUM C.

4. THE PATTERN OF PIN 1 FIDUCIAL IS FOR REFERENCE ONLY.

5. BALL PLACEMENT USE 0.50 mm SOLDER BALL.

BGA PAD SOLDER MASK OPENING = 0.40 mm.

Figure 8-1 Package outline dimension

8.2 Supplies contact assignments and functional contact assignments

Table 8-1 shows supplies contact assignments.

Table 8-1 Supplies contact assignments

Power Rail Name (BALL NAME)	Description
AVDD_ADC_1V8	ADC Analog 1.8V supply

AVDD_CSI_0V75	MIPI_CSI_VP Analog 0.75V supply
AVDD_CSI_1V8	MIPI_CSI Analog 1.8V supply
AVDD_CSIVDD_0V75	MIPI_CSIVDD Analog 0.75V supply
AVDD_DDR0_1V8	DDR0 Analog 1.8V supply
AVDD_DDR0_VDD2H	DDR0 PLL supply
AVDD_DDR0_VDDQ	DDR0 IO supply
AVDD_DDR1_1V8	DDR1 Analog 1.8V supply
AVDD_DDR1_VDD2H	DDR1 PLL supply
AVDD_DDR1_VDDQ	DDR1 IO supply
AVDD_DP_0V75	DP Analog 0.75V supply
AVDD_DP_1V8	DP Analog 1.8V supply
AVDD_DSI_1V8	MIPI_DSI Analog 1.8V supply
AVDD_DSIVDD_0V75	MIPI_DSIVDD Analog 0.75V supply
AVDD_DSIVP_0V75	MIPI_DSIVP Analog 0.75V supply
AVDD_DSIVPL_0V75	MIPI_DSIVPL Analog 0.75V supply
AVDD_EFUSE_1V8	EFUSE Analog 1.8V supply
AVDD_ETH0_1V8	ETH PHY0 Analog 1.8V supply
AVDD_ETH1_1V8	ETH PHY1 Analog 1.8V supply
AVDD_ETH_0V75	ETH Analog 0.75V Supply
AVDD_OSC_1V8	OSC Analog 1.8V supply
AVDD_OTP0_1V8	OTP Analog 1.8V supply
AVDD_PCIE0_1V8	PCIE PHY0 Analog 1.8V supply
AVDD_PCIE1_1V8	PCIE PHY1 Analog 1.8V supply
AVDD_PCIE_0V75	PCIE Analog 0.75V supply
AVDD_PLL0_0V75	PLL0 Analog 0.75V supply
AVDD_PLL0_1V8	PLL0 Analog 1.8V supply
AVDD_PLL1_0V75	PLL1 Analog 0.75V supply
AVDD_PLL1_1V8	PLL1 Analog 1.8V supply
AVDD_PLL2_0V75	PLL2 Analog 0.75V supply
AVDD_PLL2_1V8	PLL2 Analog 1.8V supply
AVDD_PLL_LJ0_0V75	PLL_LJ0 Analog 0.75V supply
AVDD_PLL_LJ0_1V8	PLL_LJ0 Analog 1.8V supply
AVDD_PLL_LJ1_0V75	PLL_LJ1 Analog 0.75V supply
AVDD_PLL_LJ1_1V8	PLL_LJ1 Analog 1.8V supply
AVDD_PLL_LJ2_0V75	PLL_LJ2 Analog 0.75V supply
AVDD_PLL_LJ2_1V8	PLL_LJ2 Analog 1.8V supply
AVDD_PLL_LJ3_0V75	PLL_LJ3 Analog 0.75V supply
AVDD_PLL_LJ3_1V8	PLL_LJ3 Analog 1.8V supply
AVDD_PVT0_1V8	PVT0 Analog power supply, 1.8V
AVDD_PVT1_1V8	PVT1 Analog power supply, 1.8V
AVDD_PVT2_1V8	PVT2 Analog power supply, 1.8V
AVDD_PVT3_1V8	PVT3 Analog power supply, 1.8V
AVDD_PVT4_1V8	PVT4 Analog power supply, 1.8V
AVDD_PVT5_1V8	PVT5 Analog power supply, 1.8V
AVDD_PVT6_1V8	PVT6 Analog power supply, 1.8V
AVDD_PVT7_1V8	PVT7 Analog power supply, 1.8V
AVDD_UFS_0V75	UFS Analog 0.75V supply
AVDD_UFS_1V8	UFS Analog 1.8V supply
AVDD_USB0_0V75	USB3.1 PHY0 core supply, 0.75V
AVDD_USB0_1V8	USB3.1 PHY0 I/O supply, 1.8V
AVDD_USB0_3V3	USB2.0 PHY0 analog supply, 3.3V
AVDD_USB0_VDDH	USB2.0 PHY0 analog supply, 3.3V
AVDD_USB1_0V75	USB3.1 PHY1 core supply, 0.75V

AVDD_USB1_1V8	USB3.1 PHY1 I/O supply, 1.8V
AVDD_USB1_3V3	USB2.0 PHY1 analog supply, 3.3V
AVDD_USB1_VDDH	USB2.0 PHY1 analog supply, 3.3V
DVDD_AON_0V75	AON core supply, 0.75V
DVDD_DDR0_0V75	DDR0 core supply, 0.75V
DVDD_DDR1_0V75	DDR1 core supply, 0.75V
DVDD_ISPCV_0V75	ISP/CV core supply, 0.75V
DVDD_SOC_0V75	SOC and NET core supply, 0.75V
DVDD_ST_0V75	ST core supply, 0.75V
VDDIO_AON_1V8	AON IO supply, 0.75V
VDDIO_RGMII	RGMII IO supply, 1.8V/3.0V
VDDIO_SDIO18_00	Internal 1.8V supply output for SDIO18_00, 1.8V
VDDIO_SDIO18_01	Internal 1.8V supply output for SDIO18_01, 1.8V
VDDIO_SDIO18_02	Internal 1.8V supply output for SDIO18_02, 1.8V
VDDIO_SDIO18_10	Internal 1.8V supply output for SDIO18_10, 1.8V
VDDIO_SDIO18_11	Internal 1.8V supply output for SDIO18_11, 1.8V
VDDIO_SDIO18_12	Internal 1.8V supply output for SDIO18_12, 1.8V
VDDIO_SDIO18_RGMII_0	Internal 1.8V supply output for SDIO18_RGMII0, 1.8V
VDDIO_SDIO18_RGMII_1	Internal 1.8V supply output for SDIO18_RGMII1, 1.8V
VDDIO_SDIO18_RGMII_2	Internal 1.8V supply output for SDIO18_RGMII2, 1.8V
VDDIO_SDIO_0	SDIO IO0 supply, 1.8V/3.0V
VDDIO_SDIO_1	SDIO IO1 supply, 1.8V/3.0V
VDDIO_SOC_1V8_0	SOC IO0 supply, 1.8V
VDDIO_SOC_1V8_1	SOC IO1 supply, 1.8V
VDDIO_ST_1V8	ST IO supply, 1.8V
DVDD_SOC_SENSE	SOC and NET core supply sense pin
DVDD_AON_SENSE	AON core supply sense pin
DVDD_ISPCV_SENSE	ISP/CV core supply sense pin
VSS_SOC_SENSE	SOC and NET core supply sense pin return path
VSS_ISPCV_SENSE	ISP/CV core supply sense pin return path
AVSS_ADC	ADC supply return path
VC_PVT0	PVT0 supply return path
VC_PVT1	PVT1 supply return path
VC_PVT2	PVT2 supply return path
VC_PVT3	PVT3 supply return path
VC_PVT4	PVT4 supply return path
VC_PVT5	PVT5 supply return path
VC_PVT6	PVT6 supply return path
VC_PVT7	PVT7 supply return path
VSS	Core supply and IO supply return path

Notes:

It is required to implement 0.5uF-1uF capacitor on VDDPST18 pin on PCB to stabilize the output voltage of LDO.

8.3 Pinmux and Ball map

The pinmux and ball map please refer to *C1200 IOMUX and Ballmap Configuration.xlsx*.

9 Revision history

Table 8-1 shows supplies contact assignments.

Table 9-1 Revision history

Revision Number	Date	Change(s)
V1.0	2023/05/21	Initial version
V1.1	2024/03/30	Informal version, some information may be adjusted.
V1.2	2024/09/30	Add additional series number description. ES version, some information may be adjusted.
V1.21	2024/10/31	Add display row in chapter 4 (Device Comparison). Fix description in ISP chapter.